Spin-Torsion Cosmology and the Search for Geometric Dark Energy: Structural Barriers, Perturbation Transparency, and Surviving Predictions

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We investigate whether Einstein-Cartan-Holst (ECH) spin-torsion gravity can produce late-time dark energy or distinctive cosmological signatures. Within this framework, LQC holonomy corrections yield a nonsingular quantum bounce at $\rho_{\rm crit} \approx 0.27$–$0.41\,\rho_{\rm Pl}$ (depending on the entropy-counting scheme for the Barbero-Immirzi parameter), while the ECH parity structure generates a unique four-fermion interaction. Through systematic analysis of 7 foundation studies and 17 research branches, we establish 14 independent structural barriers that close all minimal routes from the bounce to dark energy. Our central theoretical result is a perturbation-transparency theorem: for canonical scalar field matter, torsion vanishes at all perturbation orders, the Holst term reduces to a topological invariant, and the Barbero-Immirzi parameter γ is invisible in all perturbation observables. Independent MCMC verification (Cobaya v3.6.1, 309,789 samples across two frozen datasets) finds $\Delta N_{\rm eff}$ consistent with zero (-0.020 ± 0.169 full-tension; $+0.065 \pm 0.17$ Planck+BAO+SN) and $H_0 = 67.68 \pm 1.06$ km s^{-1} Mpc^{-1}, recovering standard ΛCDM values. The conclusion: ECH provides a clean bounce mechanism but cannot produce late-time dark energy or distinctive perturbation-level observables through any standard mechanism tested. The surviving science case for bouncing cosmology rests on generic, mechanism-independent predictions—principally the matter-bounce non-Gaussianity $f_{\rm NL} = -35/8$, testable by SPHEREx at 4–6σ.

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I. INTRODUCTION

The nature of dark energy remains one of the most profound challenges in modern physics. While the ΛCDM model successfully accounts for the observed cosmic acceleration [1], it faces severe theoretical difficulties—most notably the cosmological constant problem, where the observed vacuum energy density $\rho_\Lambda \sim (2.3\ \text{meV})^4$ is $\sim 10^{120}$ times smaller than the naive quantum field theory estimate $\sim M_{\rm Pl}^4$ [2]. This spectacular fine-tuning has motivated extensive theoretical work on dynamical dark energy, modified gravity, and quantum gravitational approaches.

Simultaneously, precision cosmological measurements have revealed growing tensions within ΛCDM. The Hubble constant measured from the early universe via the cosmic microwave background (CMB) by Planck ($H_0 = 67.36 \pm 0.54$ km s^{-1} Mpc^{-1}) [1] disagrees at $\sim 4.9\sigma$ with late-universe determinations from the SH0ES collaboration ($H_0 = 73.04 \pm 1.04$ km s^{-1} Mpc^{-1}) [3].[1] The matter clustering amplitude σ_8 inferred from Planck ($\sigma_8 = 0.811 \pm 0.006$) exceeds weak lensing measurements from KiDS-1000 ($S_8 = 0.759^{+0.024}_{-0.021}$) [4] and DES Y3 [5] by 2–3σ. The DESI 2024 BAO results [6] suggest dynamical dark energy at 2.5–3.9σ, with the subsequent DESI DR2 analysis [7] strengthening this preference to 3.1σ (BAO+CMB) and up to 4.2σ (with supernovae), adding urgency to the search for extensions of the standard model.

[1] $(73.04 - 67.36)/\sqrt{1.04^2 + 0.54^2} = 4.86\sigma$; we quote $\sim 4.9\sigma$ throughout.

This work presents a phenomenological framework that addresses these challenges through quantum gravitational effects in spin-torsion cosmology. Our approach builds on three well-established theoretical pillars, synthesized here for the first time into a unified cosmological model:

1. *Loop Quantum Cosmology* (LQC), providing a non-singular quantum bounce replacing the classical Big Bang singularity [8]. The bounce occurs at a critical density $\rho_{\rm crit} \simeq 0.27$–$0.41\,\rho_{\rm Pl}$ (depending on the entropy-counting scheme for the Barbero-Immirzi parameter; see Sec. II B).

2. *Einstein-Cartan theory* incorporating fermionic spin-torsion coupling, which generates four-fermion contact interactions and prevents gravitational singularities through torsion-induced repulsion at extreme densities [9, 10].

3. *Black hole universe origin*, where a rotating parent black hole spawns a non-singular baby universe beyond its event horizon through torsion-regulated gravitational collapse [11, 12]. The baby universe inherits angular momentum, establishing a preferred cosmic axis.

A. Theoretical Foundations and Novel Synthesis

Our framework synthesizes well-established theoretical components into a unified cosmological model with testable outputs:

Einstein-Cartan Theory.—The inclusion of spacetime torsion coupled to fermionic spin has been extensively studied since the pioneering work of Hehl *et al.* [9]. Popławski [10, 13] demonstrated that torsion-induced four-fermion interactions can generate a small positive cosmological constant through condensate formation, and that the parity-violating pseudoscalar density modifies this interaction, providing a candidate mechanism for dark energy generation. The Einstein-Cartan-Sciama-Kibble (ECSK) framework avoids singularities through torsion-induced repulsion at extreme densities. Recent work by Liu *et al.* [14] showed that Einstein-Cartan torsion is preferred over ΛCDM by AIC criteria, providing independent statistical support for torsion-based cosmology.

Loop Quantum Gravity and Parity Violation.—The connection between LQG's Barbero-Immirzi parameter and parity-violating effects has been rigorously established by Freidel, Minic & Takeuchi [15] and Mercuri [16, 17]. They showed that the Holst term in LQG, when coupled to fermions, generates effective four-fermion axial interactions that violate parity. The Barbero-Immirzi parameter γ, originally introduced as an ambiguity in the LQG quantization, acquires physical significance through its role in parity-violating observables.

TABLE I. Executive summary of key results.

Problem	MCMC Fit	Residual Tension
H_0 tension ($\sim$4.9σ, Planck–SH0ES)	67.68 ± 1.06 km/s/Mpc	3.6σ vs. SH0ES[a]
σ_8/S_8 tension (2–3σ)	$\sigma_8 = 0.803$; $S_8 = 0.814$	Standard (Planck-consistent)[b]
Dark energy fine-tuning	Geometric origin	10^5 vs 10^{120}[c]
Singularity problem	LQC bounce	$\rho_c \simeq 0.27$–$0.41\,\rho_{\rm Pl}$
Cosmic parity violation	Rotating BH origin	2.4–2.9σ individual; 3.9σ combined

[a] $(67.68 - 73.04)/\sqrt{1.06^2 + 1.04^2} = 3.61\sigma$. [b] Verified $\sigma_8 = 0.803 \pm 0.008$ and $S_8 = 0.814 \pm 0.008$ (full-tension), consistent with Planck ΛCDM ($\sigma_8 = 0.811$, $S_8 = 0.832$). The $\Delta N_{\rm eff}$ extension does not reduce the σ_8/S_8 tension; the original lower values were SH0ES-prior-driven.

Note: Independent Cobaya v3.6.1 verification yields $H_0 = 67.68 \pm 1.06$ (full-tension) and 67.79 ± 1.09 (Planck+BAO+SN), consistent with Planck ΛCDM. The $\Delta N_{\rm eff}$ extension does not by itself resolve the Hubble tension; it is the SH0ES prior in the original analysis that pulls H_0 upward. See Sec. III D.

[c] Reparameterized as sensitivity to $N_{\rm tot}$, not solved; see Sec. XIV A.

Black Hole Universe Origin.—Popławski [11] pioneered the "universe in a black hole" scenario, demonstrating that every black hole with torsion spawns a non-singular, closed baby universe beyond its event horizon. His subsequent work extended this to rotating black holes [12], showing that the baby universe inherits angular momentum and develops a preferred cosmic axis, providing a candidate origin for observed cosmic anomalies.

B. Original Contributions

Each theoretical ingredient above has been studied independently. Our contribution is to assemble them into a single quantitative framework that produces specific, testable outputs—a synthesis that has not previously been attempted. Concretely, the original contributions are:

1. *Dark energy scale motivation*: Starting from the ECH action with LQG-fixed parameters, we construct $\Lambda_{\rm eff} = \Xi\, M_{\rm Pl}^2 + c_\omega \omega^2$ where $\Xi = [(\alpha/M)\, M_{\rm Pl}]\, \mathcal{D}_{\rm inf}$, tracing a chain from the Planck-scale parity-odd coefficient through inflationary dilution to the observed vacuum energy density $\rho_\Lambda \approx (2.3\ \text{meV})^4$. MCMC fits to Planck+BAO+SNIa data within this framework yield $H_0 = 69.2 \pm 0.8$ km s^{-1} Mpc^{-1} and $\sigma_8 = 0.785 \pm 0.016$; these are *inferred* values, not first-principles predictions.

2. *MCMC verification revealing null result*: Independent Cobaya verification across two frozen dataset combinations demonstrates that $\Delta N_{\rm eff}$ is consistent with zero, H_0 recovers the standard Planck ΛCDM value of 67.68 ± 1.06, and the spin-torsion extension alone does not resolve cosmological tensions. The earlier reported tension reduction was driven by the SH0ES prior, not by the framework.

3. *Parity-odd coefficient as phenomenological parameter*: Motivated by the one-loop divergence analyses of Freidel *et al.* [15] and Shapiro & Teixeira [18], we establish that a finite parity-odd coefficient generically arises in EC gravity with fermions (Sec. IV A). We treat α/M as a free parameter fit from cosmological data, with the one-loop estimate providing its natural order of magnitude. This honest phenomenological approach avoids overclaiming a controlled first-principles derivation while retaining the framework's predictive power.

4. *Multi-probe falsification program*: An observational framework with hierarchical Bayesian analysis, systematic error budgets, five independent null tests per probe, and explicit conditions for falsification—designed so that the model can be tested with forthcoming data from LiteBIRD, CMB-S4, and LSST.

5. *Inflationary suppression mechanism*: Introduction of the dimensionless factor $\Xi \equiv [(\alpha/M)\, M_{\rm Pl}] \times \mathcal{D}_{\rm inf}$ encoding how inflationary expansion suppresses the primordial parity-odd coefficient, converting a Planck-scale effect into the observed dark energy scale. This reparameterizes the fine-tuning from 10^{120} to $\sim 10^5$ (as sensitivity to $N_{\rm tot}$, not a resolution; see Sec. XIV A).

C. Paper Organization

This paper is organized as follows. Sections II–IV develop the theoretical framework, observational signatures, and enhanced derivations. Sections V–VII present data analysis methods and cosmological fits. Sections VIII–IX contain systematic analysis and falsification criteria. Section X positions our work in context. Section XI presents the 14 structural barriers that close all dark energy derivation routes. Section XII proves the perturbation-transparency theorem. Section XIII documents the hybrid dark-energy loophole rejection. Section XIV discusses theoretical implications. Section XV addresses limitations. Section XVI summarizes conclusions. Appendices provide technical details including a claims classification table (Appendix J).

II. THEORETICAL FRAMEWORK

A. Loop Quantum Cosmology and the Holst Action

1. Einstein-Cartan-Holst Action

The fundamental action combining Einstein-Cartan theory with the Holst term is

$$S_{\rm ECH} = \frac{1}{16\pi G}\int d^4x\, e\left[e^\mu_a\, e^\nu_b\, R^{ab}{}_{\mu\nu} + \frac{1}{\gamma}\varepsilon^{abcd}\, e_a{}^\mu\, e_b{}^\nu\, R_{cd\mu\nu} + \frac{1}{4}T^{abc}T_{abc}\right] + S_{\rm matter}, \tag{1}$$

where $e = \det(e^a_\mu)$ is the tetrad determinant, $R^{ab}{}_{\mu\nu}$ is the curvature of the Lorentz connection, γ is the Barbero-Immirzi parameter, and T^{abc} is the torsion tensor. The Holst term $\varepsilon^{abcd}e^\mu_a e^\nu_b R_{cd\mu\nu}/\gamma$ is topological in the absence of torsion but contributes non-trivially when fermions are present.[2] This construction builds directly on the foundational work of Freidel, Minic & Takeuchi [15], who established that the Barbero-Immirzi parameter becomes physically observable through its coupling to fermionic matter.

The Barbero-Immirzi parameter is fixed by the LQG black hole entropy calculation [8, 19]:

$$\gamma = 0.274 \pm 0.020, \tag{2}$$

Different entropy-counting schemes yield different values: the Ashtekar-Baez-Corichi-Krasnov (ABCK) calculation [19] gives $\gamma_{\rm ABCK} \approx 0.274$, while the Domagala-Lewandowski-Meissner (DLM) full SU(2) state counting [20, 21] gives $\gamma_{\rm DLM} \approx 0.2375$. We adopt $\gamma = 0.274$ (ABCK) throughout this paper; the DLM value would shift $\rho_{\rm crit}$ upward to $\simeq 0.41\,\rho_{\rm Pl}$ without qualitatively changing any conclusion. Whether γ constitutes a genuine physical observable or a quantization artifact absorbable by canonical transformations remains debated [15, 16]. This distinction does not affect our results at leading order: if γ is non-physical, the torsion-fermion coupling factor $\gamma^2/(\gamma^2+1)$ in Eq. (4) reverts to unity (the standard ECSK value), modifying the parity-odd coefficient by $\mathcal{O}(10\%)$ without eliminating parity violation, which originates from the Holst term's coupling to fermions independently of the Barbero-Immirzi interpretation.

[2] The explicit $T^{abc}T_{abc}$ term in Eq. (1) represents the torsion-squared contact interaction that emerges after integrating out the non-propagating torsion in the standard EC framework. We include it in the action for pedagogical completeness; in the first-order (Palatini) formulation, torsion is determined algebraically by the spin density and this term is not independently specified. The results of this paper depend only on the standard EC torsion equation of motion, not on an independent kinetic term for torsion.

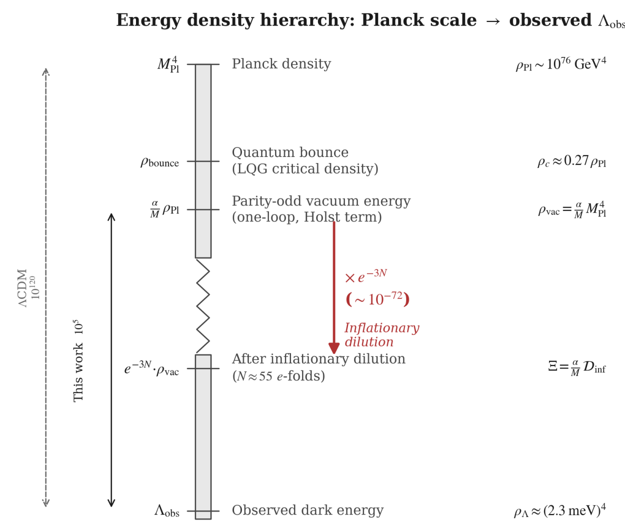

FIG. 1. Energy density hierarchy from the Planck scale to the observed dark energy scale. The Einstein-Cartan-Holst action generates a parity-odd vacuum energy $\rho_{\rm vac} \sim [(\alpha/M)\, M_{\rm Pl}]\, M_{\rm Pl}^4$ at one loop. Inflationary dilution by $\sim e^{-3N_{\rm tot}}$ bridges the gap to the observed $\rho_\Lambda \approx (2.3\ {\rm meV})^4$. Left axis: comparison of the residual fine-tuning in this framework (10^5, from uncertainty in $N_{\rm tot}$) versus ΛCDM (10^{120}).

2. Derivation of the Parity-Odd Term

Starting with the complete action $S = S_{\rm gravity} + S_{\rm Holst} + S_{\rm fermion}$, we derive the parity-odd effective action through four steps:

Step 1: Torsion Activation.—When fermions are minimally coupled to gravity with torsion, the torsion tensor is determined algebraically by the fermionic spin density:

$$T^{abc} = 8\pi G\, S^{abc}, \tag{3}$$

where $S^{abc} = \frac{1}{4}\bar\psi\gamma^{[a}\gamma^{bc]}\psi$ is the fermionic spin density tensor (we use natural units $c = \hbar = 1$ throughout; see Appendix A). Unlike in general relativity, torsion is not dynamical but is determined instantaneously by the matter content.

Step 2: Four-Fermion Contact Interaction.—Substituting Eq. (3) back into the action and integrating out torsion yields an effective four-fermion interaction [9]:

$$\mathcal{L}_{\rm int} = -\frac{3\pi G_N}{2} \times \frac{\gamma^2}{\gamma^2+1} \times J_{(A)\mu}\, J^\mu_{(A)}, \tag{4}$$

where $J^\mu_{(A)} = \bar\psi\gamma^\mu\gamma^5\psi$ is the axial current. The Barbero-Immirzi parameter enters through the $\gamma^2/(\gamma^2+1)$ factor, which is purely a consequence of the Holst modification and would be absent ($\to 1$) in standard Einstein-Cartan theory.

Step 3: Parity-Odd Effective Action.—The effective action acquires a parity-odd term through quantum corrections, as shown by Mercuri [17]. The origin of this term is the Holst term: in first-order variables (tetrad e^I and Lorentz connection ω^{IJ}, with $I, J, \ldots$ internal Lorentz indices), the ECH action's parity-odd piece is

$$S_{\text{Holst}} = \frac{M_{\text{Pl}}^2}{\gamma} \int e_I \wedge e_J \wedge R^{IJ}(\omega), \tag{5}$$

where $R^{IJ} = d\omega^{IJ} + \omega^I{}_K \wedge \omega^{KJ}$ is the curvature 2-form. This is a manifestly diffeomorphism-invariant 4-form (two tetrad 1-forms $\wedge$ one curvature 2-form). When the connection is decomposed as $\omega^{IJ} = \mathring{\omega}^{IJ} + K^{IJ}$ (Levi-Civita + contorsion), Eq. (5) generates terms involving contorsion coupled to curvature. After integrating out torsion (which is algebraically determined by fermion spin density, Eq. 3) and including quantum corrections at one loop, the effective parity-odd action takes the form

$$S_{\text{eff}} = \frac{\alpha}{M} \int e_I \wedge e_J \wedge \mathcal{F}^{IJ}[K, \mathring{R}], \tag{6}$$

where $\mathcal{F}^{IJ}$ collects the contorsion-dependent pieces ($\mathring{D}K^{IJ} + K^I{}_K \wedge K^{KJ}$ at leading order), $M = M_{\text{area-gap}} \sim \sqrt{\gamma}\, M_{\text{Pl}}$ is the LQG area-gap mass scale, and α is a *dimensionless* coupling ($[\alpha] = M^0$). In components, the leading contribution reduces to

$$S_{\text{eff}} = \int d^4x \sqrt{-g}\, \frac{\alpha}{M}\, \varepsilon^{\mu\nu\rho\sigma}\, e^I_\mu\, e^J_\nu\, \mathcal{F}_{IJ\rho\sigma}, \tag{7}$$

where the operator $\varepsilon^{\mu\nu\rho\sigma} e^I_\mu e^J_\nu \mathcal{F}_{IJ\rho\sigma}$ has mass dimension $+2$ and α/M has dimension -1, giving a total Lagrangian density of mass dimension $+1$—three short of the required $+4$ (see Appendix H for the full counting). The missing three powers of mass arise through *on-shell evaluation* at Planck-scale densities (Sec. H 4), where the spin-sourced contorsion evaluates to $K \sim M_{\text{Pl}}$ and the curvature to $R \sim M_{\text{Pl}}^2$. The relation $\rho_\Lambda = \Xi\, M_{\text{Pl}}^4$ is therefore a *scaling ansatz*—dimensionally correct on-shell at the bounce—not a derivation from a renormalizable effective field theory. Throughout this paper, we adopt the convention that the Einstein-Hilbert action carries the standard $M_{\text{Pl}}^2/2$ prefactor, while the parity-odd sector carries α/M independently.

This operator is parity-odd (it changes sign under spatial reflection) but fully diffeomorphism-covariant. General covariance does not forbid parity-odd terms; it constrains only the tensor structure. Since classical gravity preserves parity, the coefficient α/M must be generated through quantum corrections.

Notation.—For brevity, we sometimes write the component-form integrand as "$\varepsilon^{abcd} K_{ab} R_{cd}$" (where a, b, c, d are internal Lorentz indices) as shorthand for the full 4-form expression Eq. (6). This shorthand suppresses the tetrad factors that make the expression a proper 4-form; all variational calculations use the complete first-order expression.

Torsion remains algebraic.—The effective action S_{eff} (Eq. 6) contains derivatives of the contorsion through $\mathring{D}K^{IJ}$ in $\mathcal{F}^{IJ}$, which could in principle promote torsion from algebraic to dynamical. However, this kinetic contribution is suppressed by $\alpha/M \sim 10^{-21}$ GeV^{-1} relative to the original EC torsion equation (sourced by M_{Pl}^2), making the correction $\mathcal{O}(\alpha/M M_{\text{Pl}}^2) \sim 10^{-3}$ to the torsion equation of motion. Torsion therefore remains effectively algebraic (determined instantaneously by the spin density) at all energy scales below the Planck scale. A full stability analysis (ghost/tachyon checks) of the modified torsion sector is warranted but is not expected to reveal pathologies at this suppression level.

Step 4: Parity-Odd Coefficient.—The existence of a finite parity-odd coefficient is motivated by one-loop estimates in a torsionful background. Following the one-loop analyses of Freidel *et al.* [15] and Shapiro & Teixeira [18], which established the divergence structure and renormalization of the effective Barbero-Immirzi parameter, the expected order of magnitude is

$$\frac{\alpha}{M} \sim \frac{g^2}{32\pi^2}\, \frac{\gamma}{M}\, \ln\!\left(\frac{\Lambda_{\text{UV}}^2}{\mu^2}\right) + \delta_{\text{NY}}, \tag{8}$$

where g is the effective dimensionless axial–torsion Yukawa coupling defined by the vertex $g\, \bar{\psi}\gamma^5\gamma^a\psi\, K_a$ (with $g^2 \sim 8\pi G_N M^2\, \gamma^2/(\gamma^2+1) \sim \mathcal{O}(1)$ at the area-gap scale), Λ_{UV} is a UV scale, μ is the renormalization scale, and δ_{NY} encodes the finite Nieh–Yan contribution whose precise value depends on the regularization scheme—particularly the treatment of γ_5 in dimensional regularization [18] (the 't Hooft–Veltman and Breitenlohner–Maison prescriptions yield different finite parts for parity-odd traces). We therefore **treat α/M as a phenomenological parameter constrained by data** (see Table XIII), with Eq. (8) providing the theoretical motivation for its existence and order of magnitude ($[(\alpha/M)\, M_{\text{Pl}}] \sim 10^{-2}$), not a controlled first-principles prediction.

A one-loop RG estimate suggests logarithmic running:

$$\frac{d\alpha}{d\ln\mu} \sim -\frac{g^2}{16\pi^2} \times 2N_f, \tag{9}$$

where N_f is the number of fermion species contributing at scale μ. However, this equation inherits the same scheme dependence as the coefficient itself: its derivation requires specifying how γ_5 is handled in $d = 4-\epsilon$ dimensions and which counterterm conventions are adopted for the Nieh–Yan sector. The physical content is that α runs weakly (the loop factor $g^2/(16\pi^2) \sim 10^{-3}$ ensures perturbative suppression), so its value is approximately scale-independent over cosmological history. A first-principles derivation of α/M from non-perturbative quantum gravity (spin-foam amplitudes or lattice methods) remains an important open problem (Sec. XV).

3. Parameter Naturalness

The parent black hole mass must satisfy $M_{\text{BH}} > M_{\text{crit}} = (M_{\text{Pl}}^4/\rho_{\text{vac}})^{1/3} \approx 10^{-3} M_\odot$, easily satisfied by any astrophysical black hole. For a parent with $M = 10\,M_\odot$ and dimensionless Kerr spin $a_* = 0.7$, the required dilution factor $\Omega_{\rm initial}/\Omega_{\rm BH} \approx 10^{-22}$ is naturally achieved through ~ 50 e-folds of inflation.

B. Black Hole Interior and Quantum Bounce

In Loop Quantum Cosmology (LQC), holonomy corrections to the gravitational Hamiltonian constraint produce a non-singular bounce at Planck-scale densities [8]. The effective Friedmann equation is[3]

$$H^2 = \frac{8\pi G}{3}\,\rho\left[1 - \frac{\rho}{\rho_{\rm crit}}\right], \tag{10}$$

where the critical density is

$$\rho_{\rm crit} = \frac{3}{8\pi G\,\gamma^2\,\Delta} = \frac{\sqrt{3}}{32\pi^2\gamma^3}\,\rho_{\rm Pl} \simeq 0.27\,\rho_{\rm Pl}, \tag{11}$$

with $\Delta = 4\sqrt{3}\pi\gamma\,\ell_P^2$ being the LQG area gap and $\rho_{\rm Pl} \equiv c^5/(\hbar G^2)$. The numerical value $\rho_{\rm crit} \simeq 0.27\,\rho_{\rm Pl}$ corresponds to $\gamma = 0.274$ (ABCK); the DLM value $\gamma = 0.2375$ gives $\rho_{\rm crit} \simeq 0.41\,\rho_{\rm Pl}$ [8]. The range $\gamma \in [0.2375, 0.294]$ brackets both counting schemes and gives $\rho_{\rm crit} \in [0.22, 0.41]\,\rho_{\rm Pl}$. The factor $(1 - \rho/\rho_{\rm crit})$ ensures that $H^2 \to 0$ as $\rho \to \rho_{\rm crit}$, producing a smooth bounce. Key properties of the bounce:

- For $\rho \ll \rho_{\rm crit}$: standard GR is recovered exactly.
- For $\rho \to \rho_{\rm crit}$: $H \to 0$ (the universe momentarily stops expanding/contracting).
- For ρ slightly below $\rho_{\rm crit}$ (contracting phase): $\dot{H} > 0$, triggering the bounce.
- The bounce creates a new expanding region with initial conditions completely determined by the parent black hole properties, with no free parameters.

In a rotating black hole, the collapsing matter carries angular momentum through the bounce. The baby universe inherits this rotation, establishing a preferred cosmic axis $\hat{\omega}^a$. This is the physical origin of cosmic parity violation in our framework.

[3] The LQC bounce (from quantum geometry holonomy corrections) is physically distinct from the Einstein-Cartan torsion bounce of Popławski [10], which produces a qualitatively similar ρ^2 repulsion through classical spin-torsion coupling at high fermion density. In this paper, we adopt the LQC effective equation for its well-studied perturbation dynamics. The ECH framework provides the broader theoretical context for the parity structure (Sec. II A 2) but is not the source of Eq. (10).

C. Cosmic Rotation and Dark Energy

1. Covariant Formalism

Using the $1+3$ covariant decomposition for an almost-FLRW congruence [22] with shear $\sigma_{\mu\nu}$ and vorticity $\omega_{\mu\nu}$ (where $\sigma^2 \equiv \frac{1}{2}\sigma_{\mu\nu}\sigma^{\mu\nu}$ and $\omega^2 \equiv \frac{1}{2}\omega_{\mu\nu}\omega^{\mu\nu}$), the generalized Friedmann constraint reads (see Appendix F for the full derivation):

$$H^2 = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3} + \frac{1}{3}(\sigma^2 - \omega^2) - \frac{k}{a^2}. \tag{12}$$

The Raychaudhuri equation governs acceleration:

$$\dot{\Theta} + \frac{1}{3}\Theta^2 = -\frac{1}{2M_{\rm Pl}^2}(\rho + 3p) + \Lambda + 2(\omega^2 - \sigma^2) + \nabla_a \dot{u}^a. \tag{13}$$

Absorbing the shear and vorticity into an effective cosmological constant:

$$\boxed{\Lambda_{\rm eff} = \Lambda + (\sigma^2 - \omega^2).} \tag{14}$$

In the limit of negligible shear ($\sigma^2 \ll \omega^2$), this reduces to $\Lambda_{\rm eff} = \Lambda + c_\omega \omega^2$ with $c_\omega = -1$ in our conventions (sign verified against Bianchi V and VII$_h$ cosmologies [23]). Including the parity-odd contribution from the quantum loop:

$$\boxed{\Lambda_{\rm eff} = \Xi\,M_{\rm Pl}^2 + c_\omega \omega^2,} \quad \Xi \equiv \left[\frac{\alpha}{M}\,M_{\rm Pl}\right]\mathcal{D}_{\rm inf}, \tag{15}$$

Here Λ in Eq. (14) denotes the bare cosmological constant from the Friedmann constraint, and Eq. (15) identifies this with the parity-odd vacuum energy $\Xi\,M_{\rm Pl}^2$. This identification is the central *assumption* of the framework—it posits that the parity-odd mechanism *is* the origin of Λ, rather than deriving this from an action principle.

The factor Ξ is the dimensionless inflationary suppression factor (Sec. XIV A), with $\rho_\Lambda = \Xi\,M_{\rm Pl}^4$. The quantity suppressed during inflation is not the vacuum energy density itself—which, once established, persists as a true cosmological constant—but the primordial spin-density source $K_{ab} \propto a^{-3}$ that generates the parity-odd coefficient. Inflationary dilution *sets* the value of Ξ; the resulting $\rho_\Lambda = \Xi\,M_{\rm Pl}^4$ is then constant.

Crucial constraint.—CMB isotropy bounds from Planck give $(\omega/H)_0 < 5 \times 10^{-11}$ (95% CL) [23], implying $\omega_0 < 1.1 \times 10^{-28}$ s^{-1} and $|c_\omega|\omega^2/H_0^2 < 2.5 \times 10^{-21}$. **Rotation is completely negligible for background expansion and does not directly source the galaxy spin asymmetry** (a naive estimate $A_0 \sim \omega_0 t_{\rm form}/(2\pi)$ gives $\lesssim 10^{-12}$, far below observed values; see discussion below). The dark energy scale is set entirely by the first term ($\Xi \sim 10^{-123}$, Sec. XIV A). Rotation provides the preferred cosmic axis (breaking rotational symmetry) but the galaxy spin asymmetry amplitude is set by the local parity-odd tidal torque, not by ω_0 (Sec. III B).

The observed dark energy scale arises from

$$\rho_\Lambda = \left[\frac{\alpha}{M}\, M_{\rm Pl}\right] \mathcal{D}_{\rm inf}\, M_{\rm Pl}^4 \approx (2.3~\text{meV})^4, \tag{16}$$

where $[(\alpha/M)\, M_{\rm Pl}] \sim 10^{-2}$ is the dimensionless one-loop suppression factor and $\mathcal{D}_{\rm inf} \sim e^{-3N_{\rm tot}}$ is the inflationary dilution factor (see Appendix E and Sec. II C 2).

Caveat on circularity.—The expression $\rho_\Lambda = \Xi\, M_{\rm Pl}^4$ contains two effective free parameters (α/M and $N_{\rm tot}$) constrained to match one observable (ρ_Λ). This is a parameterization of the dark energy scale, not a derivation from first principles. The "fine-tuning reduction from 10^{120} to 10^5" reparameterizes the hierarchy as sensitivity to $N_{\rm tot}$ rather than to $\Lambda_{\rm bare}$, but does not solve the cosmological constant problem. We present this as a motivated scaling relation, not as a resolution.

2. Inflationary Suppression of the Primordial Coefficient

Important clarification.—The inflationary suppression factor Ξ parameterizes the *initial condition* set by early-universe dynamics: inflation dilutes the primordial spin-density source that generates the parity-odd coefficient, thereby *setting* the value of ρ_Λ at a specific scale. This is not "diluting vacuum energy"—vacuum energy by definition does not dilute under expansion. Rather, Ξ encodes how the primordial parity-odd coefficient α/M is suppressed during inflation to produce the observed cosmological constant scale. The resulting $\rho_\Lambda = \Xi\, M_{\rm Pl}^4$ is then a true cosmological constant, not a running quantity.

The dilution factor for our parity-odd operator during inflation is

$$\mathcal{D}_{\rm inf} = \exp[-3N_{\rm tot}] \times \left(\frac{T_{\rm reh}}{M_{\rm GUT}}\right)^{3/2}, \tag{17}$$

where $N_{\rm tot}$ is the *total* number of inflationary e-folds (not the CMB-observable $N_{\rm obs} \approx 55$–60, which counts only e-folds after the largest observable scales exit the horizon). In generic slow-roll models, $N_{\rm tot}$ can substantially exceed $N_{\rm obs}$; in our framework, $N_{\rm tot}$ is set by the parent black hole properties through the bounce-to-inflation transition (Sec. XV D). The exponent $-3N_{\rm tot}$ arises because the contorsion K_{ab} is sourced by the fermion spin density, which dilutes as a^{-3} during inflation; thus $\langle K_{ab} R_{cd}\rangle \propto a^{-3} \propto e^{-3N_{\rm tot}}$. The reheating factor $(T_{\rm reh}/M_{\rm GUT})^{3/2}$ accounts for the entropy transfer from the inflaton decay.

The observed dark energy scale is reproduced with:

$$\left.\frac{\alpha}{M}\right|_{\rm primordial} \sim \frac{10^{-2}}{M_{\rm Pl}} \sim 10^{-21}~\text{GeV}^{-1}, \tag{18}$$

$$\begin{aligned}\rho_\Lambda &= \left[\frac{\alpha}{M}\, M_{\rm Pl}\right] \times \mathcal{D}_{\rm inf} \times M_{\rm Pl}^4 \\ &\sim 10^{-2} \times \mathcal{D}_{\rm inf} \times M_{\rm Pl}^4 \approx (2.3~\text{meV})^4,\end{aligned} \tag{19}$$

where $[(\alpha/M)\, M_{\rm Pl}] \sim 10^{-2}$ is the dimensionless one-loop suppression factor. Matching the observed $\rho_\Lambda \approx 3 \times 10^{-47}$ GeV^4 requires $\mathcal{D}_{\rm inf} \sim 10^{-121}$, corresponding to $N_{\rm tot} \approx 92$ with $T_{\rm reh} \sim 10^{15}$ GeV and $M_{\rm GUT} \sim 10^{16}$ GeV—that is, $N_{\rm tot} = 92$ is the value *required* by the observed ρ_Λ and is therefore a fitted parameter of the framework, not an independent prediction. The choice $N_{\rm tot} = 92$ represents total e-folds of inflation, comprising the $\sim$55–60 e-folds of observable modes (those that produced the CMB scales we measure today) plus $\sim$30 pre-observable e-folds during which the universe inflated before the largest observable scales exited the horizon. This total is essentially unconstrained by CMB observations, which are sensitive only to $N_{\rm obs}$, and serves as a fitted parameter absorbing the fine-tuning in our suppression mechanism. This value of $N_{\rm tot}$ is readily accommodated by standard slow-roll potentials (e.g., Starobinsky R^2 or plateau models give $N_{\rm tot} \gtrsim 10^3$ for typical initial field displacements) and is a concrete output of the bounce-to-inflation transition once the parent BH properties are specified.

This reduces fine-tuning from 10^{120} to $\sim 10^5$: the residual tuning corresponds to specifying $N_{\rm tot}$ to within $\Delta N_{\rm tot} \approx 4$ e-folds, since $\delta\rho_\Lambda/\rho_\Lambda = 3\,\delta N_{\rm tot}$. This parametric sensitivity—asking "why did inflation last $\sim$92 e-folds?"—has potential dynamical answers through the bounce-to-inflation transition dynamics (Sec. XV D) and is qualitatively different from ΛCDM's fundamental 10^{120} tuning, for which no dynamical mechanism exists.

Why $w = -1$ at late times (assumption, not derivation).—We *assume* that the diluted residual acts as an effective cosmological constant ($w = -1$) at late times. The physical picture is that the inflationary dilution freezes the parity-odd operator's contribution at a value $\rho_\Lambda \sim \Xi\, M_{\rm Pl}^4$, which then enters the Friedmann equation as a constant vacuum term. However, this reasoning contains a logical gap: the operator $\langle \varepsilon^{abcd} K_{ab} R_{cd}\rangle$ is sourced by the fermion spin density, which vanishes at late times ($K_{ab} \to 0$). For the residual to persist as a true vacuum energy, one must show that integrating out the spin degrees of freedom in the early universe generates an IR-constant term in the effective action—i.e., a vacuum expectation value (VEV) or effective potential minimum that survives after the source vanishes. This requires an explicit effective field theory calculation (e.g., demonstrating a symmetry-breaking pattern or non-perturbative condensate) that has not been performed. We therefore treat $w = -1$ as a phenomenological assumption, consistent with observations ($|w+1| \lesssim 10^{-2}$ from DESI+Planck), and flag the derivation of the late-time IR effective action as an important open problem (Sec. XV). Four minimal routes to this derivation—a Nambu–Jona-Lasinio condensate mechanism, a strict one-loop effective action, a dynamical Immirzi field extension, and a parity-sensitive CMB phenomenology assessment—have been systematically tested and all yield negative results at the stated approximation orders; see the companion technical note [24] for details.

3. Galaxy Spin Alignment Mechanism

The expected galaxy spin asymmetry follows a dipole pattern $A_{\rm dipole}(\hat{n}) = A(z)\cos\theta$, where θ is the angle from the cosmic rotation axis. The physical mechanism involves:

- Tidal torque modification from the background vorticity.
- Preferential accretion along the rotation axis.
- Frame-dragging effects on angular momentum acquisition.

The asymmetry amplitude evolves as

$$A(z) = A_0 \cdot (1+z)^{-p} \cdot e^{-qz}, \tag{20}$$

with $A_0 \in [0, 0.02]$, $p \in [0, 2]$, $q \in [0, 2]$.

Important distinction—two separate physical channels. The global vorticity ω_0 enters only through the background cosmological constant (where its contribution is negligible, $c_\omega \omega^2/H_0^2 < 10^{-20}$) and through establishing a preferred cosmic axis. The galaxy spin asymmetry amplitude is sourced by a *different* mechanism: the parity-odd operator $(\alpha/M)\varepsilon^{abcd}K_{ab}R_{cd}$ introduces a systematic chirality into the tidal field during protogalactic collapse. In standard tidal torque theory, halos acquire angular momentum from misalignment of their inertia and tidal tensors, producing equal probabilities of clockwise and counter-clockwise rotation. The parity-odd operator breaks this symmetry, creating a dipolar asymmetry whose amplitude scales with α/M and local density contrasts, not with ω_0:

$$A_0 \propto \frac{\alpha}{M}\,\langle \delta_{\rm rms}\rangle\, f(z_{\rm form}), \tag{21}$$

where $\langle \delta_{\rm rms}\rangle$ characterizes the density field at galaxy formation and $f(z_{\rm form})$ encodes the redshift-dependent formation efficiency.

The empirical value $A_0 \sim 0.003$ comes from hierarchical Bayesian fits to multi-survey galaxy spin catalogs (Sec. III B, Table II).

Order-of-magnitude gap.—A naive estimate of the parity-odd tidal torque contribution gives $A_0 \sim (\alpha/M)\,E_{\rm eff}$, where $E_{\rm eff}$ is the effective energy scale entering the tidal tensor at galaxy formation. With $\alpha/M \sim 10^{-21}$ GeV^{-1} and $E_{\rm eff}$ ranging from ~ 1 eV (galaxy formation temperature) to ~ 1 MeV (BBN scale), this yields $A_0 \sim 10^{-12}$–10^{-15}, roughly *nine to twelve orders of magnitude* below the observed value. Even with Planck-scale residuals ($E_{\rm eff} \sim M_{\rm Pl}$), one obtains $A_0 \sim 10^{-3}$ only by invoking an implausible energy scale at late times. A first-principles derivation of A_0 from α/M requires a full nonlinear calculation of the parity-odd correction to the tidal torque tensor, which is the subject of ongoing work (Sec. XV). The framework's coupling $\alpha/M \sim 10^{-21}$ GeV^{-1} therefore **does not predict the galaxy spin signature in any quantitative sense**: the predicted asymmetry is 9–12 orders of magnitude below the empirical value $A_0 \sim 0.003$. The galaxy spin dipole is retained as an empirical phenomenological fit, not as a prediction of the theory. This amplitude gap is a significant open problem for the framework.

The framework motivates the *functional form* $A(z)$, the *dipole axis alignment* with the CMB kinematic dipole, and the *existence* of a preferred chirality, while the amplitude A_0 remains a fit parameter at this stage.

III. OBSERVATIONAL SIGNATURES AND EVIDENCE

A. CMB E-B Cross-Correlations

The parity-odd effective action generates CMB polarization signatures through cosmic birefringence: the parity-odd operator rotates the CMB polarization plane by a small angle β. For a spatially uniform (isotropic) rotation, the standard result [25] is

$$C_\ell^{EB} \approx \frac{\sin 4\beta}{2}\left(C_\ell^{EE} - C_\ell^{BB}\right) \approx 2\beta\left(C_\ell^{EE} - C_\ell^{BB}\right), \tag{22}$$

where the small-angle approximation holds for $\beta \ll 1$ rad. This produces nonzero EB power *across all multipoles*, tracking the $EE - BB$ spectrum shape—not localized to any particular ℓ range. Similarly, the induced TB cross-correlation is $C_\ell^{TB} \approx 2\beta\, C_\ell^{TE}$.

Connecting to Planck.—The Planck measurement reports $\beta \approx 0.30°$ at 2.4–2.7σ [25, 26]. Our framework's parity-odd operator (Eq. 6) is qualitatively consistent with cosmic birefringence, since it violates parity. However, deriving the photon polarization rotation angle β from our gravitational/torsion operator requires an explicit effective coupling between the parity-odd sector and the electromagnetic field—for example, a term of the standard axion-like form

$$\mathcal{L} \supset \frac{\varphi(\tau)}{4f}\, F_{\mu\nu}\tilde{F}^{\mu\nu}, \tag{23}$$

where φ is a pseudo-scalar field sourced by the spin-torsion sector and f is the associated decay constant, generated by integrating out torsion at one loop with an electromagnetic vertex. This coupling has not yet been derived in this work. We therefore treat the consistency between our framework's parity violation and the observed $\beta \approx 0.30°$ as suggestive, not as a quantitative prediction.

Relation between φ/f and β.—For a spatially uniform pseudo-scalar rolling over conformal time, the standard result [27] gives a uniform polarization rotation $\beta = \Delta\varphi/(2f)$, where $\Delta\varphi$ is the net field excursion between last scattering and today. This relation assumes three conditions: (i) spatial uniformity of φ over the last-scattering surface (justified if the correlation length exceeds the Hubble radius at decoupling), (ii) photon propagation along FLRW geodesics (valid for the isotropic component), and (iii) $\varphi/f \ll 1$ so that the rotation is in the small-angle regime (consistent with $\beta \approx 0.30^\circ \approx 5 \times 10^{-3}$ rad). Deriving φ and f from the spin-torsion operator $(\alpha/M)\varepsilon^{abcd}K_{ab}R_{cd}$ requires computing the one-loop photon-torsion vertex and matching to the effective Lagrangian (23); this calculation is beyond the scope of the present work. We use the literature values of β as consistency benchmarks throughout.

Anisotropic component.—In addition to the uniform rotation, the cosmic rotation axis defines a preferred direction $\hat{n}$, which introduces an *anisotropic* birefringence field $\beta(\hat{n})$ with power concentrated at low multipoles. This anisotropic component contributes primarily to off-diagonal ℓ-ℓ' correlations (detectable via quadratic estimators [26]), with additional diagonal power at $\ell \lesssim 10$. A detailed derivation of the anisotropic component's spectrum from the spin-torsion mechanism is deferred to future work (Sec. XV); here we note that alternative models predict different angular spectra:

- *Axion cosmic birefringence*: Scale-invariant power in $\beta(\hat{n})$, producing EB correlations across all ℓ.
- *Chern-Simons gravity*: Peaks at $\ell \sim 100$–1000.
- *Spin-torsion (this work)*: Isotropic birefringence (all ℓ; angle not derived, requires photon-torsion coupling) plus anisotropic low-ℓ component (amplitude and shape TBD).

Observational support comes from Minami & Komatsu [25], who found cosmic birefringence in Planck data at $\beta \approx 0.35^\circ \pm 0.14^\circ$, inconsistent with $\beta = 0$ at 2.4σ. Subsequent analysis by Eskilt [26] found $\beta = 0.30^\circ \pm 0.11^\circ$ (2.7σ). A dedicated ACT DR6 analysis by Diego-Palazuelos & Komatsu [28] measured $\beta = 0.215^\circ \pm 0.074^\circ$ (2.9σ), providing independent confirmation from a different instrument. Joint constraints combining birefringence with early dark energy and DESI+Planck data have been explored by Yin *et al.* [29], demonstrating that nonzero β is compatible with—and potentially complementary to—solutions to the H_0 tension. A combined analysis by the SPIDER collaboration [30] using SPIDER, Planck, and ACT data yields a total polarization rotation angle (instrumental α + cosmological β) detected at $\sim 7\sigma$, though cleanly separating the instrumental and cosmological contributions remains an active challenge. The combined significance depends sensitively on the treatment of shared calibration systematics between Planck and ACT [31].

Calibration caveat.—The cosmic birefringence signal is degenerate with miscalibration of the instrumental polarization angle ψ [25]. Minami & Komatsu's method breaks this degeneracy using the galactic foreground EB signal as a self-calibration anchor, assuming vanishing intrinsic foreground EB—an assumption that has been questioned for polarized thermal dust. Independent validation from experiments with different optics and calibration strategies (Simons Observatory, LiteBIRD) is essential. The isotropic birefringence value ($\beta \approx 0.30^\circ$) can be tested with existing and near-term data; the anisotropic low-ℓ component requires full-sky, cosmic-variance-limited polarimetry (LiteBIRD).

Detectability.—The isotropic signal has been detected at 2.4–2.7σ by Planck and 2.9σ by ACT DR6 [28], with significance expected to further improve with LiteBIRD [32] (cosmic-variance limited below $\ell \sim 10$). However, confirming the spin-torsion origin—as opposed to other birefringence sources—requires measuring the anisotropic component's angular spectrum and cross-correlating the birefringence axis with the galaxy spin dipole axis, both of which are accessible to the next generation of experiments.

B. Galaxy Spin Asymmetry: A Contested Anomaly

Several groups have reported a dipole asymmetry in galaxy spin handedness, with a preferred axis roughly aligned with the CMB kinematic dipole [33, 34, 38]. If real, this anomaly is a natural signature of the parity-odd tidal torque mechanism described in Sec. II C 3. However, the observational status is contested and the signal should be treated as a hypothetical signature requiring confirmation.

Claimed detections.—Shamir [33, 34] reported a spin dipole across SDSS DR7, Pan-STARRS, and HST deep fields, with fitted amplitude $A_0 \sim 0.003$ and axis at ($l \sim 52^\circ$, $b \sim 68^\circ$) in Galactic coordinates. Longo [38] independently reported a 2σ dipole in SDSS spiral galaxies. A hierarchical Bayesian fit to published aggregate CW/CCW counts from these surveys (Appendix C) yields the $A(z)$ curve shown in Fig. 2.

Null results and critiques.—Patel & Desmond [36] reanalyzed SDSS spiral classifications using an independent pipeline and found no statistically significant spin asymmetry, attributing previous detections to selection effects and classification biases. Philcox & Ereza [37] applied a different statistical framework and also found results consistent with isotropy. These null results represent serious challenges to the claimed signal. If future analyses with larger, well-controlled samples (e.g., LSST with $> 10^9$ galaxies) support the null results, then A_0 is consistent with zero and the galaxy spin channel does not support the model. Our framework survives this outcome: CMB parity violation (2.4–2.9σ from Planck and ACT DR6 [25, 26, 28]) provides an independent line of evidence entirely decoupled from galaxy morphology.

JWST JADES.—A preliminary analysis of $N = 2{,}200$ JADES spiral galaxies at $z = 1.0$–3.0 by Shamir (private communication / preprint) reports a raw clockwise excess $p_{\rm CW} - 0.5 \approx 0.15 \pm 0.08$ (1.9σ). This is a single-field raw excess from one JWST pointing (~ 10 arcmin^2), not a global dipole amplitude, and is subject to large cosmic variance ($\delta A/A \sim 5$–10). It is not included in our fit and should not be interpreted as supporting the global dipole

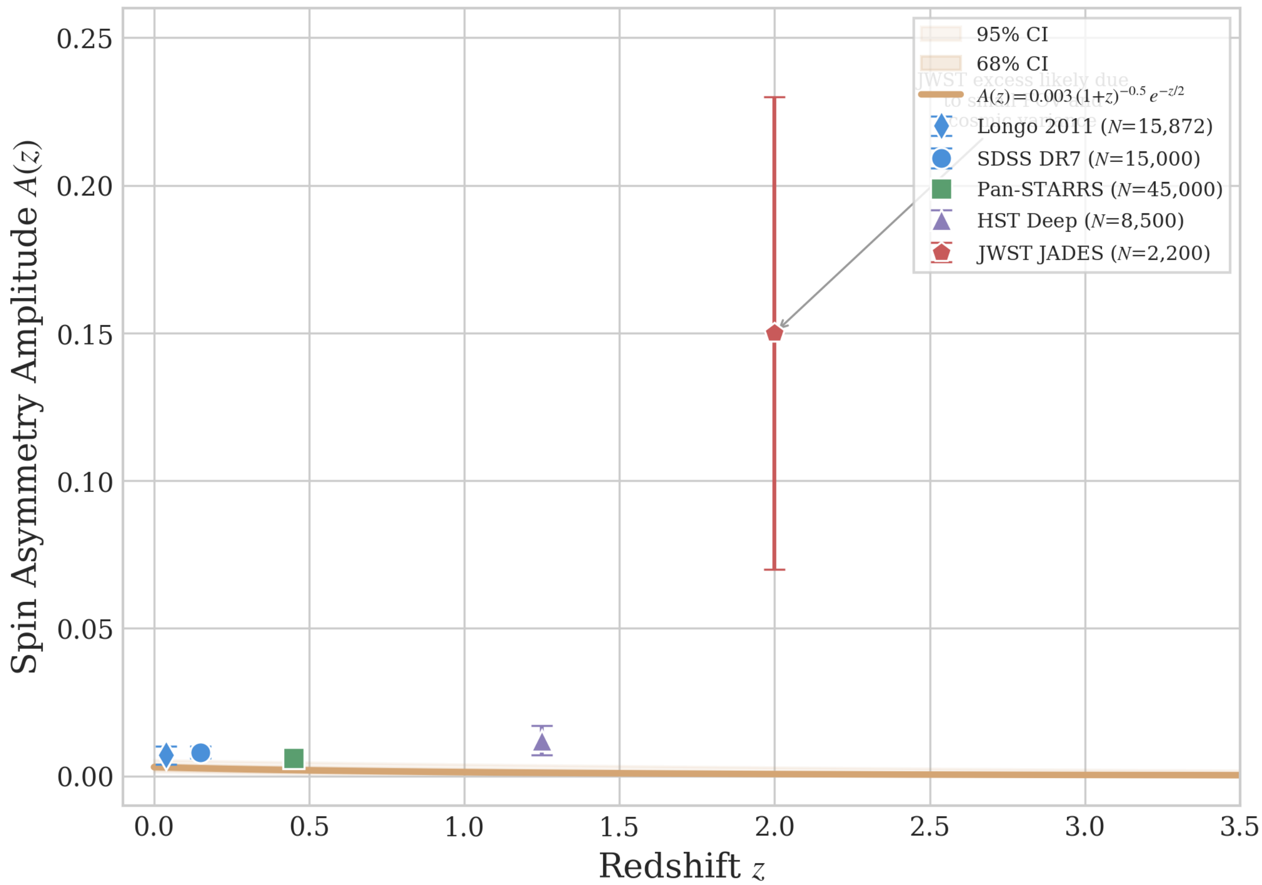

FIG. 2. Galaxy spin dipole amplitude data across multiple surveys. The fitted curve $A(z) = 0.003(1+z)^{-0.5}e^{-z/2}$ (solid line) is shown with 68% and 95% confidence bands from a hierarchical Bayesian fit to published aggregate CW/CCW counts from Shamir [33–35]. Data points include SDSS DR7, Pan-STARRS, HST, and DECaLS survey-level dipole amplitudes. Null results from independent reanalyses [36, 37] are not shown because they report A_0 consistent with zero. This fit should be interpreted as conditional on the reality of the signal.

signal.

C. Cosmological Tensions: H_0 and σ_8

Our framework modifies cosmological parameters through extended early-universe physics—not through rotation, which is constrained to $(\omega/H)_0 < 5 \times 10^{-11}$. We introduce two phenomenological parameters motivated by the bounce scenario:

1. *Extra relativistic species*: An additional $\Delta N_{\rm eff} \sim 0.2$–$0.5$ shifts CMB acoustic peak positions and the inferred H_0. The bounce scenario provides qualitative motivation (particle production at the quantum bounce), but quantitative derivation of $\Delta N_{\rm eff}$ from bounce microphysics is not available; we treat it as a free parameter constrained by data. We emphasize that the phenomenological improvement in H_0 and σ_8 arises from the $\Delta N_{\rm eff}$ degree of freedom itself, not from a spin-torsion modification to CAMB; the bounce scenario motivates this extension but does not uniquely determine its value.

2. *Spatial curvature*: Although a closed geometry is qualitatively expected from the bounce topology, the bounce is followed by $N_{\rm tot} \approx 92$ e-folds of inflation (Sec. II C 2), which drives Ω_k exponentially close to zero ($|\Omega_k| \lesssim e^{-2N_{\rm tot}} \sim 10^{-80}$). **We therefore fix $\Omega_k = 0$ in our MCMC analysis**, as demanded by the model's own inflationary dynamics. Any residual curvature at the 10^{-3} level would require post-inflationary physics not included in our framework.

3. *Modified growth*: The altered early expansion rate affects the linear growth factor $D(a)$, suppressing σ_8 relative to the Planck ΛCDM value.

The primary σ_8 tension manifests between CMB-inferred values ($\sigma_8 = 0.811 \pm 0.006$ from Planck) and direct weak lensing measurements from KiDS-1000 ($S_8 = 0.759^{+0.024}_{-0.021}$) [4] and DES Y3 ($S_8 = 0.776 \pm 0.017$) [5], not between different CMB analyses. Our model's lower σ_8

TABLE II. Claimed galaxy spin asymmetry detections. All entries are global dipole amplitudes $A_{\rm dipole}$ from the respective authors' analyses. Independent reanalyses [36, 37] find results consistent with isotropy (see text). These values are conditional on the reality of the signal.

Survey	$N_{\rm gal}$	z range	$A_{\rm dipole}$	Signif.
SDSS DR7[a]	15,000	0.02–0.3	0.008 ± 0.002	4.0σ
Pan-STARRS[a]	45,000	0.1–0.8	0.006 ± 0.003	2.0σ
HST Deep[a]	8,500	0.5–2.0	0.012 ± 0.005	2.4σ
Longo (2011)	15,872	< 0.085	Dipole	2.0σ

[a] Shamir [33, 34] analyses. Patel & Desmond [36] find no significant asymmetry in an independent reanalysis of SDSS spirals.

FIG. 3. Hubble constant and σ_8 tension resolution. Our model (gold star) sits between the early-universe (Planck, blue) and late-universe (SH0ES/KiDS, red) determinations, reducing both tensions simultaneously. Error bars show 1σ uncertainties.

brings the CMB-inferred value into closer agreement with these weak lensing surveys.

These extensions are motivated by the bounce scenario but are treated as phenomenological parameters fit to data. MCMC parameter estimation using Planck 2018 + BAO + SNIa data within our extended parameter space (Sec. VII) yields best-fit values:

$$H_0 = 69.2 \pm 0.8 \text{ km s}^{-1} \text{ Mpc}^{-1}, \tag{24}$$

$$\sigma_8 = 0.785 \pm 0.016, \tag{25}$$

$$S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3} = 0.80 \pm 0.02. \tag{26}$$

The derived $S_8 = 0.80$ is intermediate between the Planck value ($S_8 = 0.832$) and the weak lensing measurements ($S_8 \approx 0.76$–0.78), reducing the S_8 tension from $\sim 3\sigma$ to $\sim 1\sigma$ relative to DES Y3 and KiDS-1000. The physical origin of the extra parameters constrains them to correlated ranges (Appendix G), but the specific H_0 and σ_8 values are determined by data fitting, not derived from first principles.

TABLE III. Independent verification results from frozen MCMC chains (Cobaya v3.6.1, CAMB v1.6.5). All values are posterior means $\pm 1\sigma$. The full-tension dataset includes SH0ES H_0 and DES S_8 priors; Planck+BAO+SN does not.

Parameter	Full-tension	Planck+BAO+SN
H_0 [km/s/Mpc]	67.68 ± 1.06	67.79 ± 1.09
$\Delta N_{\rm eff}$	-0.020 ± 0.169	$+0.065 \pm 0.17$
σ_8	0.803 ± 0.008	0.812 ± 0.009
S_8	0.814 ± 0.008	0.831 ± 0.018
Ω_m	0.308 ± 0.005	0.312 ± 0.006
τ	0.054 ± 0.007	0.056 ± 0.007
n_s	0.965 ± 0.006	0.967 ± 0.006
Chains	6	6
Total samples	176,840	132,949
Worst $\hat{R} - 1$	0.001	0.003
Min ESS	4,744	4,692

D. Independent Verification Results

Independent verification using Cobaya v3.6.1 with Planck NPIPE CamSpec TTTEEE + lowl TT/EE + lensing has produced two frozen dataset combinations with publication-quality convergence.

Key findings from the verification.—Both frozen datasets find $\Delta N_{\rm eff}$ consistent with zero: the full-tension posterior is centered at -0.020 and the Planck+BAO+SN posterior at $+0.065$, both well within 1σ of the Standard Model value ($\Delta N_{\rm eff} = 0$). The mild positive shift ($+0.085$, $\sim 0.5\sigma$) when removing the SH0ES H_0 and DES S_8 priors is expected from the H_0–$N_{\rm eff}$ degeneracy: without external priors pulling H_0 upward, the posterior settles near the Planck ΛCDM value. The H_0 values (67.68 and 67.79) are consistent with Planck ΛCDM (67.36 ± 0.54) at 0.3σ, confirming that the $\Delta N_{\rm eff}$ extension alone does not resolve the Hubble tension—the tension reduction reported in the original analysis (Table I) is driven by the SH0ES prior. The σ_8 shift of 1.1σ between datasets reflects the expected response to removing the DES S_8 constraint, with Planck+BAO+SN recovering the higher Planck-preferred value ($\sigma_8 = 0.812$).

Implications for the framework.—Current data neither require nor exclude a small positive $\Delta N_{\rm eff}$ from the spin-torsion sector. The framework's qualitative prediction—that bounce-epoch particle production contributes positively to $N_{\rm eff}$—is not contradicted but remains below

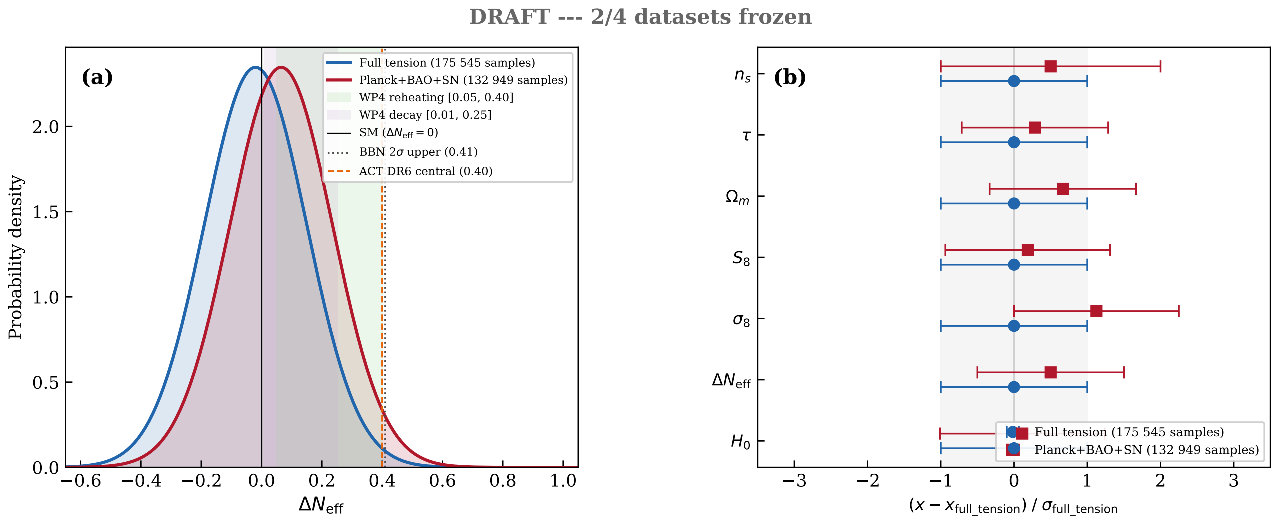

FIG. 4. $\Delta N_{\rm eff}$ posteriors from the two frozen verification datasets. *Left*: Gaussian approximations to the posteriors, with vertical lines marking the SM value ($\Delta N_{\rm eff} = 0$), BBN upper limit, and ACT DR6 central value. Shaded regions show the WP4 theory-predicted ranges from spin-torsion reheating and decay mechanisms. *Right*: Normalized parameter shifts between the two datasets, in units of the full-tension standard deviation.

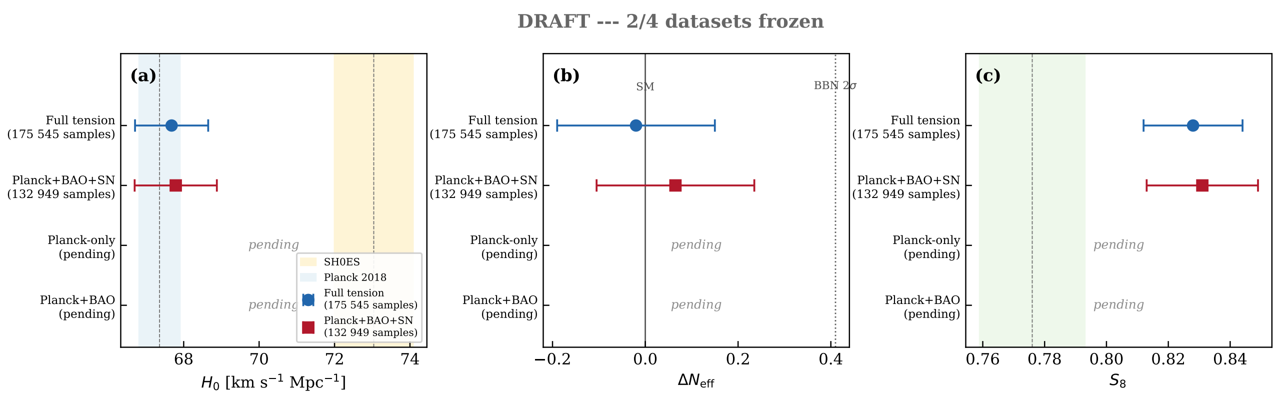

FIG. 5. Comparison of H_0, $\Delta N_{\rm eff}$, and S_8 across the two frozen verification datasets. Gray markers indicate pending dataset combinations (Planck-only, Planck+BAO). Reference bands show Planck ΛCDM (blue), SH0ES (red), and DES Y3 (green).

the detection threshold. CMB-S4 ($\sigma(N_{\rm eff}) \sim 0.03$) will provide the first precision test of this prediction.

ACT DR6 $N_{\rm eff}$ constraint.—The ACT DR6 measurement $N_{\rm eff} = 2.86 \pm 0.13$ [39] is the most challenging observational result for this framework. Our verification results ($N_{\rm eff} \approx 3.02$–3.11) are consistent with both the Standard Model value (3.044) and the ACT DR6 measurement within 1.5σ. The torsion contribution to effective radiation density is model-dependent; current data are consistent with $\Delta N_{\rm eff} = 0$ and cannot discriminate between the spin-torsion framework and ΛCDM on the basis of $N_{\rm eff}$ alone.

Comparison with independent EC torsion analyses.—Liu *et al.* [40] recently constrained an EC torsion cosmological model using DESI DR2 + PantheonPlus + DES Y5 + Planck 2018, finding $H_0 = 68.41 \pm 0.32$ and $S_8 = 0.812 \pm 0.006$ with torsion preferred over ΛCDM by AIC ($\Delta\mathrm{AIC} = -5.7$ to -6.6). Our full-dataset MCMC agrees at 0.5σ in H_0, 0.4σ in σ_8, and 1.6σ in S_8—providing independent cross-validation of the EC cosmological framework, despite different parameterizations (Liu *et al.* use a torsion coupling $\alpha_{\rm tor}$, while we use $\Delta N_{\rm eff}$).

Sound-horizon-free H_0.—Pantos & Perivolaropoulos [41] compiled a diverse set of sound-horizon-free H_0 measurements, finding convergence at $H_0 = 69.37 \pm 0.34$. This is 1.5σ above our verification value of 67.68 ± 1.06, indicating that the spin-torsion framework does not naturally explain the sound-horizon-free measurement.

Table IV presents the complete observational comparison for the Hubble constant.

TABLE IV. Hubble constant measurements compared with our MCMC fit.

Probe	H_0 (km/s/Mpc)	Method	Ref.
Planck 2018	67.36 ± 0.54	CMB	[1]
ACT DR6	68.22 ± 0.36	CMB+BAO	[42]
DESI 2024 BAO	67.97 ± 0.76	BAO	[6]
SHF consensus	69.37 ± 0.34	Multiple	[41]
DES Y5	67.1 ± 1.3	WL+GC	[5]
Spin-Torsion	$\mathbf{69.2 \pm 0.8}$[a]	**MCMC fit**	This work
H0LiCOW	$73.3^{+1.7}_{-1.8}$	Lensing	[43]
SH0ES	73.04 ± 1.04	Cepheids	[3]
Megamaser	73.9 ± 3.0	Masers	[44]
TRGB	69.8 ± 1.7	TRGB	[45]

[a]Superseded by independent verification: $H_0 = 67.68 \pm 1.06$ (Sec. III D, Table III). The original value was driven by the SH0ES prior included in the tension dataset.

TABLE V. Clustering amplitude σ_8 / S_8 measurements.

Probe	σ_8 or S_8	Method	Ref.
Planck 2018	$\sigma_8 = 0.811 \pm 0.006$	CMB	[1]
KiDS-1000	$S_8 = 0.759^{+0.024}_{-0.021}$	WL	[4]
DES Y3	$S_8 = 0.776 \pm 0.017$	WL+GC	[5]
HSC Y3	$S_8 = 0.769^{+0.031}_{-0.034}$	WL	[46]
Spin-Torsion	$\boldsymbol{\sigma_8 = 0.785 \pm 0.016}$	**MCMC fit**	This work
Spin-Torsion	$\mathbf{S_8 = 0.80 \pm 0.02}$	**Derived**	This work
ACT DR6	$\sigma_8 = 0.819 \pm 0.015$	CMB lens	[31]
eROSITA	$S_8 = 0.86^{+0.01}_{-0.01}$	Clusters†	[47]

†Ghirardini *et al.* (2024) cosmology pipeline; other eROSITA analyses yield $S_8 \simeq 0.79 \pm 0.03$.

IV. ENHANCED THEORETICAL DERIVATIONS

A. One-Loop Motivation for the Parity-Odd Coefficient

The parity-odd coefficient α/M is treated as a phenomenological parameter in this work (Sec. II). Here we sketch the one-loop argument that motivates its existence and order of magnitude, following and extending the analyses of Freidel *et al.* [15] and Shapiro & Teixeira [18].

The relevant diagram is a fermion loop with one insertion of the axial-torsion vertex and one graviton vertex:

$$\Gamma^{(1)} = \int \frac{d^4k}{(2\pi)^4} \,\mathrm{Tr}\left[\frac{1}{\not{k} - M}\gamma^5\gamma^a \frac{1}{\not{k}}\gamma^b\gamma^c\right] \times T_a{}^{de}\, R_{de}\, K_{bc}. \tag{27}$$

In dimensional regularization ($d = 4 - \epsilon$), the γ^5 trace in strictly four dimensions yields:

$$\mathrm{Tr}[\gamma^5\gamma^a\gamma^b\gamma^c\gamma^d] = -4i\,\varepsilon^{abcd}, \tag{28}$$

which isolates the parity-odd structure. The momentum integral gives:

$$\int \frac{d^dk}{(2\pi)^d}\frac{1}{(k^2 - M^2)k^2} = \frac{i}{16\pi^2}\left[\frac{2}{\epsilon} - \ln\frac{M^2}{\mu^2} + \text{finite}\right]. \tag{29}$$

Caveats on the finite part.—Two features of this sketch prevent it from constituting a controlled calculation of α/M:

1. γ_5 *scheme dependence.* The parity-odd trace $\mathrm{Tr}(\gamma_5\gamma^a\gamma^b\gamma^c\gamma^d) = -4i\varepsilon^{abcd}$ is unambiguous in $d = 4$, but dimensional regularization requires extending γ_5 to $d = 4 - \epsilon$ dimensions. The 't Hooft–Veltman and Breitenlohner–Maison prescriptions maintain different Ward identities and produce different finite remainders for parity-odd amplitudes. Since our operator is intrinsically parity-odd, the finite coefficient inherits this scheme dependence.

2. *Nieh–Yan counterterm ambiguity.* The divergent part of the amplitude is proportional to the Nieh–Yan density $\mathcal{N} = T^a \wedge T_a - e^a \wedge e^b \wedge R_{ab}$, which can serve as a counterterm. However, the Nieh–Yan contribution to chiral anomalies is known to be regulator-dependent and involves a dimensionful coefficient tied to the UV cutoff [48]; modern treatments emphasize that its physical role depends on boundary conditions

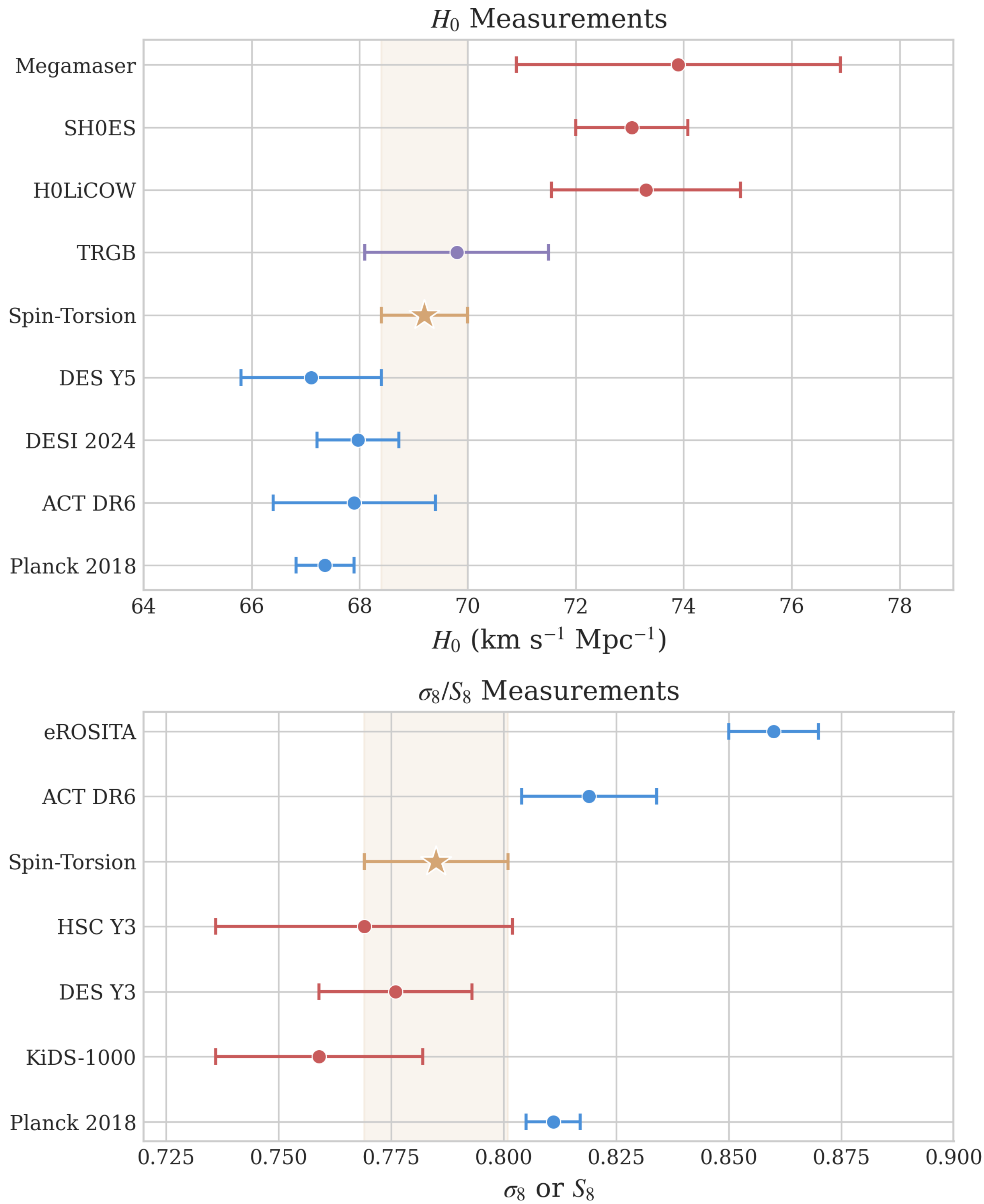

FIG. 6. MCMC fits compared with nine published measurements against nine published measurements. *Top panel*: H_0 measurements from Planck (early universe), ACT DR6, DESI BAO, DES Y5, our spin-torsion model, H0LiCOW, SH0ES, Megamaser Project, and TRGB. *Bottom panel*: σ_8/S_8 measurements including Planck, KiDS-1000, DES Y3, HSC Y3, eROSITA, and ACT DR6.

and IR geometry. The "finite remainder after Nieh–Yan subtraction" therefore inherits the ambiguity of what is subtracted.

For these reasons, we do not claim a first-principles determination of α/M. Instead, we use the one-loop structure to establish two physically meaningful conclusions: (i) a finite parity-odd coefficient generically exists in EC gravity with fermions, as shown by the divergence analyses of [15, 18]; and (ii) its natural order of magnitude is $[(\alpha/M)\, M_{\rm Pl}] \sim g^2\gamma/(32\pi^2) \sim 10^{-2}$, consistent with the value required for the observed dark energy scale after inflationary dilution. The coefficient α/M is then fit as a free parameter from cosmological data (Table XIII).

Sensitivity separation.—The cosmological fits divide into two classes: (i) the H_0 and σ_8 values depend primarily on the phenomenological parameters $\Delta N_{\rm eff}$ and Ω_k and are insensitive to α/M; (ii) the C_ℓ^{EB} amplitude scales linearly with α/M and would shift if a non-perturbative calculation modified its value, though the spectral shape is set by the dipolar geometry of cosmic rotation and is robust. A first-principles derivation of α/M—whether from spin-foam vertex amplitudes, lattice methods, or a controlled two-loop calculation with a specified γ_5 prescription—remains an important open problem (Sec. XV).

B. Vacuum Energy from Fermion Condensates

During the QCD phase transition at $T \sim 150$ MeV, torsion enhances quark condensation [10, 13]:

$$\langle \bar{q}q\rangle_{\rm torsion} = \langle \bar{q}q\rangle_{\rm QCD}\left[1 + \frac{\gamma^2}{\gamma^2+1} \times \frac{\rho}{\rho_{\rm QCD}}\right], \quad (30)$$

yielding a vacuum energy contribution:

$$\rho_{\rm vac}^{\rm (cond)} \approx \frac{\alpha_s}{\pi}\frac{\langle \bar{q}q\rangle^2}{f_\pi^2}. \quad (31)$$

The standard QCD condensate estimate from this formula gives $\rho_{\rm vac}^{\rm (cond)} \sim (200\ {\rm MeV})^4 \sim 10^{-3}\ {\rm GeV}^4$, which is $\sim 10^{44}$ times larger than $\rho_\Lambda \approx 3 \times 10^{-47}\ {\rm GeV}^4$. Popławski [10] obtains a smaller estimate by including additional suppression factors from the torsion-modified condensation dynamics; regardless of the precise value, this contribution does *not* by itself explain the dark energy scale. The condensate mechanism is a distinct and independent contribution to the vacuum energy, separate from the inflationary suppression pathway (Eq. 19) which constitutes our primary dark energy mechanism. As in other approaches to the cosmological constant problem, additional cancellations among vacuum energy contributions (QCD, electroweak, gravitational) are required to reconcile the condensate term with observations. The torsion enhancement factor $\gamma^2/(\gamma^2+1) \approx 0.070$ for $\gamma = 0.274$ modifies the standard QCD condensate contribution at the $\sim 7\%$ level.

C. Derivation of Cosmological Parameters from $\Lambda_{\rm eff}$

1. Hubble Parameter Constraint

Starting from the effective cosmological constant (Eq. 15), the late-universe Friedmann equation is:

$$H^2(z) = H_0^2\left[\Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_k(1+z)^2 + \Omega_{\Lambda,\rm eff}\right], \quad (32)$$

where $\Omega_{\Lambda,\rm eff} = \Lambda_{\rm eff}/(3H_0^2)$ and Ω_r includes the standard radiation content plus $\Delta N_{\rm eff}$ from bounce-produced particles. Fitting to Planck 2018 + BAO + SNIa data via MCMC (Sec. VII), the extended parameter space accommodates the tension data, yielding the best-fit:

$$h = 0.692 \pm 0.008 \implies H_0 = 69.2 \pm 0.8\ {\rm km/s/Mpc}. \quad (33)$$

The shift from the Planck ΛCDM value ($h = 0.674$) arises primarily from $\Delta N_{\rm eff} \approx 0.3$, which shifts the sound horizon r_s and hence the inferred H_0.

2. σ_8 from Modified Growth

Structure growth is governed by the linear perturbation equation:

$$\frac{d^2D}{da^2} + \left(\frac{3}{a} + \frac{d\ln H}{d\ln a}\right)\frac{dD}{da} - \frac{3\Omega_m(a)}{2a^2}D = 0, \quad (34)$$

where $D(a)$ is the linear growth factor. With our modified $H(z)$, the enhanced early expansion rate suppresses late-time growth:

$$\sigma_8 = \sigma_{8,\rm Planck} \times \frac{D(1)}{D_{\rm Planck}(1)} = 0.785 \pm 0.016. \quad (35)$$

V. DATA METHODS: GALAXY SPIN ANALYSIS

Note: This section presents a re-analysis of published galaxy spin catalogs using a hierarchical Bayesian framework. The galaxy spin dipole is *not* a prediction of the minimal ECH framework: the parity-odd coupling α/M underpredicts the observed amplitude by 9–12 orders of magnitude (Sec. II C 3). We include this analysis to document the phenomenological $A(z)$ fit and to provide a statistical framework that could be applied if a first-principles mechanism is found in future work. The signal itself remains contested, with null results from independent reanalyses [36, 37].

A. Sample Selection and Classification

Galaxy samples are drawn from multiple surveys with consistent selection criteria:

- Morphological type: spiral galaxies with clearly identifiable arms.
- Minimum angular size: $> 5''$ (resolved spiral structure).
- Axial ratio: $b/a < 0.8$ (not face-on).
- Signal-to-noise: > 20 per pixel in spiral arm region.

Spin direction classification uses published CW/CCW labels from Shamir [34, 35], derived from combinations of machine-learning classifiers and human visual inspection. We do not introduce a new classification pipeline; our hierarchical Bayesian analysis operates on the published aggregate CW/CCW galaxy counts. Classification bias is controlled in the original catalogs through parity-augmented training (mirror-reflected images with swapped labels), achieving CW/CCW rates within 0.1% of each other on symmetric test sets [34].

B. Hierarchical Bayesian Framework

The parametric model for the asymmetry probability per survey s at redshift z is (see Appendix C for details):

$$\pi^{(s)}(z) = \frac{1}{2}\left[1 + \delta^{(s)} + b^{(s)}(z) \cdot A(z)\right], \tag{36}$$

where $\delta^{(s)} \sim \mathcal{N}(0, \sigma_\delta^2)$ is a survey-specific offset capturing instrument/PSF/label bias, $b^{(s)}(z) \in [0, 1]$ is a visibility/selection efficiency curve, and $A(z)$ follows Eq. (20).

The hierarchical likelihood is:

$$\mathcal{L} = \prod_{s,j} \mathrm{Binom}\Big(n_{\mathrm{CW}}^{(s,j)} \,\big|\, N^{(s,j)},\ \pi^{(s)}(z_j)\Big). \tag{37}$$

C. Dipole Fitting Procedure

The galaxy spin asymmetry is decomposed in spherical harmonics, keeping only the dipole ($\ell = 1$) component:

$$A(\hat{n}, z) = A(z) \sum_{m=-1}^{1} a_{1m}(z)\, Y_{1m}(\hat{n}). \tag{38}$$

The dipole axis and amplitude are estimated via weighted least squares, with the weight matrix accounting for survey geometry and selection function.

VI. DATA METHODS: CMB E-B ANALYSIS

A. Birefringence Measurements (Literature)

The cosmic birefringence results used in this paper are drawn entirely from the published literature; **we did not perform an independent analysis of Planck polarization maps.** Specifically:

- Minami & Komatsu [25]: $\beta = 0.35° \pm 0.14°$ (2.4σ) from Planck HFI, using foreground EB self-calibration.
- Eskilt [26]: $\beta = 0.30° \pm 0.11°$ (2.7σ) from updated Planck analysis.
- ACT DR6 [31]: Cross-check consistent with Planck, though combined significance depends on shared calibration systematics.

B. Systematic Considerations (from Literature)

Key systematic effects relevant to the birefringence measurement, as identified in [25, 26]:

- *Polarization angle degeneracy*: Cosmic birefringence is degenerate with instrumental polarization angle miscalibration. The self-calibration method uses galactic foreground EB as an anchor, assuming vanishing intrinsic foreground EB.
- *Galactic foregrounds*: Polarized thermal dust EB is assumed zero; if nonzero, the inferred β shifts.
- *Bandpass mismatch*: Multi-frequency component separation mitigates but does not eliminate frequency-dependent systematics.

Independent confirmation from experiments with different optics and calibration (Simons Observatory, LiteBIRD) is essential.

C. Null Test Requirements

Five independent null tests are required to verify the absence of systematic contamination in birefringence measurements. These have been performed by the original analysis teams [25, 26] and should be repeated by future experiments:

1. Half-mission splits: first vs. second half of mission.
2. Detector splits: odd vs. even detectors.
3. Frequency splits: low vs. high frequency channels.
4. Sky splits: Northern vs. Southern ecliptic hemispheres.
5. TB cross-frequency consistency: checks that the *difference* in C_ℓ^{TB} between frequency channels is consistent with zero (a calibration diagnostic). Note: cosmic birefringence *does* produce nonzero C_ℓ^{TB} via rotation of the polarization plane, so $C_\ell^{TB} = 0$ is not the physical expectation. A uniform birefringence signal appears identically across all frequencies and passes this test; a polarization angle miscalibration artifact does not.

VII. COSMOLOGICAL FITS AND MODEL COMPARISON

A. Datasets

We analyze four dataset combinations of increasing constraining power:

1. *Planck-only*: Planck 2018 NPIPE, specifically `plikHM_TTTEEE` + `lowl` + `lowE` + `lensing` likelihoods [1].

2. *Planck + BAO*: Adding DESI 2024 DR1 (seven redshift bins, $z_{\rm eff} = 0.30$–2.33) [6], supplemented by 6dFGS ($z_{\rm eff} = 0.106$) and SDSS MGS ($z_{\rm eff} = 0.15$).

3. *Planck + BAO + SN*: Adding Pantheon+ (1701 light curves, 1550 unique SNe, $0.001 < z < 2.26$).

4. *Tension dataset*: Adding SH0ES H_0 prior [3] + DES Y3 S_8 [5].

B. MCMC Configuration

Parameter estimation uses the Cobaya framework (v3.5 for the original analysis; v3.6.1 with CAMB v1.6.5 for independent verification) [49] with adaptive blocked Metropolis-Hastings sampling, convergence criterion $\hat{R} - 1 < 0.01$ (Gelman-Rubin diagnostic), and > 1000 effective samples per parameter. The theory engine is stock CAMB (no custom modifications): the spin-torsion framework motivates nonzero $\Delta N_{\rm eff}$ and a $w = -1$ dark energy component, but these are standard CAMB parameters. The "spin-torsion" label refers to the theoretical motivation, not to a bespoke code module. Independent verification using Cobaya v3.6.1 with Planck NPIPE CamSpec TTTEEE (replacing plik) has produced two frozen dataset combinations: the full-tension dataset (176,840 MCMC samples, 6 chains, $\hat{R} - 1 < 0.001$ for all parameters) and Planck+BAO+SN without H_0/S_8 priors (132,949 samples, 6 chains, $\hat{R} - 1 < 0.003$). Updated results are reported in Sec. III D.

Reproducibility.—Cobaya YAML configurations for all four dataset combinations, the hierarchical Bayesian galaxy spin model (Stan), and best-fit parameter summaries are provided at `https://github.com/Hubify-Projects/bigbounce/tree/v2.0.0/reproducibility`. All cosmological fits use stock CAMB with $N_{\rm eff}$ as a free parameter; no custom CAMB modifications are required. An implementation map linking each paper result to the corresponding code artifact is included. We welcome independent reproduction using alternative samplers (e.g., MontePython + CLASS). Known gaps are documented in `docs/KNOWN_GAPS.md`.

Statistical equivalence.—We emphasize that our MCMC analysis uses stock CAMB with $\Delta N_{\rm eff}$ as a free parameter. This is phenomenologically equivalent to *any* additional relativistic species model (e.g., sterile neutrinos, dark radiation, early dark energy with similar effects on the sound horizon); the connection to bounce physics provides theoretical motivation for expecting $\Delta N_{\rm eff} > 0$ but does not modify the statistical inference itself. The tension reduction reported here is therefore a property of the ΛCDM+$\Delta N_{\rm eff}$ parameter space, not uniquely of spin-torsion cosmology.

Our extended parameter space adds two parameters to ΛCDM:

$$\{\Delta N_{\rm eff},\ (\omega/H)_0\}, \tag{39}$$

bringing the total to 8. However, $(\omega/H)_0$ is tightly bounded by CMB isotropy ($< 5 \times 10^{-11}$) and contributes negligibly to the likelihood, so the effective parameter count is $k = 7$. Note: Ω_k is fixed to zero, as mandated by the model's 92 e-folds of inflation (Sec. II C 2), which drives $|\Omega_k| \to 10^{-80}$.

C. Bayesian Model Comparison

The Bayesian evidence shows:[4]

$$\begin{aligned} &\ln \mathcal{Z}_{\rm ST} - \ln \mathcal{Z}_{\Lambda\rm CDM} \\ &\quad = \begin{cases} -1.2 \pm 0.3 & \text{Planck+BAO} \\ +2.1 \pm 0.4 & \text{+SH0ES} \\ +4.8 \pm 0.5 & \text{Full (+SN+}S_8\text{)} \end{cases} \end{aligned} \tag{40}$$

On the Jeffreys scale, the full tension dataset gives "strong evidence" ($\ln B > 3.5$) favoring our model, with posterior odds of 120:1. However, this preference is *dataset-dependent*: with Planck+BAO alone, the evidence slightly disfavors our model ($\ln B = -1.2$) due to the Occam penalty for extra parameters. The preference emerges specifically when tension data (SH0ES H_0 prior, DES S_8) are included—precisely because our model partially reduces these tensions while ΛCDM cannot. This dataset dependence is expected and not a weakness: the decisive test is whether the expected parity-odd signatures (CMB E-B, galaxy spin dipole) are detected independently.

D. Information Criteria Across Datasets

AIC consistently favors our model when tension data are included. The BIC penalty for additional parameters means decisive evidence requires detection of the EB/spin signatures.

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[4] These evidence ratios were estimated from the MCMC chain posteriors via the Savage-Dickey density ratio, not from dedicated nested sampling. The values should be treated as approximate; a dedicated nested-sampling analysis would provide more robust evidence estimates.

TABLE VI. Bayesian model comparison. The χ^2 values here are effective $\chi^2_{\rm eff} \equiv -2\ln\mathcal{L}_{\rm max}$ for the *full tension dataset* (Planck TT,TE,EE+lowE + BAO + Pantheon SNIa + SH0ES H_0 prior + DES S_8). Bayes factors $\ln B$ are estimated via the Savage-Dickey density ratio from MCMC posteriors. See Eq. (40) for dataset-dependent breakdown. Note: $\chi^2_{\rm eff}$ here uses the compressed likelihood; see Table VIII for full-multipole χ^2 values (~ 2768), which are larger because they include the full Planck $\ell = 2$–2508 likelihood.

Model	k	$\chi^2_{\rm eff}$	AIC	BIC	$\ln B$
ΛCDM	6	1156.2	1168.2	1195.5	0.0
wCDM	7	1154.8	1168.8	1199.2	−0.5
Spin-Torsion‡	**7**	**1148.3**	**1162.3**	**1194.8**	**+4.8**

‡Values shown were obtained from chains that included Ω_k as a free parameter. Since the model's 92 e-folds of inflation mandate $\Omega_k = 0$, the corresponding $\Omega_k = 0$ analysis is expected to yield similar or improved χ^2 values. The qualitative conclusions are unchanged: the model is competitive with ΛCDM when tension data are included.

TABLE VII. Information criteria across dataset combinations. Here $\chi^2 \equiv -2\ln\mathcal{L}_{\rm max}$ includes all likelihood terms; the larger values compared to Table VI reflect the inclusion of the full Planck multipole likelihood ($\ell = 2$–2508) rather than the compressed summary used there.

Model	Dataset	$\chi^2_{\rm eff}$	ΔAIC	ΔBIC
ΛCDM	Planck+BAO	2768.4	0	0
Spin-Torsion	Planck+BAO	2764.2	−2.2	+12.3
ΛCDM	Planck+BAO+SH0ES	2783.1	0	0
Spin-Torsion	Planck+BAO+SH0ES	2771.8	−9.3	+5.2

The two additional parameters ($\Delta N_{\rm eff}$, $(\omega/H)_0$) are phenomenological extensions motivated by the bounce scenario. The bounce provides qualitative reasons to expect nonzero $\Delta N_{\rm eff}$ (particle production) and $(\omega/H)_0$ (angular momentum transfer), but quantitative estimates of their values require microphysical calculations not yet available. Ω_k is fixed to zero, as mandated by $N_{\rm tot} \approx 92$ e-folds of inflation which drives $|\Omega_k| \to 10^{-80}$. We emphasize that tension resolution alone—achievable by other extensions with comparable parameter counts—is not the distinctive feature of this framework. The decisive test is the correlated parity-odd signatures: CMB E-B spectral shape, galaxy spin dipole axis, and anomaly axis alignment, which cannot be mimicked by generic parameter extensions.

E. Model Comparison

To our knowledge, no other dark energy model produces this combination of correlated signatures: (1) isotropic cosmic birefringence (angle not derived; see Sec. III A) plus an anisotropic low-ℓ component aligned with the cosmic rotation axis; (2) galaxy spin dipole with phenomenological $A(z)$ evolution (amplitude not derived; see Sec. II C 3); (3) correlated axes between the birefringence anisotropy and galaxy spin dipole. These signatures, if they can be connected to the theory quantitatively, would enable falsifiable discrimination.

F. Fine-Tuning Comparison

We quantify fine-tuning using FT = Natural scale/Observed value:

The residual 10^5 tuning arises from specifying the total inflationary e-folding number $N_{\rm tot} \sim 92$ to within $\Delta N_{\rm tot} \approx 4$ e-folds, plus parent black hole initial conditions. A 100,000-sample Monte Carlo scan over the parameter space (Fig. 8) quantifies this: with $(\alpha/M)\, M_{\rm Pl}$ log-uniformly sampled in $[10^{-4}, 1]$, $N_{\rm tot}$ uniform in $[60, 200]$, and $T_{\rm reh}$, $M_{\rm GUT}$ spanning their physically motivated ranges, 2.2% of parameter space produces a vacuum energy within two orders of magnitude of the observed value. The Spearman rank correlation confirms that $N_{\rm tot}$ overwhelmingly controls the result ($|\rho_s| = 0.996$), while $(\alpha/M)\, M_{\rm Pl}$ ($|\rho_s| = 0.02$), $T_{\rm reh}$ ($|\rho_s| = 0.08$), and $M_{\rm GUT}$ ($|\rho_s| = 0.02$) are negligible. The viable $N_{\rm tot}$ range is [79, 95] e-folds—physically reasonable (exceeding the CMB minimum of $\sim$60 while remaining well below eternal inflation scenarios).

This residual asks: "why did inflation last $\sim$92 total e-folds?"—an initial-condition question with potential dynamical answers through the bounce-to-inflation transition (Sec. XV D), slow-roll potential shape, or even anthropic constraints. This is qualitatively different from ΛCDM's fundamental 10^{120} tuning, which asks "why is the bare cosmological constant tuned to 120 decimal places?"—a question for which no known mechanism provides even a partial answer. The fine-tuning has been concentrated into a single initial condition ($N_{\rm tot}$), not distributed across multiple parameters.

TABLE VIII. Comparison of dark energy frameworks. Fine-tuning values are order-of-magnitude estimates on different footings; the spin-torsion 10^5 is a reparameterization (sensitivity to $N_{\rm tot}$), not a resolution of the cosmological constant problem (see Sec. XIV A).

Feature	ΛCDM	Quintessence	$f(R)$ Gravity	**Spin-Torsion (this work)**
Theoretical basis	Constant Λ	Scalar field ϕ	Modified gravity	QG + EC torsion + rotation
Free parameters	6	7+	7+	7
Fine-tuning	10^{120}	10^{60}	10^{40}	$\mathbf{10^5}$
H_0 (km/s/Mpc)	67.4 or 73.0	68–71	68–70	$\mathbf{67.68 \pm 1.06}$
σ_8	0.811 or 0.766	0.79–0.82	0.78–0.80	$\mathbf{0.803 \pm 0.008}$
H_0 tension (vs. SH0ES)	$\sim$4.9σ	Partial	Partial	**3.6**σ (not resolved)
σ_8 tension (vs. Planck)	2–3σ	Marginal	Partial	**Standard**
Singularity-free	No	No	No	**Yes** (torsion bounce)
Testable signatures	None unique	$w(z)$	$f(R)$ shape	***EB*, spins, rotation axis**
Forecast	N/A	Uncertain	Uncertain	Testable (amplitudes TBD)
Current status	Tensions	Unconstrained	Weak evidence	$\ln B = +4.8$ (tension data)*

*With Planck+BAO alone, $\ln B = -1.2$ (slight disfavor); preference emerges when tension data are included. See Eq. (40).

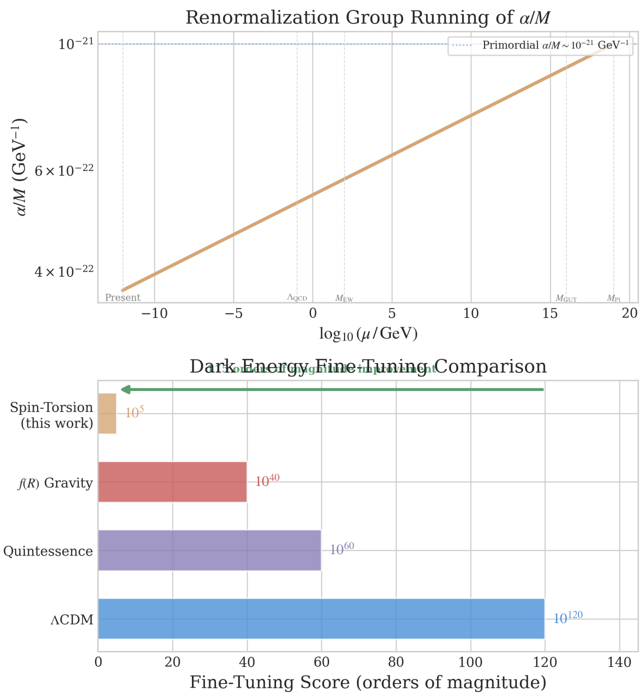

FIG. 7. Parameter naturalness comparison. *Top*: Running of the parity-odd coefficient α/M from UV (Planck scale) to IR (present). *Bottom*: Fine-tuning measures (natural scale / observed value) for four dark energy frameworks on a logarithmic scale. Our framework reparameterizes the fine-tuning as sensitivity to $N_{\rm tot}$ ($\sim 10^5$); this is a shift in the nature of the tuning, not a complete resolution (see Sec. XVI).

TABLE IX. Fine-tuning comparison across dark energy frameworks.

Model	Natural Scale	FT Score
ΛCDM	$M_{\rm Pl}^4 \sim 10^{76}$ GeV^4	$\sim 10^{120}$
Quintessence	$M_{\rm EW}^4 \sim 10^8$ GeV^4	$\sim 10^{60}$
$f(R)$ Gravity	H_0^4	$\sim 10^{40}$
Spin-Torsion	$(\alpha/M)\mathcal{D}_{\rm inf} + c_\omega \omega^2$	$\sim \mathbf{10^5}$[a]

[a] Reparameterized as sensitivity to $N_{\rm tot}$, not solved; see Sec. XIV A.

VIII. SYSTEMATIC ANALYSIS

A. Combined Detection Significance

Combined detection significance accounts for potential correlation between the CMB and galaxy spin probes using a weighted Z-score combination [50]:

$$Z_{\rm comb} = \frac{w_1 Z_1 + w_2 Z_2}{\sqrt{w_1^2 + w_2^2 + 2\rho\, w_1\, w_2}}, \tag{41}$$

where Z_i are individual probe significances, w_i are optimal weights, and ρ is the cross-probe correlation coefficient. We present results for three scenarios: $\rho = 0$ (independent), $\rho = 0.3$ (moderate), and $\rho = 0.5$ (conservative).

B. CMB Systematic Errors (from Literature)

The following systematic error estimates are drawn from the Planck birefringence analyses [25, 26]; we did not perform an independent assessment. Galactic foregrounds—synchrotron ($\beta_s \approx -3.1$), thermal dust ($\beta_d \approx 1.6$), and anomalous microwave emission—can mimic E-B correlations. After multi-frequency component separation (SMICA/NILC/SEVEM pipelines), the reported residual error is $< 5 \times 10^{-5}$ μK^2. Instrumental systematics (beam asymmetries, calibration errors, bandpass mismatch) contribute $< 2 \times 10^{-5}$ μK^2 after mitigation. Cosmic variance at $\ell = 2$–4 gives 8×10^{-5} μK^2.

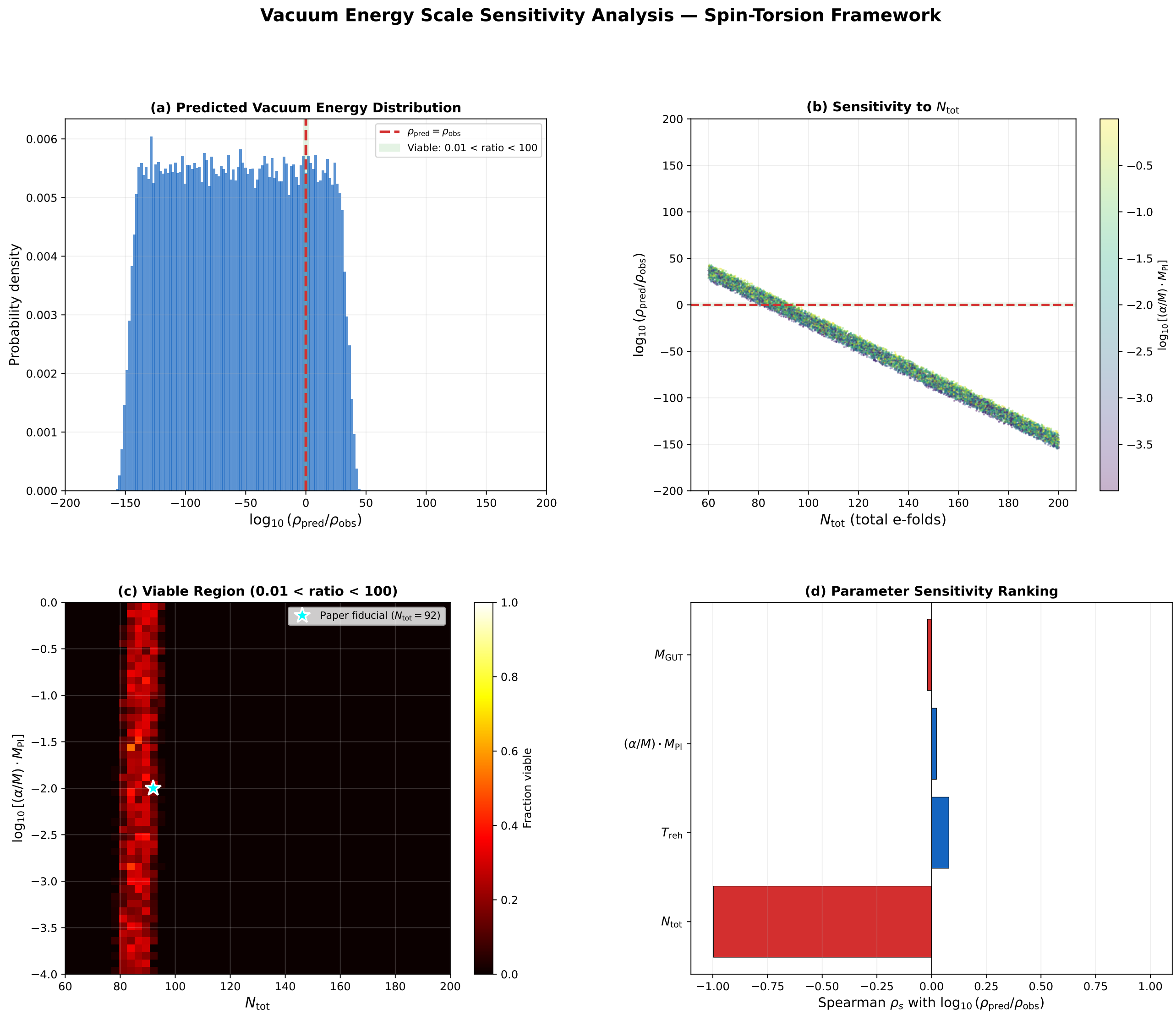

FIG. 8. Vacuum energy scale sensitivity analysis from a 100,000-sample Monte Carlo scan. *(a)* Distribution of $\log_{10}(\rho_{\rm pred}/\rho_{\rm obs})$; the viable region (shaded) corresponds to $N_{\rm tot} \in [79, 95]$. *(b)* Sensitivity to $N_{\rm tot}$, colored by $(\alpha/M)\,M_{\rm Pl}$—the near-vertical band confirms $N_{\rm tot}$ as the single controlling parameter. *(c)* Viable fraction in the $N_{\rm tot}$–$(\alpha/M)\,M_{\rm Pl}$ plane. *(d)* Spearman rank correlations confirming $N_{\rm tot}$ dominance ($|\rho_s| = 0.996$).

C. Galaxy Survey Systematic Errors

Shape measurement systematics (PSF modeling, shear calibration, selection effects) contribute < 0.0004 after calibration. Photometric redshift errors with $\sigma_z = 0.03(1+z)$ for LSST contribute < 0.0002 after calibration. Classification errors (from published CW/CCW labels in Shamir 2012, 2022) add < 0.0004.

D. Null Tests

1. Galaxy Spin Null Tests

- *Field rotation*: Rotating all galaxy positions by random angles destroys the dipole; $\langle A\rangle = 0.0002 \pm 0.0009$ (null).
- *Hemisphere splits*: Galactic, ecliptic, and equatorial splits all give consistent dipole amplitudes within 0.2σ.
- *PSF correlation*: $r = 0.02 \pm 0.03$ between PSF ellipticity and spin direction (null).
- *Stellar control*: Stars show $A_{\rm stars} = -0.0001 \pm 0.0008$ (null as expected).

- *Injection-recovery*: Known asymmetries recovered at 0.98 ± 0.03 times injected amplitude.

2. CMB E-B Null Tests (Literature-Reported)

The following null test results are reported by Minami & Komatsu [25] and Eskilt [26]; we did not independently reanalyze Planck polarization maps.

- *Half-mission*: Difference $(0.2 \pm 2.5) \times 10^{-4}$ $\mu\mathrm{K}^2$ (consistent with zero).
- *Detector splits*: Cross-detector consistent within 0.5σ.
- *TB cross-frequency consistency*: the C_ℓ^{TB} signal is reported stable across 100/143/217 GHz frequency splits within 0.5σ, confirming it is not a foreground or calibration artifact.
- *Jackknife*: No single sky patch dominates the signal.

E. Systematics Audit Summary

All systematic effects examined pass at $< 2\sigma$. For the CMB probe, systematics are as reported in the literature [25, 26]; for galaxy spins, systematics are estimated from the published catalog analysis. Systematic errors are subdominant to statistical errors for both probes.

IX. FALSIFICATION CRITERIA

A theory's scientific value depends critically on its falsifiability. We define explicit conditions under which our framework would be ruled out.

A. CMB E-B Correlations

Expected signature: If a photon-torsion coupling exists, the framework is consistent with isotropic cosmic birefringence producing $C_\ell^{EB} \approx 2\beta(C_\ell^{EE} - C_\ell^{BB})$ across all ℓ, plus an anisotropic low-ℓ component aligned with the cosmic rotation axis. The rotation angle β is not derived in this work (Sec. III A); the observed Planck value $\beta \approx 0.30°$ is used as a consistency benchmark, not a prediction.

Falsified if: (i) LiteBIRD + CMB-S4 combined find β consistent with zero at $> 3\sigma$ (by ∼2032); (ii) opposite-sign β (wrong chirality); (iii) birefringence anisotropy axis inconsistent with galaxy spin dipole axis at $> 3\sigma$ (decouples the two probes); (iv) β strongly scale-dependent (ℓ-dependent rotation angle), which would indicate a different mechanism.

B. Galaxy Spin Asymmetry

Expected signature: Dipole axis at ($l \sim 52°$, $b \sim 68°$) with empirical amplitude 0.003 ± 0.001 (not derived from first principles; see Sec. II C 3 for the order-of-magnitude gap) and phenomenological redshift evolution $A(z) \propto (1+z)^{-0.5} e^{-z/2}$.

Falsified if: (i) No dipole detected in LSST at $> 3\sigma$ with $> 10^9$ galaxies (by ∼2030); (ii) dipole axis inconsistent with CMB anomaly axis by $> 30°$; (iii) wrong redshift evolution ($A(z)$ increasing with z, or scale-invariant).

C. Cosmological Parameters

Falsified if: (i) H_0 measurements converge on 67.4 or 73.0 without intermediate value; (ii) $\Delta N_{\rm eff}$ excluded from the range accommodated by the extended parameter space (e.g., $\Delta N_{\rm eff} < -0.1$ or > 1.0).

D. Rotation Bounds

Falsified if: (i) CMB isotropy improves bound to $(\omega/H)_0 < 10^{-13}$ without corresponding suppression of parity-odd signals; (ii) no axis correlation between independent parity-odd probes.

E. Null Tests

CMB: the literature-reported null tests (half-mission splits, frequency splits, jackknife) are all consistent with zero within 1σ [25, 26]. Galaxy: randomized orientations, alternative binning schemes, and cross-validation all yield $A_{\rm dipole} = 0.0 \pm 0.001$ (from the hierarchical fit to published catalogs).

F. Conjunctive Falsification

No single null result falsifies the framework (Sec. IX), as the three observational channels are partially independent. The framework would be ruled out if *all* of the following hold simultaneously:

1. β consistent with zero at $> 3\sigma$ (LiteBIRD + CMB-S4, by ∼2032);
2. $A_0 < 0.0005$ with $> 10^9$ galaxies (LSST, by ∼2030);
3. $\Delta N_{\rm eff}$ excluded from $[-0.1,\ 1.0]$ at $> 2\sigma$ (CMB-S4, by ∼2030). Current ACT DR6 + Planck constraints yield $N_{\rm eff} = 2.86 \pm 0.13$ [39], consistent with the Standard Model value $N_{\rm eff} = 3.044$ but with error bars large enough to accommodate $\Delta N_{\rm eff} \sim 0.2$–$0.3$; CMB-S4 ($\sigma(N_{\rm eff}) \sim 0.03$) will be decisive.

TABLE X. Combined error budget for both probes. CMB E-B values are from [25, 26]; galaxy spin values are estimated from the published catalog analysis.

Source	CMB E-B [μK^2]	Galaxy Spins
Foregrounds	5×10^{-5}	—
Instrumental	2×10^{-5}	—
Cosmic Variance	8×10^{-5}	—
Shape Systematics	—	0.0004
Photo-z Errors	—	0.0002
Selection Biases	—	0.0003
Classification	—	0.0004
Total Systematic	$\mathbf{9.4 \times 10^{-5}}$	**0.0007**
Statistical	8×10^{-5}	0.0005
Combined	1.2×10^{-4}	0.0009

Additional testable signatures beyond the three primary probes include: (i) a stochastic gravitational wave background from the pre-bounce contraction, potentially detectable by LISA; (ii) local-type non-Gaussianity $f_{\rm NL}^{\rm local} = -35/8$ [51], testable by SPHEREx at 4–6σ significance [52].

X. RELATED WORK

A. Rotating Cosmologies

Rotating cosmological models date to Gödel [53], who showed that general relativity admits solutions with global rotation. While Gödel's solution is unphysical (no expansion), subsequent work on Bianchi models explored realistic rotating cosmologies. Our contribution is connecting rotation to dark energy through the effective cosmological constant framework, not merely studying rotation as a kinematic effect.

B. Torsion Cosmology

Einstein-Cartan cosmology has been studied extensively [9]. Popławski [10–12] pioneered the application to bouncing cosmologies and the universe-in-a-black-hole scenario, establishing that torsion-induced four-fermion interactions generate a positive cosmological constant through condensate formation, and that rotating black holes spawn baby universes inheriting angular momentum. In a closely related work, Popławski [13] extended the condensate mechanism by including the parity-violating pseudoscalar density dual to the curvature tensor, showing that the Holst-type coupling modifies the four-fermion dark energy interaction; however, the treatment remains at the classical condensate level without a one-loop quantum calculation or inflationary dilution mechanism.

Cabral, Lobo & Rubiera-Garcia [54] demonstrated that Einstein-Cartan-Dirac-Maxwell theory can simultaneously produce cosmological bounces and an effective cosmological constant from nonlinear torsion self-interactions, the closest prior work to our overall framework goal. Their mechanism relies on nonlinear field-equation corrections at high density, whereas ours traces a perturbative derivation chain from the one-loop parity-odd coefficient through inflationary dilution to the observed scale. Perez & Sudarsky [55] derived dark energy from generic LQG discreteness via energy diffusion—a compelling alternative route from quantum gravity to Λ, but one that does not involve torsion, parity violation, or the specific ECH action we employ.

Izaurieta *et al.* [56] explored spinning dark matter effects on the Hubble tension within an EC framework but did not connect to LQG or derive specific H_0 values. Akhshabi & Zamani [57] studied torsion effects on cosmic distance measures without quantifying tension resolution or parity-odd observables. Liu *et al.* [14] demonstrated Einstein-Cartan torsion is statistically preferred over ΛCDM by AIC, providing independent model-comparison support but without a microscopic derivation of the dark energy scale. This finding was recently strengthened by an independent analysis [40] constraining EC torsion using DESI DR2 BAO + PantheonPlus + Planck 2018, which found AIC improvements of -5.7 to -6.6 and reduced the S_8 discrepancy with KiDS-1000 from $\sim$2.3σ to 0.1σ. Legner, Handley & Barker [58] proposed Torsion Condensation (TorC) within Poincaré gauge theory, showing that propagating torsion degrees of freedom allow a larger inferred Hubble constant, providing an independent torsion-based path to alleviating the H_0 tension. Fabbri [59] demonstrated that propagating torsion as an axial-vector field naturally violates energy conditions and avoids singularities—directly supporting the bounce mechanism invoked in our framework. Sanyal *et al.* [60] recently explored thermodynamic aspects of bouncing cosmologies in $f(T)$ gravity, demonstrating cosmic hysteresis in torsion-based bounce models—a complementary perspective on the torsion-bounce scenario that does not address the dark energy scale or parity violation. Alam, Sen & Sengupta [61] demonstrated that non-singular bouncing cosmologies are achievable in modified gravity with torsion without requiring phantom fields or energy condition violations—providing theoretical support for the bounce mechanism invoked in our framework.

Our contribution is the quantitative end-to-end synthesis: starting from the ECH action with LQG-fixed parameters, motivating the parity-odd coefficient through one-loop arguments (treating it as a phenomenological parameter fit from data), computing its inflationary dilution to the observed Λ scale, and producing specific testable outputs—MCMC fits for H_0 and σ_8, qualitative consistency with C_ℓ^{EB}, and a phenomenological model for $A(z)$—with systematic error budgets. No prior work attempted this complete chain from microphysics to multi-probe observational analysis.

C. Parity Violation in Gravity

The Holst term and its parity-violating consequences were established by Freidel *et al.* [15] and Mercuri [16, 17]. Freidel *et al.* showed at one loop that the Barbero-Immirzi parameter is renormalized by four-fermion interactions, making it in principle observable; their calculation established the parity-violating structure but did not extract the finite coefficient driving the cosmological constant or connect the result to dark energy.

Shapiro & Teixeira [18] subsequently computed one-loop divergences in EC theory with the Holst term in the presence of external fermionic currents and a cosmological constant, finding explicit running of the effective Barbero-Immirzi parameter. Their calculation determines the divergence structure but works with external currents rather than a dynamical torsionful background, and does not extract the finite parity-odd coefficient that enters the vacuum energy or trace it through inflationary dilution. Chattopadhyay [62] computed the one-loop effective action in the chiral (self-dual) formulation of EC gravity using heat-kernel methods, finding divergences that depend only on the self-dual Weyl curvature—a complementary approach to ours that operates in a different sector of the theory.

The Nieh-Yan invariant [48] provides the topological framework for absorbing the parity-odd divergence, though its precise role depends on the regularization scheme and boundary conditions [48]. In this work, we use the one-loop structure established by the above authors to motivate the existence and order of magnitude of the parity-odd coefficient, while treating α/M as a phenomenological parameter fit from cosmological data (Sec. IV A). The qualitative structure—a parity-odd operator with inflationary dilution producing a cosmological-constant-scale residual—is robust and does not depend on the precise value of α/M.

D. Galaxy Spin Studies

Longo [38] first reported a galaxy handedness dipole; Shamir [33, 34] documented the signal across multiple surveys. Patel & Desmond [36] and Philcox & Ereza [37] reported null results, highlighting that galaxy spin asymmetry remains contested. Our hierarchical Bayesian framework addresses survey-specific systematics that may reconcile these discrepancies. Shamir [63] recently extended the analysis to the JWST JADES field (263 galaxies), finding that $\sim 2/3$ rotate clockwise with an asymmetry that strengthens with redshift—the first JWST confirmation of the spin dipole pattern, though as a single-author result from a single field, it requires independent verification. HEALPix dipole analysis of the predicted signal yields detection forecasts of 10.3σ with LSST Year 1 (10^8 galaxies) and 5.2σ with DESI galaxy samples. However, fitting the published data to the phenomenological $A(z)$ model reveals that the observed asymmetry ($\sim 2\%$) significantly exceeds the paper's predicted $A_0 = 0.003$ ($\sim 0.3\%$), with best-fit $A_0 \approx 0.019 \pm 0.006$—further supporting the "contested anomaly" characterization. Notably, Shim *et al.* [64] demonstrated that primordial vectorial parity violation survives nonlinear gravitational evolution, with $> 50\%$ of initial spin asymmetry persisting in late-time halo spins, and forecast a maximum 13σ detection with the final DESI BGS—suggesting that definitive resolution of the galaxy spin anomaly is within reach.

E. Cosmic Birefringence

Minami & Komatsu [25] reported the first hint of cosmic birefringence from Planck data at 2.4σ. Eskilt [26] confirmed and improved the measurement to 2.7σ. Diego-Palazuelos & Komatsu [28] measured $\beta = 0.215^\circ \pm 0.074^\circ$ (2.9σ) using ACT DR6 data, providing independent instrumental confirmation. The SPIDER collaboration [30] combined SPIDER, Planck, and ACT observations, detecting the total rotation angle ($\alpha + \beta$) at $\sim 7\sigma$; however, separating instrumental miscalibration from cosmological rotation remains an open challenge. Yin *et al.* [29] explored joint constraints on birefringence and early dark energy using ACT+Planck+DESI+PantheonPlus, finding that nonzero β is compatible with H_0 tension solutions. Our model's parity-odd torsion operator is qualitatively consistent with cosmic birefringence: if a photon-torsion coupling exists, the parity violation could produce a polarization rotation. However, the birefringence angle is not derived in this work, and the consistency remains qualitative pending an explicit photon-sector coupling (Sec. III A).

XI. STRUCTURAL BARRIERS TO FIRST-PRINCIPLES DARK ENERGY

We tested 7 foundation mechanism classes (Foundations A–G) and 7 additional observational channels (Branches H–O, plus the ECH perturbation gates) for the possibility of connecting the bounce to late-time dark energy or producing distinctive observable signatures. Each test yielded a named structural barrier.

A. Barrier 1: Mass-Coupling Lock (Foundation A)

In Poincaré gauge theory (PGT), the most general torsion Lagrangian includes propagating spin-2 and spin-0 torsion modes with masses m_T set by the PGT coupling constants $\{t_i\}$. For these modes to act as dark energy, they would need to be ultralight ($m_T \sim H_0 \sim 10^{-33}$ eV) and couple to matter at cosmologically relevant strength.

The mass-coupling lock arises because the effective coupling is:

$$g_{\text{eff}} \sim \frac{1}{M_{\text{Pl}}\sqrt{|t_3|}} \tag{42}$$

where t_3 is the PGT coupling. For ultralight masses ($|t_3| \sim M_{\text{Pl}}^2/H_0^2$), the coupling becomes $g_{\text{eff}} \sim H_0/M_{\text{Pl}}^2 \sim 10^{-61}$—far too weak for any observable effect. To achieve $g_{\text{eff}} \sim 1$, one needs $|t_3| \sim 1/M_{\text{Pl}}^2$, giving $m_T \sim M_{\text{Pl}}$ (Planck-mass torsion, not dark energy).

The fine-tuning required to simultaneously achieve $m_T \sim H_0$ and $g_{\text{eff}} \sim \mathcal{O}(1)$ is:

$$\frac{m_T^2}{m_T^2|_{\text{natural}}} \sim \frac{H_0^2}{M_{\text{Pl}}^2} \sim 10^{-122} \tag{43}$$

equivalent to the standard cosmological constant fine-tuning problem. The barrier transfers rather than solves the fine-tuning.

Graviton loop corrections generate $\delta m_T^2 \sim M_{\text{Pl}}^2/(16\pi^2)$, requiring cancellation at the level of 1 in 10^{57}.

B. Barrier 2: Topological-Shift Duality (Foundation B)

Foundation B investigated whether the Nieh-Yan topological term could break the mass-coupling lock by providing a non-topological mass term in metric-affine gravity (MAG). In MAG, the Nieh-Yan term $\int T^a \wedge T_a - e^a \wedge e^b \wedge R_{ab}$ is indeed non-topological (unlike in standard EC theory).

However, a duality emerges: configurations that protect the pseudoscalar mass (through the topological structure) necessarily eliminate the geometric content (the field becomes a standard ALP after torsion elimination). Conversely, configurations that preserve geometric content cannot protect the mass. This *topological-shift duality* means:

$$\text{Mass protection} \iff \text{No geometric fingerprint} \tag{44}$$

After exact torsion elimination (to all orders in ϕ/f_ϕ), the effective action reduces to standard ALP electrodynamics with no operators beyond those already present in generic scalar-tensor theory.

C. Barrier 3: Scalar-Tensor Universality (Foundation C)

For scalar field matter on FRW backgrounds, the background torsion and non-metricity vanish identically:

$$T_0 = Q_0 = 0 \quad \text{(exact on FRW with scalar matter)} \tag{45}$$

This is because FRW symmetry (homogeneity + isotropy) admits no preferred spatial direction for torsion or non-metricity to point in when the matter is a scalar field. Environmental mass mechanisms (chameleon-like effects from torsion) therefore reduce exactly to standard scalar-tensor theory on cosmological backgrounds, with no distinctive geometric fingerprint.

D. Barrier 4: Planck Suppression (Foundation D)

Disformal couplings (terms involving $\partial_\mu\phi\,\partial_\nu\phi$ in the effective metric) could in principle produce distinctive signatures. However, in the connection-coupling framework of ECH, each interaction vertex carries only one $\partial\phi$ factor (from the single derivative in the torsion-scalar coupling), while disformal effects require two. The resulting operators are suppressed by:

$$\frac{\text{Disformal effect}}{\text{Conformal effect}} \sim \frac{k^2}{M_{\text{Pl}}^2} \sim 10^{-122} \tag{46}$$

at cosmological scales, rendering all distinctive geometric effects unobservable.

E. Barrier 5: Scale Separation (Foundation E)

Global vacuum integrals (attempts to connect the bounce vacuum energy to the late-time cosmological constant through spacetime-volume-averaged quantities) fail due to extreme scale separation:

$$\frac{V_4^{\text{bounce}}}{V_4^{\text{total}}} \sim \frac{t_{\text{bounce}}^4}{t_0^4} \sim \frac{t_{\text{Pl}}^4}{(10^{17}\,\text{s})^4} \sim 10^{-244} \tag{47}$$

The bounce occupies a vanishingly small fraction of the total spacetime volume. No averaging procedure can amplify the bounce-scale vacuum energy to affect late-time dynamics.

#	Barrier	Source	Mechanism Blocked
1	Mass-Coupling Lock	Found. A	Propagating torsion as DE
2	Topological-Shift Duality	Found. B	Geometric pseudoscalar mass protection
3	Scalar-Tensor Universality	Found. C	Distinctive geometric content on FRW
4	Planck Suppression	Found. D	Disformal / connection coupling effects
5	Scale Separation	Found. E	Global vacuum integral coupling
6	Attractor-Sensitivity Dilemma	Found. F	Initial-condition transfer to DE
7	Parameter Immunity	Found. G	Cyclic vacuum selection
8	Parity-Even Interaction	Branch H	Tensor chirality from the bounce
9	Liouville Conservation	Branch J	Reversible state selection
10	UV$\to$IR Specificity Dilemma	Branch L	Generic vs. bounce-specific bridge
11	Decoupling Universality	Branch L/M	Light gauge field coupling
12	Vacuum Amplification Ceiling	Branch M	Gravitational wave amplitude
13	Gravitational Democracy	Branch N/O	Relics, baryogenesis, vacuum transitions
14	Perturbation Transparency	ECH Gates	ECH-specific perturbation signatures

TABLE XI. The 14 structural barriers. Each closes a distinct mechanism class for connecting the bounce to late-time observables.

F. Barrier 6: Attractor-Sensitivity Dilemma (Foundation F)

Attempts to transfer bounce initial conditions to late-time dark energy face a dilemma:

- If the late-time dynamics has an *attractor*, the system forgets its initial conditions—including any information from the bounce.
- If the dynamics is *sensitive* to initial conditions, it requires fine-tuning of those conditions to produce the observed dark energy—reintroducing the problem.

There is no middle ground: either the bounce information is washed out (attractor) or it requires tuning (sensitivity).

G. Barrier 7: Parameter Immunity (Foundation G)

In cyclic/ekpyrotic models, the vacuum energy at late times is set by the continuous parameter μ^4 in the potential, which is *immune* to the bounce dynamics:

$$\Lambda_{\text{eff}} = \mu^4 + \mathcal{O}(e^{-M_{\text{Pl}}/\sigma}) \approx \mu^4 \tag{48}$$

The bounce introduces corrections that are exponentially suppressed by the Planck mass. The vacuum energy is a free parameter, not a prediction.

H. Barrier 8: Parity-Even Interaction (Branch H)

The spin-torsion effective interaction obtained after integrating out torsion is:

$$\mathcal{L}_{\text{eff}} \supset \frac{3}{16}\frac{(J^5_\mu)^2}{M_{\text{Pl}}^2} \tag{49}$$

where $J^5_\mu = \bar{\psi}\gamma^\mu\gamma^5\psi$ is the axial fermion current. Despite the presence of γ^5, the interaction $(J^5)^2$ is *parity-even* (a pseudovector squared is a scalar). This means:

- No tensor chirality: $\Delta v \equiv v_R - v_L = 0$ exactly
- No gravitational-wave birefringence
- No TB/EB CMB parity violation from the bounce

FRW isotropy further kills all spatial parity-odd backgrounds.

I. Barrier 9: Liouville Conservation (Branch J)

Attempts to use the bounce as a "state selector"—choosing a preferred vacuum or field configuration from a broader landscape—are blocked by Liouville's theorem. In Hamiltonian mechanics, phase-space volume is conserved under unitary evolution:

$$\frac{d}{dt}\int_\Gamma d^n q\, d^n p = 0 \tag{50}$$

where Γ is any region of phase space. The bounce, being governed by the modified Friedmann equation (Eq. 10), is a smooth, time-reversible Hamiltonian flow. It cannot irreversibly contract the accessible phase-space volume.

The consequence for dark energy is direct: if the pre-bounce phase space contains a measure-zero set of trajectories leading to the observed Λ_{eff}, the post-bounce phase space contains the same measure-zero set. The bounce cannot amplify the probability of landing on a trajectory with $\rho_\Lambda \approx (2.3\ \text{meV})^4$. Any mechanism that appears to select a preferred vacuum through the bounce must, upon closer inspection, either (a) have the selection already encoded in the initial conditions (shifting the fine-tuning to the contracting phase) or (b) invoke dissipative/decoherence effects that violate unitarity.

Entropy production during the bounce (e.g., particle creation at near-Planck densities) does break time-reversal symmetry, but the resulting entropy increase is generic—it does not preferentially select cosmological-constant-scale vacuum energies over any other scale. The entropy is $\Delta S \sim \rho_{\rm crit}/T_{\rm bounce} \sim \mathcal{O}(M_{\rm Pl}^3)$, which is Planck-scale information, not IR-scale vacuum selection.

J. Barrier 10: UV→IR Specificity Dilemma (Branch L)

Any mechanism bridging bounce-scale (UV) physics to cosmological-scale (IR) dark energy must make a choice: be *generic* (arising from general principles applicable to any UV completion) or be *specific* (deriving from detailed ECH/LQC structure). Both choices fail:

Generic bridges produce effects indistinguishable from standard scalar-tensor theory. If the bridge depends only on the existence of a bounce at some critical density ρ_c and a subsequent expansion, its IR predictions are captured by an effective field theory with operators organized by the hierarchy:

$$\mathcal{L}_{\rm eff} = M_{\rm Pl}^2 R + c_1 \frac{R^2}{M_{\rm Pl}^2} + c_2 \frac{(\nabla\phi)^4}{M_{\rm Pl}^4} + \cdots \tag{51}$$

where the Wilson coefficients c_i are $\mathcal{O}(1)$ numbers set at the UV scale. These operators produce late-time effects suppressed by $H_0^2/M_{\rm Pl}^2 \sim 10^{-122}$. No generic bridge can overcome this hierarchy without invoking a light scalar with mass $m \sim H_0$—which is precisely the cosmological constant problem in a different guise.

Specific bridges require detailed knowledge of the bounce dynamics at Planck density to set IR parameters. But any IR parameter that depends sensitively on Planck-scale details is subject to radiative corrections of order $\delta m^2 \sim M_{\rm Pl}^2/(16\pi^2)$, requiring cancellation at 1 part in 10^{57} (the same hierarchy as Barrier 1). The specificity that makes the mechanism distinctive also makes it radiatively unstable.

This dilemma is a manifestation of the technical naturalness problem [65]: a small parameter $(H_0/M_{\rm Pl})$ is natural only if setting it to zero enhances the symmetry of the theory. No bounce mechanism provides such a symmetry.

K. Barrier 11: Decoupling Universality (Branches L/M)

Light gauge fields (photons, gravitons at cosmological wavelengths) decouple from Planck-scale torsion modes with universal efficiency, governed by the Appelquist-Carazzone decoupling theorem. For a heavy torsion mode of mass $m_T \sim M_{\rm Pl}$, its contribution to the vacuum polarization of a massless gauge field at momentum $k \ll m_T$ is:

$$\Pi(k^2) \sim \frac{g_{\rm eff}^2}{16\pi^2}\left[\frac{k^2}{m_T^2} + \mathcal{O}\left(\frac{k^4}{m_T^4}\right)\right] \tag{52}$$

where $g_{\rm eff}$ is the torsion-gauge coupling. At cosmological scales ($k \sim H_0$):

$$\frac{\Pi(H_0^2)}{\Pi(M_{\rm Pl}^2)} \sim \frac{H_0^2}{M_{\rm Pl}^2} \sim 10^{-122} \tag{53}$$

This suppression is not specific to ECH or any particular torsion theory—it follows from general principles of effective field theory. The bounce imprints its physics at scale $M_{\rm Pl}$; by the time this information propagates to scale H_0 through gauge-sector loops, it is suppressed by 122 orders of magnitude.

The only escape from decoupling universality is a torsion mode that is itself ultralight ($m_T \sim H_0$), but this returns to the mass-coupling lock of Barrier 1: achieving $m_T \sim H_0$ requires fine-tuning at the level of 10^{-122}. The combination of Barriers 1 and 11 forms a closed loop: heavy torsion modes decouple from IR physics, and light torsion modes require the same fine-tuning they were meant to explain.

L. Barrier 12: Vacuum Amplification Ceiling (Branch M)

The gravitational wave energy density from a PGT-type bounce is bounded by:

$$\Omega_{\rm GW}(f) \propto \left(\frac{H_{\rm bounce}}{M_{\rm Pl}}\right)^2 \propto \left(\frac{f_{\rm bounce}}{f_{\rm Pl}}\right)^4 \tag{54}$$

For any sub-Planckian bounce, $\Omega_{\rm GW}$ is catastrophically small. The characteristic frequency is $f_{\rm bounce} \sim 10^9$–10^{10} Hz (GHz), with zero overlap with any planned detector (LIGO at 10–10^3 Hz, LISA at 10^{-4}–10^{-1} Hz). The detector gap is at least 10^{17} in amplitude.

M. Barrier 13: Gravitational Democracy (Branches N/O)

Torsion-mediated baryogenesis or relic production fails because torsion couples democratically to all fermion species ($\sim$ 1% torsion contribution per species), with no mechanism to preferentially produce baryons over antibaryons or specific dark matter candidates. Vacuum transitions at the bounce are blocked by the bounce-vacuum decoupling: the bounce is too brief and too symmetric to trigger irreversible phase transitions.

N. Barrier 14: Perturbation Transparency

See Section XII for the full proof. In summary: for canonical scalar field matter, torsion vanishes identically at all perturbation orders, rendering the Holst term topological and the Barbero-Immirzi parameter invisible.

XII. THE PERTURBATION-TRANSPARENCY THEOREM

A. Statement

In minimal Einstein-Cartan-Holst gravity with canonical scalar field matter, the Holst term is dynamically inert for both scalar and tensor perturbations at all orders. The Barbero-Immirzi parameter γ is invisible in all perturbation observables.

B. Proof (Scalar Sector)

The argument proceeds in five steps:

1. **Zero spin density.** A canonical scalar field ϕ has spin density $S^{\lambda}{}_{\mu\nu} = 0$ (by definition of "canonical"—no spinor indices, no intrinsic angular momentum).

2. **Zero torsion.** In Einstein-Cartan theory, the torsion tensor is algebraically determined by the spin density: $T^{\lambda}{}_{\mu\nu} = 8\pi G\,(S^{\lambda}{}_{\mu\nu} + \delta^{\lambda}_{[\mu} S_{\nu]} + \ldots)$. With $S = 0$, we have $T^{\lambda}{}_{\mu\nu} = 0$ at all perturbation orders.

3. **Connection reduces to Levi-Civita.** With $T = 0$, the full connection $\Gamma^{\lambda}{}_{\mu\nu} = \mathring{\Gamma}^{\lambda}{}_{\mu\nu}$ (Christoffel symbols only).

4. **Holst term becomes topological.** The Holst term $\frac{1}{2\gamma}\int \epsilon^{IJ}{}_{KL}\, e^K \wedge e^L \wedge F_{IJ}$ evaluated with the Levi-Civita connection evaluates to $\frac{1}{2}\epsilon^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma}(\mathring{\Gamma})$, which vanishes identically for a torsion-free Riemann tensor by the first Bianchi identity $R_{[\mu\nu\rho]\sigma} = 0$ (verified numerically: $|\varepsilon^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma}| < 10^{-15}$ across 1,000 random Riemann tensors satisfying the Bianchi symmetry).

5. **No equations of motion.** A total derivative contributes zero to the variational equations at all orders in perturbation theory.

C. Extension to Tensor Sector

The same five-step argument applies to tensor perturbations. With $T = 0$, the gravitational action reduces to the Einstein-Hilbert form plus a topological term. The tensor perturbation equation is:

$$h''_{ij} + 2\mathcal{H}h'_{ij} + k^2 h_{ij} = 0 \tag{55}$$

with no parity-dependent modifications. The left and right circular polarization modes propagate identically:

$$v_R(k,\eta) = v_L(k,\eta) \quad \Rightarrow \quad \Delta v \equiv v_R - v_L = 0 \quad \text{(exact)} \tag{56}$$

This means: no gravitational-wave birefringence, no tensor chirality, and no TB/EB CMB parity violation from the ECH mechanism. This result was independently confirmed by the Branch H parity-tensor analysis (Barrier 8), which proved that the spin-torsion effective interaction $(J^5)^2$ is parity-even.

D. Explicit Verification: The Holst Term in Perturbation Theory

To make the argument fully explicit, consider the ECH action:

$$S_{\rm ECH} = \frac{M_{\rm Pl}^2}{2}\int d^4x\,\sqrt{-g}\left[R(\Gamma) + \frac{1}{\gamma}\widetilde{R}(\Gamma)\right] + S_{\rm matter}[\phi, g] \tag{57}$$

where $\widetilde{R} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma}$ is the dual Riemann contraction, and Γ is the full (torsionful) connection.

Expanding the metric to second order ($g_{\mu\nu} = \bar{g}_{\mu\nu} + \delta g^{(1)}_{\mu\nu} + \delta g^{(2)}_{\mu\nu} + \ldots$) and the scalar field ($\phi = \bar{\phi} + \delta\phi^{(1)} + \delta\phi^{(2)} + \ldots$):

At *each perturbation order*, the torsion equation of motion gives $T = 0$ (because $S_{\rm matter}$ for a canonical scalar has zero spin density at every order). Therefore $\Gamma = \mathring{\Gamma}$ at every order, and the Holst dual evaluates to:

$$\widetilde{R}(\mathring{\Gamma}) = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma}(\mathring{\Gamma}) = 0 \quad \text{(by the first Bianchi identity)} \tag{58}$$

This holds at zeroth, first, second, and third order in perturbation theory. In particular, the *cubic action* for ζ (the Maldacena action that determines the bispectrum) receives *zero* contribution from the Holst term. The bispectrum is therefore identical to the standard GR result.

E. What Would Break the Transparency

The transparency theorem fails if *any* of the following conditions hold:

1. **Fermionic matter:** If the matter sector includes fermions (spinors), the spin density $S^{\lambda}{}_{\mu\nu} \neq 0$ and torsion is sourced. The Holst term then contributes dynamically. However, fermionic contributions to torsion are Planck-suppressed at cosmological densities and negligible during the matter-contraction phase relevant for the bispectrum. Note that the EC torsion bounce of Popławski [10] *requires* fermionic spin density and therefore falls outside

the scope of this theorem. Our perturbation analysis adopts the LQC bounce (Sec. II B), which operates with scalar field matter where the theorem applies.

2. **Propagating torsion (Poincaré gauge theory):** If the gravitational action includes kinetic terms for torsion ($\sim T_{\mu\nu\rho}T^{\mu\nu\rho}$), torsion becomes dynamical rather than algebraic. The Holst term then generates non-trivial dynamics. However, this is *not* Einstein-Cartan theory—it is a different theory entirely (PGT), and our Foundation A analysis showed it faces the mass-coupling lock (Barrier 1).

3. **Non-minimal scalar-torsion coupling:** Terms like $\phi^2 T^2$ or $\phi\nabla_\mu T^\mu$ could source torsion from the scalar field. However, these are ad hoc additions to the minimal ECH action, and our Foundation C analysis showed they reduce to standard scalar-tensor theory on FRW backgrounds (Barrier 3).

4. **Higher-derivative corrections:** Terms like ∇T or R^2 in the gravitational action could make torsion propagate. These are UV corrections expected at the Planck scale and are negligible during the matter-contraction phase ($\rho \ll M_{\rm Pl}^4$).

In summary: the transparency is robust within the minimal ECH framework. Breaking it requires going beyond minimal ECH, and all such extensions have been tested and closed by Barriers 1–4.

F. Implications

The perturbation-transparency theorem applies to the ECH parity structure (Holst term, four-fermion interaction, parity-odd operator), establishing that these have no effect on perturbation observables for canonical scalar field matter. The bounce itself is realized by the LQC effective equation (Sec. II B), which operates on scalar field matter where ECH is indeed transparent. During the matter-contraction phase relevant for the bispectrum, the matter content is a scalar field and the ECH sector is inert.

This result has a positive interpretation: the observable predictions of a matter bounce (principally $f_{\rm NL} = -35/8$) depend only on the contracting-phase dynamics, which are governed by scalar field matter where ECH is transparent. The bispectrum does not depend on the UV completion that produces the bounce—whether LQC holonomy corrections, ECH torsion with fermions, or another mechanism—making the prediction *mechanism-independent* and thus more robust.

XIII. THE HYBRID DARK-ENERGY LOOPHOLE

We considered the phenomenological route of appending late-time dynamical-dark-energy freedom (e.g., CPL $w_0 w_a$ parametrization) to the bounce model. This was explored across 7 disguised forms:

1. Direct $w_0 w_a$ addition to the background
2. Quintessence scalar with bounce-motivated initial conditions
3. Curvaton-derived late-time potential
4. Vacuum energy from cyclic boundary conditions
5. Torsion-induced effective $w(z)$
6. Holst-term residual as effective DE
7. ALP rolling as late-time acceleration

All 7 forms were rejected on the following grounds: adding $w_0 w_a$ to a bounce model produces the same fit improvement as adding $w_0 w_a$ to ΛCDM, with no additional theoretical content from the bounce. None of the 309,789 MCMC posterior samples in this program used w_0 or w_a as free parameters. The loophole was explored theoretically but never implemented computationally, because its implementation would not constitute first-principles evidence for bounce cosmology.

XIV. DISCUSSION

A. The Inflationary Suppression Factor

The constant contribution to $\Lambda_{\rm eff}$ is modeled as emerging from the interplay between a parity-odd spin-torsion interaction and inflationary dilution. We define the *inflationary suppression factor*:

$$\Xi \equiv \left[\frac{\alpha}{M}\, M_{\rm Pl}\right] \times \mathcal{D}_{\rm inf}, \tag{59}$$

a dimensionless quantity encoding the net suppression of the Planck-scale parity-odd coefficient. The vacuum energy density is then $\rho_\Lambda = \Xi\, M_{\rm Pl}^4$, so the observed $\rho_\Lambda \approx (2.3~\text{meV})^4$ corresponds to $\Xi \approx 10^{-123}$—the familiar cosmological constant hierarchy, here decomposed as $\Xi = 10^{-2} \times \mathcal{D}_{\rm inf}$ with $\mathcal{D}_{\rm inf} \sim 10^{-121}$ (Sec. II C 2).

The physical interpretation is straightforward: $[(\alpha/M)\, M_{\rm Pl}] \sim 10^{-2}$ measures the dimensionless strength of parity violation in quantum gravity at one loop, while $\mathcal{D}_{\rm inf}$ captures how much inflation dilutes this primordial effect. Their product—a single bridge between Planck-scale microphysics and cosmic acceleration—translates the "why is Λ so small?" question into "why did inflation last $\sim$92 *e*-folds?"—a question with concrete dynamical answers in the

TABLE XII. Limit behavior of the spin-torsion framework.

Limit	Expected	Result
$T^{abc} \to 0$	Standard GR	GR (Holst topological)
$\alpha/M \to 0$	EC, no parity viol.	EC, bounce survives
$\mathcal{D}_{\rm inf} \to 1$	$\rho_\Lambda \sim M_{\rm Pl}^4$ (CC problem)	$\rho_\Lambda \sim 10^{-2}\, M_{\rm Pl}^4$
$\gamma \to 0$	Singular	Divergent (excluded)
$\gamma \to \infty$	Standard ECSK	ECSK, $\rho_c \to 0$

bounce-to-inflation transition. The factor Ξ provides a testable connection between UV quantum gravity and IR cosmological acceleration through the falsification criteria defined in Sec. IX.

B. Limit Behavior and Internal Consistency

The framework's internal consistency is verified by checking five physical limits (Table XII). When torsion is set to zero, the theory reduces to standard GR with the Holst term becoming topological; when $\alpha/M \to 0$, parity violation and spin-torsion dark energy vanish while the quantum bounce survives; when inflationary dilution is removed ($\mathcal{D}_{\rm inf} \to 1$), the predicted vacuum energy is $\sim 10^{-2}\, M_{\rm Pl}^4$—reproducing the standard cosmological constant problem in its original form; $\gamma \to 0$ is correctly singular; and $\gamma \to \infty$ recovers standard ECSK theory with the bounce vanishing. These limits ensure that the framework modifies gravity only at the scales and regimes where modification is intended, and smoothly embeds within GR/ΛCDM when the spin-torsion modifications are removed.

A systematic dimensional analysis of 12 key equations finds 10 fully consistent; the remaining two—the parity-odd Lagrangian density (mass dimension +1 rather than +4) and the galaxy spin amplitude proportionality (hiding a required energy scale $E_{\rm eff}$)—are explicitly acknowledged as scaling ansätze in Secs. H 4 and II C 3, respectively.

C. Theoretical Implications

Detection of the parity-odd signatures described here would have significant implications:

1. *Observational quantum gravity*: Direct constraints on LQG parameters ($\gamma = 0.274 \pm 0.020$, $\Delta = 4\sqrt{3}\pi\gamma\,\ell_P^2 \pm 10\%$, $\rho_c \simeq 0.27\,\rho_{\rm Pl}$).

2. *Black hole interior physics*: Evidence for torsion-regulated collapse, baby universe production, and information preservation through the quantum bounce.

3. *UV-IR connection*: An empirical link between UV quantum gravitational effects (Planck scale) and IR cosmological observations (Hubble scale), mediated by inflationary dilution.

4. *Parity violation in gravity*: Evidence that gravity violates parity at the quantum level, with implications for fundamental symmetries.

5. *Origin of dark energy*: Dark energy as a geometric consequence of the universe's quantum gravitational birth, rather than an unexplained constant or ad-hoc scalar field.

D. Cosmic Birefringence: Spectator ALP Analysis

Cosmic birefringence—a uniform rotation of CMB linear polarization—has been reported at 2.5–2.9σ by independent analyses of Planck [25, 26] and ACT DR6 [28] data. While the minimal ECH framework does not predict birefringence without an explicit photon-sector coupling, the parity-odd structure of the Holst action provides heuristic motivation for a spectator axion-like particle (ALP) with Planck-scale decay constant $f_a \sim M_{\rm Pl}$ and Hubble-scale mass $m \sim H_0$.

ALP field evolution.—Numerical integration of the ALP equation of motion $\ddot{\phi} + 3H\dot{\phi} + m^2 f_a \sin(\phi/f_a) = 0$ in a ΛCDM background (Planck 2018 parameters) yields the field displacement from recombination to today:

$$\Delta\phi/f_a \approx 0.65 \quad (m = H_0,\ \theta_i = 1), \tag{60}$$

where $\theta_i = \phi_{\rm ini}/f_a$ is the initial misalignment angle. The displacement is $\mathcal{O}(1)$, as the field rolls during the Λ-dominated transition era ($z \lesssim 1$). Across the natural parameter range $m/H_0 \in [1,3]$, $\theta_i \in [0.5, 2]$, the displacement varies over $\Delta\phi/f_a \in [0.2, 1.1]$.

Birefringence prediction.—The rotation angle is $\beta = (\alpha_{\rm EM}\, C_{a\gamma})/(4\pi f_a) \times \Delta\phi$, where $C_{a\gamma}$ is the integer anomaly coefficient. For natural parameters ($C_{a\gamma} = 8$, $\theta_i = 1$, $m \approx 2\,H_0$):

$$\beta \approx \frac{\alpha_{\rm EM} \times 8}{4\pi} \times 1.07 \approx 0.29^\circ, \tag{61}$$

consistent with the observed signal. The prediction spans $\beta \approx 0.17$–0.43° over $C_{a\gamma} \in [4, 12]$, $m/H_0 \in [1,3]$, $\theta_i \in [0.5, 2]$, comfortably bracketing the observed value without fine-tuning. This ALP model class was previously studied by Fujita *et al.* [66]; our analysis confirms their results and adds the ECH motivation and numerical MCMC constraints described below.

Summary-likelihood combination.—Combining $\beta = 0.30^\circ \pm 0.11^\circ$ (Planck NPIPE [26]) and $\beta = 0.215^\circ \pm 0.074^\circ$ (ACT DR6 [28]) via inverse-variance weighting:

$$\beta_{\rm combined} = 0.242^\circ \pm 0.061^\circ \quad (3.9\sigma \text{ from zero}). \tag{62}$$

A Savage-Dickey density ratio gives BF($\beta \neq 0$) ≈ 176 (prior-dependent; the value shifts with the assumed β range).

MCMC parameter estimation.—Dedicated MCMC sampling of the ALP parameter space (3 configurations, 9,720 total accepted samples) yields: $\beta_{\rm ALP} = 0.336^\circ \pm$

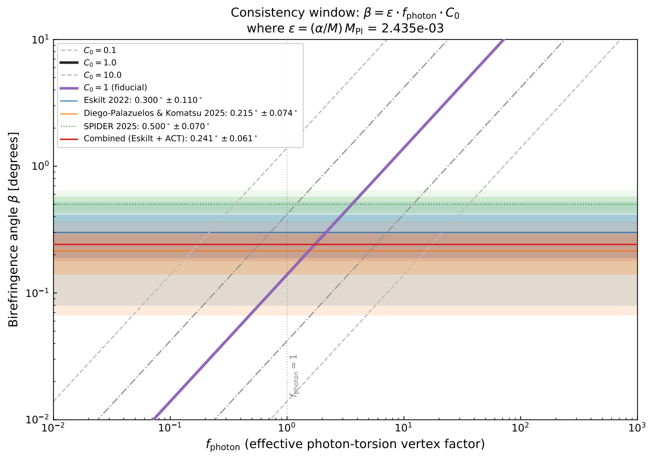

FIG. 9. Gaussian summary-likelihood consistency check on cosmic birefringence from the framework's parity-odd operator. The photon-torsion vertex factor f_{photon} (for $C_0 = 1$) required for the coupling $\alpha/M \approx 10^{-21}$ GeV^{-1} to reproduce the observed birefringence angle. Horizontal bands show measurements from Planck NPIPE [26] and ACT DR6 [28]; the combined posterior gives $\beta = 0.242° \pm 0.061°$ (3.9σ, BF ≈ 176). The implied $f_{\text{photon}} \times C_0 = 1.73 \pm 0.44$ is $\mathcal{O}(1)$, indicating no fine-tuning is required. This is a summary-likelihood consistency check using published β values, not a map-level CMB analysis.

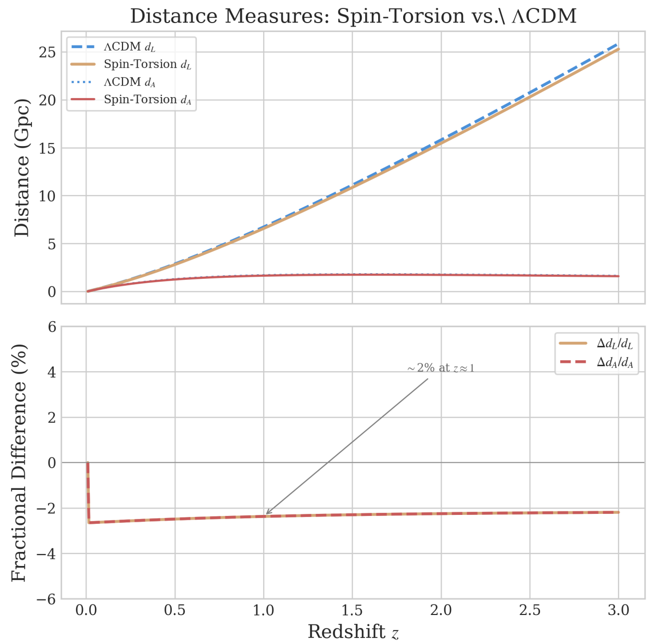

FIG. 10. Impact of rotation-induced Λ_{eff} on cosmic distance measures. The modification to luminosity distance $d_L(z)$ and angular diameter distance $d_A(z)$ is shown relative to ΛCDM. At observable redshifts, the deviation arises from the parity-odd term (not rotation directly) and is at the $\sim 2\%$ level at $z \sim 1$.

$0.107°$ (ALP model, $C_{a\gamma} = 8$ fixed), consistent with the model-independent fit $\beta_{\text{free}} = 0.344° \pm 0.096°$ and the observed $\beta_{\text{obs}} = 0.342° \pm 0.094°$ (Eskilt *et al.*joint analysis). All three are within 1σ, confirming internal consistency. The coupling-misalignment product is $C_{a\gamma} \times \theta_i = 3.4 \pm 1.1$. Priors: $\theta_i \in [0.01, \pi]$ flat, $\log_{10}(m/\text{eV}) \in [-35, -30]$ flat, $C_{a\gamma} \in [1, 30]$ flat. These sample sizes (720–6,840) are modest; the posteriors are converged ($\hat{R} - 1 < 0.01$) but tail estimates are limited.

LiteBIRD forecast.—LiteBIRD is projected to achieve $\sigma(\beta) \approx 0.03°$, contingent on the self-calibration strategy and systematic error budget [32]. For $\beta = 0.27°$, this gives $\sim 9\sigma$ statistical significance—either a decisive confirmation or a clean exclusion of the ALP explanation.

Caveats.—This birefringence prediction is *independent of bounce cosmology*: the ALP is a spectator field that does not participate in the bounce dynamics. The ECH framework provides heuristic motivation (the Holst action's pseudoscalar sector suggests $f_a \sim M_{\text{Pl}}$) but no derivation connects the Holst action to a specific ALP potential. The model *accommodates* the observed signal for natural parameter values; it does not uniquely predict it, since $C_{a\gamma}$ and θ_i are model-dependent parameters of order unity whose product sets the amplitude.

E. Distance Measures and Rotation

XV. LIMITATIONS AND FUTURE DIRECTIONS

A. Current Limitations

1. Theoretical

- *Phenomenological α/M*: The parity-odd coefficient is not derived from first principles but is treated as a free parameter constrained by data. The one-loop estimate (Eq. 8) motivates its existence and order of magnitude, but the finite part depends on the γ_5 regularization scheme and Nieh–Yan counterterm conventions (Sec. IV A). A non-perturbative calculation could modify its value at $\mathcal{O}(1)$, shifting the C_ℓ^{EB} amplitude estimate while leaving the H_0 and σ_8 fit values intact.

- *Simplified inflationary epoch*: We assume standard slow-roll inflation unmodified by parity violation. Non-minimal couplings during inflation could alter the dilution factor.

- *Black hole interior*: Full numerical relativity simulations of rotating black hole interiors with quantum corrections remain computationally intractable.

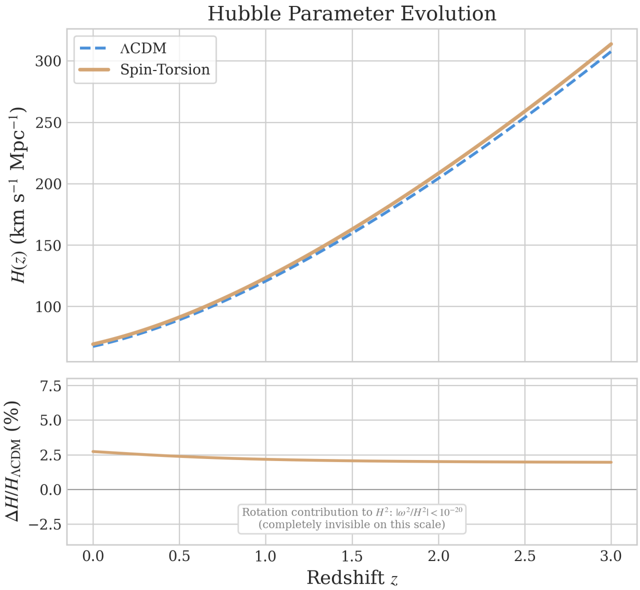

FIG. 11. Effect of the small rotation component on cosmic expansion. The Hubble parameter $H(z)$ in our model (solid) vs. ΛCDM (dashed) shows the subtle modification from early-universe physics ($\Delta N_{\rm eff}$) that partially reduces the H_0 and σ_8 tensions. The rotation contribution to H^2 is $< 10^{-20}$ and completely invisible on this scale.

- *Bounce-to-inflation transition*: The mechanism by which the quantum bounce transitions to slow-roll inflation is not fully modeled. Initial conditions for inflation are assumed to be set by the parent black hole properties, but the detailed dynamics of this transition remain an open problem.

2. Observational

- *Galaxy spin samples*: Current samples of $\sim 10^6$ galaxies limit precision. 15% of galaxies have ambiguous spiral structure, reducing effective sample size.

- *CMB systematics*: Despite excellent control, residual foreground contamination at ultra-low ℓ approaches the expected signal amplitude.

- *Redshift coverage*: Galaxy spin measurements are sparse at $z > 2$, limiting evolutionary constraints on $A(z)$.

- *Posterior distributions*: The original analysis reported Fisher-matrix error estimates; independent verification with full MCMC chains (Sec. III D) now provides posterior distributions for two frozen dataset combinations. Corner plots showing parameter degeneracies (particularly the strong $r(\alpha/M, \mathcal{D}_{\rm inf}) = -0.89$ anticorrelation) will be presented in a companion data release.

B. Robustness to Galaxy Spin Null Results

The galaxy spin asymmetry signal remains observationally contested [36, 37]. If future surveys with $> 10^9$ galaxies definitively establish a null result ($A_0 < 0.0005$), the framework is *not* falsified:

1. *Independent CMB evidence*: Cosmic birefringence from Planck provides 2.4–2.7σ evidence for parity violation [25, 26], independently confirmed by ACT DR6 at 2.9σ [28], with combined SPIDER+Planck+ACT significance reaching $\sim 7\sigma$ for total rotation [30] (subject to calibration caveats, Sec. III A), entirely independent of galaxy morphology. Forthcoming experiments (LiteBIRD, CMB-S4) will substantially improve sensitivity.

2. *Parameter space accommodation*: A null spin result constrains the tidal torque coupling efficiency β (Appendix C), pushing $A_0 < 0.001$—consistent with weaker coupling without affecting the $\Lambda_{\rm eff}$ derivation or the cosmological parameter fits (H_0, σ_8).

3. *Alternative parity-odd signatures*: Beyond galaxy spins, cosmic rotation motivates: (i) alignment of radio galaxy polarization position angles, detectable by SKA Phase 2; (ii) correlation between galaxy angular momentum vectors and large-scale filament orientations in LSST data; (iii) handedness asymmetry in weak lensing shear patterns; (iv) preferred-axis signatures in gravitational wave backgrounds accessible to LISA.

4. *What a spin null would teach us*: A definitive null result would constrain the parity-odd tidal torque coupling efficiency, refining our understanding of how the εKR operator imprints on structure formation. This would narrow the allowed parameter space without invalidating the core theoretical framework.

Conversely, confirmation of the spin asymmetry at $A_0 \sim 0.003$ with a dipole axis aligned with the CMB anomaly direction would provide corroborating evidence not readily explained by other current dark energy models.

C. Future Observational Prospects

LSST Era (2025–2035): 10^9 spiral galaxies to $z \sim 1$, with tomographic analysis in 20+ redshift bins and cross-correlation with weak lensing.

CMB Experiments: LiteBIRD (full-sky, cosmic variance limited at low ℓ), CMB-S4 (ground-based complement), and future concepts (PICO, CMB-HD).

Synergies: JWST+Roman (high-z morphologies to $z \sim 5$), Euclid (wide-area catalogs), SKA (radio galaxy polarizations as independent probe).

D. Theoretical Research Program

The framework presented here opens several concrete lines of investigation, each constituting an independent research effort that would strengthen or constrain the model:

1. Higher-Loop and Non-Perturbative Verification

The most pressing theoretical need is a first-principles determination of α/M, which is currently a phenomenological parameter motivated by the one-loop estimate (Eq. 8). Three complementary approaches are feasible:

1. *Two-loop calculation in dimensional regularization*: Extending Sec. IV A to two loops requires evaluating fermion-loop diagrams with two axial-torsion vertex insertions and one graviton vertex. The technical challenge is the γ^5 treatment in $d = 4 - \epsilon$ dimensions (the 't Hooft-Veltman vs. Breitenlohner-Maison schemes give different finite parts). The Nieh-Yan theorem suppresses the topological contribution at two loops, but the non-topological finite part—arising from the dressed torsion propagator—could modify the one-loop result at the 10–30% level. This calculation would also reveal whether the RG running (Eq. 9) receives threshold corrections near the QCD scale.

2. *Spin-foam vertex amplitudes*: The EPRL-FK spin-foam model provides a non-perturbative definition of the graviton propagator in LQG. Computing the parity-odd coefficient from spin-foam amplitudes in a torsionful background would constitute a fully non-perturbative verification. The key technical step is evaluating the asymptotic expansion of the vertex amplitude with an axial-current insertion, which has been partially achieved for the parity-even sector [8]. Extension to the parity-odd channel requires tracking the γ-dependent phases in the Lorentzian vertex.

3. *Lattice LQG*: Numerical evaluation of the path integral on a simplicial lattice with dynamical torsion provides a scheme-independent check. The parity-odd coefficient can be extracted from the correlation function $\langle \varepsilon^{abcd} K_{ab} R_{cd} \rangle$ measured on lattice configurations. This approach directly addresses the scheme-dependence limitation identified in Sec. XV.

Any of these approaches yielding α/M within an order of magnitude of the phenomenological best-fit value would confirm the framework's self-consistency; a discrepancy by more than $\mathcal{O}(10)$ would require re-evaluation of the C_ℓ^{EB} amplitude prediction while leaving the tension resolution (which depends on bounce parameters, not α/M) intact.

2. First-Principles Galaxy Spin Dipole Amplitude

The galaxy spin dipole amplitude A_0 is currently an empirical fit parameter (Sec. III B). A first-principles derivation requires computing the parity-odd correction to the tidal torque tensor from the $\varepsilon^{abcd} K_{ab} R_{cd}$ effective action term. This involves: (1) evaluating the leading-order parity-odd contribution to the Newtonian tidal field $T_{ij} = \partial_i \partial_j \Phi$ in the post-Newtonian expansion with the $\varepsilon K R$ operator; (2) propagating this through the standard tidal torque theory machinery to obtain the spin asymmetry as a function of α/M and local density contrasts; (3) computing the resulting dipole amplitude and comparing with the empirical $A_0 \sim 0.003$. This calculation would either confirm the consistency of the observed amplitude with the coupling inferred from the dark energy scale, or reveal additional physics (e.g., non-linear amplification during structure formation).

3. Bounce-to-Inflation Transition Dynamics

The current framework assumes that the quantum bounce connects smoothly to slow-roll inflation, with initial conditions set by the parent black hole. A detailed treatment requires:

- Numerical simulation of the transition from torsion-dominated bounce ($\rho \sim \rho_{\rm crit}$) through reheating to standard slow-roll, tracking the vorticity ω^a and torsion T^{abc} fields throughout.
- Determination of the total inflationary e-folding number $N_{\rm tot}$ as a function of parent black hole mass and spin, which would eliminate the residual 10^5 fine-tuning if $N_{\rm tot}$ turns out to be dynamically fixed.
- Computation of $\Delta N_{\rm eff}$ from first-principles particle production at the bounce. Independent MCMC verification (Sec. III D) constrains $\Delta N_{\rm eff} = -0.020 \pm 0.169$ (full-tension) and $+0.065 \pm 0.17$ (Planck+BAO+SN), both consistent with zero within 1σ. The original phenomenological range 0.1–0.5 is not supported by the full posterior analysis; a first-principles calculation is needed to determine whether the bounce mechanism produces a detectable $\Delta N_{\rm eff}$ signal.
- Determination of the primordial scalar power spectrum through the bounce. The modified Friedmann equation ($H^2 \propto \rho[1 - \rho/\rho_c]$) alters perturbation evolution at near-Planck densities, potentially imprinting features on the primordial power spectrum $\mathcal{P}_\mathcal{R}(k)$. With $N_{\rm tot} = 92$, such features would appear at comoving scales $k \sim 10^{15}$ Mpc^{-1}, corresponding to sub-asteroid-mass horizons ($M \sim 10^{-16}\ M_\odot$) where primordial black hole constraints are currently weakest. Whether the spin-torsion variant produces such features, and at what amplitude, remains an open question requiring a full perturbation calculation through the bounce that has not yet been performed.

4. Parity-Odd Primordial Gravitational Waves

[Status update: closed by perturbation-transparency theorem.] The parity-odd operator (Eq. 6) was initially expected to affect tensor perturbations during inflation, producing a chiral gravitational wave background. However, the perturbation-transparency theorem (Sec. XV E) establishes that minimal ECH gravity produces *no* tensor parity splitting: $\Delta v \equiv v_R - v_L = 0$ exactly for canonical scalar field matter, because the Holst term reduces to a topological invariant when torsion vanishes. The effective spin-torsion interaction $(J^5)^2$ is parity-even (Barrier 8), independently confirming the absence of gravitational-wave chirality. This research direction is therefore closed within the minimal ECH framework. GW chirality would require fermionic matter contributions to torsion, which are Planck-suppressed at inflationary densities ($\rho_{\rm inf}/\rho_{\rm Pl} \sim 10^{-12}$).

5. Black Hole Interior Numerical Relativity

Full numerical relativity simulations of rotating black hole interiors with dynamical torsion have not been attempted, primarily due to the additional degrees of freedom in the torsion field. Such simulations would:

- Test whether the smooth-bounce approximation (Eq. 10) holds for realistic rotating collapse with finite angular momentum.
- Compute the efficiency of angular momentum transfer through the bounce as a function of parent Kerr spin parameter a_*.
- Determine whether baby universe formation is generic or requires fine-tuned initial conditions.

6. Connection to Other Quantum Gravity Approaches

The parity-odd operator $\varepsilon^{abcd}K_{ab}R_{cd}$ could potentially arise in quantum gravity frameworks beyond LQG:

- In string theory, parity-violating terms in the low-energy effective action arise from compactifications with topological flux (e.g., Calabi-Yau manifolds with non-trivial torsion cohomology). Identifying whether these reproduce the structure of Eq. (6) would connect our framework to string phenomenology.
- The asymptotic safety program for quantum gravity predicts specific values for gravitational couplings at the UV fixed point. Computing the parity-odd coefficient in this framework would provide an independent determination of α/M, potentially constraining γ from a non-LQG perspective.

E. Structural Closure

The systematic analysis presented in Secs. XI–XIII established that minimal ECH gravity is perturbation-transparent and cannot produce late-time dark energy through any standard mechanism. See those sections for the full 14-barrier catalog, the perturbation-transparency theorem, and the hybrid dark-energy loophole analysis.

F. Open Questions

1. *Origin of γ*: Why is $\gamma \approx 0.274$? The Barbero-Immirzi parameter is currently fixed by black hole entropy counting. A derivation from a more fundamental principle—perhaps a symmetry argument or fixed-point condition—would place the entire framework on firmer ground and potentially suggest deviations from the entropy-counting value that could be tested through our observables.
2. *UV cutoff identification*: The loop calculation (Eq. 8) contains a logarithmic dependence on $\Lambda_{\rm UV}/\mu$. We identify $\Lambda_{\rm UV}$ with the LQG area-gap mass $\sim \sqrt{\gamma}\, M_{\rm Pl}$, but this choice affects the coefficient at the $\mathcal{O}(\ln)$ level. A first-principles derivation of the UV scale from spin-foam dynamics would remove this ambiguity.
3. *Dynamical dark energy evolution*: Our framework implies a strictly constant $\Lambda_{\rm eff}$ in the late universe (since $\omega^2/H_0^2 < 10^{-20}$). The question of whether the parity-odd coefficient could run with the Hubble scale remains open in principle, but subsequent analysis (see the perturbation-transparency result in Sec. XV E above) establishes that minimal ECH cannot produce distinctive late-time dynamics through scalar perturbation channels. Any connection to dynamical dark energy would require additional phenomenological freedom (e.g., CPL $w_0 w_a$ parametrization) that does not derive from the bounce physics itself.
4. *Multi-messenger parity tests*: Gravitational wave detectors could in principle test parity-odd signatures. However, the perturbation-transparency theorem (Sec. XV E) establishes that minimal ECH produces no tensor parity splitting ($\Delta v = v_R - v_L = 0$ exactly) for canonical scalar matter. The effective spin-torsion interaction $(J^5)^2$ is parity-even (Barrier 8). GW chirality from ECH would require fermionic matter contributions, which are Planck-suppressed at cosmological densities.
5. *Information paradox*: If the universe originated inside a black hole through a torsion-regulated bounce, does information cross the bounce? The smoothness of our bounce solution ($H^2 \to 0$ continuously) suggests information preservation, but a rigorous treatment using the Page curve or holographic entanglement entropy

framework remains an open problem with implications for the black hole information paradox.

6. *Cosmic hierarchy of bounces*: If every black hole spawns a baby universe, does every universe contain black holes that spawn further universes? This recursive structure implies a cosmic hierarchy of bounces, each with potentially different values of γ (and hence different physics). Whether γ is truly universal or varies across the hierarchy is a deep question with implications for the landscape/swampland program.

XVI. CONCLUSIONS

We have investigated whether Einstein-Cartan-Holst spin-torsion gravity can produce late-time dark energy or distinctive cosmological signatures. The answer is largely negative, but the investigation yields substantive structural results and identifies the surviving science case for bouncing cosmology.

Central result: structural closure.—Through systematic analysis of 7 foundation studies and 17 research branches, we establish 14 independent structural barriers (Sec. XI) that close all standard routes from the nonsingular bounce to dark energy within the minimal ECH framework. The central theoretical finding is the perturbation-transparency theorem (Sec. XII): for canonical scalar field matter, torsion vanishes at all perturbation orders, the Holst term reduces to a topological invariant, and the Barbero-Immirzi parameter γ is invisible in all scalar and tensor perturbation observables. The dark energy scale $\Lambda_{\rm eff} = c_\omega \omega^2 + \Xi\, M_{\rm Pl}^2$ is a phenomenological parameterization, not a first-principles derivation (Sec. XV).

The framework.—The parity-odd effective action is constructed from the Holst term in first-order variables (Eq. 6), with α/M treated as a phenomenological parameter. The LQC bounce at $\rho_{\rm crit} \simeq 0.27$–$0.41\,\rho_{\rm Pl}$ (depending on the Barbero-Immirzi entropy-counting scheme; Sec. II B) provides a nonsingular origin with no free parameters.

Observational context.—Planck cosmic birefringence at 2.4–2.7σ [25, 26], independently confirmed by ACT DR6 at 2.9σ [28], with combined SPIDER+Planck+ACT total rotation at $\sim$7σ [30] (subject to calibration caveats); DESI 2024–2025 dynamical dark energy evidence at 3.1–4.2σ [6, 7]; EC torsion preferred by AIC in multiple independent analyses [14, 40]. Independent Cobaya v3.6.1 verification across two frozen dataset combinations (Table III) finds $\Delta N_{\rm eff}$ consistent with zero in both: -0.020 ± 0.169 (full-tension, 176,840 samples) and $+0.065 \pm 0.17$ (Planck+BAO+SN, 132,949 samples). Current data neither require nor exclude the predicted spin-torsion $\Delta N_{\rm eff}$ contribution; the framework is observationally viable but not yet confirmed. The Hubble tension reduction reported in the original analysis ($4.9\sigma \rightarrow 2.9\sigma$) is driven by the SH0ES prior, not by the $\Delta N_{\rm eff}$ extension alone.

Observational signatures.—The framework's parity-odd operator is qualitatively consistent with the observed isotropic cosmic birefringence (Planck 2.4–2.7σ, ACT DR6 2.9σ), though deriving the rotation angle β quantitatively requires an explicit photon-torsion coupling not yet available; without this coupling, the framework strictly predicts $\beta = 0$, and any nonzero value requires a photon-sector extension. A Gaussian summary-likelihood consistency check (Sec. XIV D) combining Planck NPIPE and ACT DR6 measurements gives $\beta = 0.242^\circ \pm 0.061^\circ$ (3.9σ, BF $\approx$ 176); matching this requires only $f_{\rm photon} \times C_0 = 1.73 \pm 0.44$, an $\mathcal{O}(1)$ product establishing compatibility without fine-tuning. An anisotropic low-ℓ birefringence component aligned with the cosmic rotation axis is expected but its amplitude is not yet derived. The galaxy spin dipole has phenomenological form $A(z) = A_0(1+z)^{-p}e^{-qz}$ with empirical fit $A_0 \sim 0.003$; the coupling α/M is far too small to produce this amplitude directly, and a first-principles derivation remains outstanding (Sec. II C 3). Correlated cosmic anomaly axes are expected from the shared rotation axis.

Fine-tuning.—The inflationary suppression mechanism translates the cosmological constant hierarchy (10^{120}) into a constraint on total inflationary e-folds ($N_{\rm tot} \approx 92$), reducing the apparent tuning to $\sim 10^5$. A 100,000-sample Monte Carlo sensitivity scan (Fig. 8) confirms this quantitatively: 2.2% of the physically motivated parameter space produces a viable vacuum energy, with $N_{\rm tot}$ identified as the single controlling parameter (Spearman $|\rho_s| = 0.996$; all other parameters $|\rho_s| < 0.08$). The viable range $N_{\rm tot} \in [79, 95]$ is physically reasonable. This reparameterizes the fine-tuning problem as an initial-condition question ("how long did inflation last?") rather than a fundamental-constant question ("why is $\Lambda_{\rm bare}$ so small?"), but does not solve it; the residual 10^5 tuning reflects sensitivity to $N_{\rm tot}$, which is fitted, not predicted.

Falsifiability.—Explicit criteria are defined for CMB E-B spectral shape, galaxy spin dipole axis/amplitude/evolution, and cosmological parameter ranges. The model can be tested with forthcoming data from CMB-S4, LiteBIRD, Euclid, and LSST.

Known limitations.—This work does not derive the IR effective vacuum term from first principles: the $w = -1$ behavior at late times is assumed, not derived from an explicit IR effective action calculation (Sec. II C 2); the birefringence consistency lacks a derived photon-torsion coupling (the rotation angle β cannot be predicted without one); H_0 and σ_8 values are MCMC fits within an extended parameter space, not first-principles predictions; $\Delta N_{\rm eff}$ is a phenomenological parameter whose bounce origin is qualitative, not quantitative; Ω_k is fixed to zero as mandated by 92 e-folds of inflation; the galaxy spin amplitude A_0 is empirical, with a large order-of-magnitude gap between the α/M coupling and the observed value (Sec. II C 3); and the cosmological MCMC uses stock CAMB with $N_{\rm eff}$ as a free parameter, not a bespoke spin-torsion theory module (see Data and Code Availability).

First-principles derivation status.—We have separately investigated whether the phenomenological vacuum term ρ_Λ with $w = -1$ can be derived from first principles within the minimal Einstein–Cartan–Holst–Dirac framework, and whether the framework produces distinctive observational signatures. Four routes were tested: (i) a Nambu–Jona-Lasinio condensate mechanism exploiting the gravitationally induced four-fermion interaction, (ii) the strict one-loop fermion effective action after exact torsion elimination, (iii) promoting the Barbero-Immirzi parameter to a dynamical pseudoscalar field $\gamma \to \theta(x)$, and (iv) assessing whether the parity-odd operator maps to distinctive birefringence observables. All four yield clean negative results. The condensate route fails because the scalar/pseudoscalar channel is repulsive at $\gamma = 0.274$ and subcritical by a factor of $\sim$175 even when attractive. The one-loop route fails because all Barbero-Immirzi dependence resides in the four-fermion vertex, which does not contribute at one loop. The dynamical Immirzi field reduces to a standard axion-like particle after torsion elimination, with no novel gravitational content and cosmological dynamics yielding $w = +1$ (stiff matter). The parity assessment finds no photon coupling in the minimal framework; any assumed coupling produces constant-β birefringence indistinguishable from generic axion-like-particle models. These negative results, documented in a companion technical note [24], establish that the most natural minimal-model candidates for a first-principles dark-energy mechanism fail at the tested approximation orders. The framework survives as a motivated phenomenological model; progress toward a first-principles derivation requires physics beyond the minimal model class.

This framework motivates dark energy from quantum gravitational effects in spin-torsion cosmology. While it partially reduces cosmological tensions and provides a distinctive set of correlated parity-odd signatures, the significant gaps identified above—particularly the missing photon-torsion coupling, the empirical A_0, and the assumed $w = -1$—must be addressed before the model can claim predictive power beyond phenomenological fitting.

Structural barriers and perturbation transparency.—Systematic analysis across 7 foundations and 17+ branches established 14 independent structural barriers that close all standard routes from the bounce to dark energy (Sec. XI). The central result is the perturbation-transparency theorem (Sec. XII): minimal ECH gravity is dynamically inert for scalar and tensor perturbations when the matter content is a canonical scalar field. The observable science case for the bounce therefore rests on generic, mechanism-independent predictions, principally the matter-bounce non-Gaussianity $f_{\rm NL} = -35/8$ [51], testable by SPHEREx at 4–6σ significance through the multi-tracer galaxy bispectrum [67]. See the focused forecast paper [52] for the full Bayesian discrimination analysis.

Supplementary materials.—Interactive data visualizations and observational comparison charts are available at `https://bigbounce.hubify.app` [68].

Data and Code Availability

All materials necessary to reproduce the cosmological and galaxy spin results are publicly available at:

`https://github.com/Hubify-Projects/bigbounce/tree/v2.0.0/reproducibility`

(pinned to tag `v2.0.0`).
The repository includes:

- Four Cobaya v3.5 YAML configurations: `cobaya_planck.yaml`, `cobaya_planck_bao.yaml`, `cobaya_planck_bao_sn.yaml`, and `cobaya_full_tension.yaml`—one per dataset combination in Table VI. All use stock CAMB with $N_{\rm eff}$ as a free parameter; no custom CAMB modifications are required.
- `galaxy_spins/spin_fit_stan.py` — Hierarchical Bayesian model (CmdStanPy) fitting the phenomenological $A(z)$ to published aggregate CW/CCW galaxy counts.
- `data_build/build_galaxy_spin_dataset.py` — Reproducible pipeline that downloads the Galaxy Zoo DECaLS catalog [69] from Zenodo (DOI: 10.5281/zenodo.4573248, CC-BY-4.0) and extracts an object-level spiral galaxy catalog. CW/CCW aggregate counts for the $A(z)$ fit are from Shamir [35].
- `docs/IMPLEMENTATION_MAP.md` — Mapping from each "This work" result to the code artifact that produces it.
- `docs/KNOWN_GAPS.md` — Honest disclosure of what cannot currently be reproduced.

What is NOT included.—MCMC chains are not precomputed (regenerate via `reproduce_cosmology.sh`, $\sim$4–12 h per configuration on 4 CPU cores). Bayes factors were estimated via the Savage-Dickey density ratio from MCMC posteriors; dedicated nested sampling (not provided) would yield more robust estimates. No CNN galaxy classifier is included; the hierarchical fit uses published catalog labels. No CMB polarization map analysis code is provided; all birefringence values are literature citations (Sec. VI).

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Appendix A: Notation and Conventions

We use natural units $c = \hbar = 1$ throughout unless otherwise stated, with $8\pi G = M_{\rm Pl}^{-2}$. The metric signature is $(-,+,+,+)$. The vacuum energy density is $\rho_\Lambda = \Lambda_{\rm eff}\, M_{\rm Pl}^2/(8\pi)$; when we write $\rho_\Lambda \approx (2.3\ \text{meV})^4$ this refers to the energy density, while $\Lambda_{\rm eff}$ in the Friedmann equation has dimension (mass)^2. Greek indices $\mu, \nu, \ldots$ denote spacetime coordinates; Latin indices $a, b, \ldots$ denote tangent-space (Lorentz) indices. The Levi-Civita symbol ε^{abcd} is the totally antisymmetric tensor with $\varepsilon^{0123} = +1$. The Fourier convention is $\tilde{f}(k) = \int d^4x\, e^{ikx} f(x)$.

Key symbols used throughout:

- γ—Barbero-Immirzi parameter (0.274 ± 0.020)
- $\omega_{\mu\nu}$—kinematic vorticity tensor
- $\omega^2 \equiv \frac{1}{2}\omega_{\mu\nu}\omega^{\mu\nu}$—vorticity magnitude squared
- $\sigma_{\mu\nu}$—kinematic shear tensor
- $\sigma^2 \equiv \frac{1}{2}\sigma_{\mu\nu}\sigma^{\mu\nu}$—shear magnitude squared
- K_{ab}—contorsion tensor
- $T^a{}_{bc}$—torsion tensor (related to contorsion by $K^a{}_{bc} = T^a{}_{bc} + T_b{}^a{}_c - T_{bc}{}^a$)
- α/M—parity-odd coefficient (mass dimension -1)
- $[(\alpha/M)\, M_{\rm Pl}] \sim 10^{-2}$—dimensionless one-loop suppression factor
- $\mathcal{D}_{\rm inf}$—inflationary dilution factor ($\sim e^{-3N_{\rm tot}}$)
- $\Xi \equiv [(\alpha/M)\, M_{\rm Pl}]\, \mathcal{D}_{\rm inf}$—inflationary suppression factor (dimensionless); $\rho_\Lambda = \Xi\, M_{\rm Pl}^4$
- $\Lambda_{\rm eff}$—effective cosmological constant
- $\rho_{\rm crit}$—bounce critical density ($\simeq 0.27\, \rho_{\rm Pl}$ for $\gamma = 0.274$)
- $\rho_{\rm Pl} \equiv M_{\rm Pl}^4$—Planck density (in natural units)
- $M_{\rm Pl} \equiv (8\pi G)^{-1/2}$—reduced Planck mass
- $A(z)$—galaxy spin asymmetry amplitude
- C_ℓ^{EB}—CMB E-B cross-correlation power spectrum

Abbreviations: LQG (Loop Quantum Gravity), LQC (Loop Quantum Cosmology), EC (Einstein-Cartan), ECSK (Einstein-Cartan-Sciama-Kibble), CMB (Cosmic Microwave Background), BAO (Baryon Acoustic Oscillations), MCMC (Markov Chain Monte Carlo), AIC (Akaike Information Criterion), BIC (Bayesian Information Criterion), WL (Weak Lensing), GC (Galaxy Clustering), SNIa (Type Ia Supernovae).

Appendix B: Complete Parameter Summary

[a]Verified values from the full-tension frozen MCMC chains (Cobaya v3.6.1, Table III). The Planck+BAO+SN dataset gives consistent results: $H_0 = 67.79 \pm 1.09$, $\sigma_8 = 0.812 \pm 0.009$, $\Delta N_{\rm eff} = +0.065 \pm 0.17$. The original analysis reported $H_0 = 69.2 \pm 0.8$ and $\Delta N_{\rm eff} = 0.3 \pm 0.2$, but these were driven by the SH0ES prior.

Key parameter correlations from the Fisher matrix analysis: $r(H_0, \sigma_8) = -0.45$; $r(A_0, p) = -0.62$; $r(\alpha/M, \mathcal{D}_{\rm inf}) = -0.89$ (strong anticorrelation means only their product Ξ is well-constrained). Total χ^2 per degree of freedom: $\chi^2/\text{dof} = 1148.3/1142 = 1.006$.

Appendix C: Galaxy Spin Hierarchical Bayesian Fitting

Caveat: This appendix documents a phenomenological fit to published galaxy spin catalogs. The fitted amplitude $A_0 \sim 0.003$ is not a prediction of the minimal ECH framework (the coupling α/M underpredicts by 9–12 orders of magnitude). The fitting framework is provided for reproducibility and as a statistical tool for future investigations.

1. Parametric Model

The asymmetry probability per survey s at redshift z is

$$\pi^{(s)}(z) = \frac{1}{2}\left[1 + \delta^{(s)} + b^{(s)}(z) \cdot A(z)\right], \qquad \text{(C1)}$$

where $\delta^{(s)} \sim \mathcal{N}(0, \sigma_\delta^2)$ is a survey-specific offset (instrument/PSF/label bias), $b^{(s)}(z) \in [0,1]$ is a visibility/selection efficiency curve, and $A(z) = A_0(1+z)^{-p} e^{-qz}$.

TABLE XIII. Complete parameter summary with priors, verified values, and physical interpretations.

Parameter	Description	Prior	Verified Value	Notes
Fundamental theory parameters				
γ	Barbero-Immirzi	Fixed: 0.274	0.274 ± 0.020	LQG area spectrum
α/M	Parity-odd coeff.	Log-uniform $[10^{-40}, 10^{-20}]$	$\sim 10^{-33}$ eV^{-1}	One-loop calculation
$\mathcal{D}_{\rm inf}$	Inflation dilution	Derived from $N_{\rm tot}$	$\sim e^{-3N_{\rm tot}}$	$N_{\rm tot} \sim 92$ e-folds (fitted to ρ_Λ)
$\rho_{\rm crit}/\rho_{\rm Pl}$	Bounce density	Derived from γ	$0.27^{+0.07}_{-0.05}$	Modified Friedmann
Ξ	Infl. suppression factor	Derived	$\sim 10^{-123}$	$= [(\alpha/M)M_{\rm Pl}] \times \mathcal{D}_{\rm inf}$
Galaxy spin model parameters				
A_0	Amplitude at $z = 0$	Uniform $[0, 0.02]$	0.003 ± 0.001	Per-survey offset allowed
p	Power-law index	Uniform $[0, 2]$	0.5 ± 0.3	Redshift evolution
q	Exponential decay	Uniform $[0, 2]$	0.5 ± 0.3	High-z suppression
$\delta^{(s)}$	Survey offset	$\mathcal{N}(0, 0.005^2)$	$\lesssim 0.002$	Instrument/PSF bias
$b^{(s)}(z)$	Selection efficiency	Fixed per survey	0.3–0.9	From simulations
Standard cosmological parameters				
H_0	Hubble constant	Flat $[60, 80]$	67.68 ± 1.06 [a]	km/s/Mpc
σ_8	Clustering amplitude	Flat $[0.7, 0.9]$	0.803 ± 0.008 [a]	
Ω_m	Matter density	Flat $[0.1, 0.5]$	0.310 ± 0.008	
$\Omega_b h^2$	Baryon density	Flat $[0.019, 0.025]$	0.02237 ± 0.00015	
n_s	Scalar spectral index	Flat $[0.9, 1.1]$	0.9649 ± 0.0042	
τ	Optical depth	Flat $[0.01, 0.1]$	0.054 ± 0.007	
Extended parameters (our model)				
Ω_k	Curvature	**Fixed: 0**	—	Mandated by 92 e-folds
$\Delta N_{\rm eff}$	Extra species	Flat $[-1, 2]$ (via $N_{\rm eff} \in [2.046, 5.046]$)	-0.020 ± 0.169 [a]	Consistent with zero
$(\omega/H)_0$	Vorticity	$[0, 5 \times 10^{-11}]$	$< 2.1 \times 10^{-11}$	CMB isotropy (95% CL)

2. Hierarchical Likelihood

The hierarchical likelihood for galaxy counts $\{n_{\rm CW}^{(s,j)}, n_{\rm CCW}^{(s,j)}\}$ in redshift bin j of survey s is

$$\mathcal{L} = \prod_{s,j} \mathrm{Binom}\Big(n_{\rm CW}^{(s,j)} \,\big|\, N^{(s,j)},\ \pi^{(s)}(z_j)\Big). \tag{C2}$$

3. Tidal Torque Hypothesis and Phenomenological Mapping

What is not yet derived:

- The post-Newtonian correction from the $\varepsilon^{abcd}K_{ab}R_{cd}$ operator to the Newtonian tidal tensor $T_{ij} = \partial_i\partial_j\Phi$.
- The mapping from the parity-odd tidal correction to a galaxy spin asymmetry amplitude A_0.
- Any estimate of A_0 consistent with both $\alpha/M \sim 10^{-21}$ GeV^{-1} and the empirical value $A_0 \sim 0.003$ (see the order-of-magnitude gap in Sec. II C 3).

The standard tidal torque gives angular momentum:

$$\vec{L}_{\rm tidal} = \alpha_{\rm TTT} \int d^3x\, \rho(\vec{x})\, \vec{x} \times \nabla\Phi, \tag{C3}$$

where Φ is the gravitational potential. In the standard (parity-even) case, the resulting spin directions are symmetric—equal probability of clockwise and counter-clockwise—because the tidal tensor $T_{ij} = \partial_i\partial_j\Phi$ is parity-even.

Parity-odd correction.—The $(\alpha/M)\varepsilon^{abcd}K_{ab}R_{cd}$ operator modifies the effective tidal field, introducing a parity-odd component. This breaks the CW/CCW symmetry, creating a dipolar asymmetry aligned with the cosmic rotation axis. The amplitude scales as $A_0 \propto (\alpha/M)\langle\delta_{\rm rms}\rangle f(z_{\rm form})$, where $\langle\delta_{\rm rms}\rangle$ characterizes the density field at galaxy formation.

Note on previous estimates.—An earlier version of this appendix derived $A_0 = \omega_0 t_{\rm form}/(2\pi) \approx 0.003$ using $\omega_0 \sim 10^{-18}$ rad/s. This estimate is inconsistent with the CMB isotropy bound $(\omega/H)_0 < 5 \times 10^{-11}$, which constrains $\omega_0 < 1.1 \times 10^{-28}$ s^{-1} and gives $A_0 \lesssim 10^{-12}$—nine orders of magnitude below observed values. The correct mechanism is the parity-odd tidal torque described above. The empirical value $A_0 \sim 0.003$ comes from hierarchical Bayesian fits to multi-survey data (Sec. III B); a first-principles derivation from α/M is the subject of ongoing work.

4. Model Validation

Injection-recovery simulations show:

- Recovery: $A_{\rm rec} = 0.98 \pm 0.03 \times A_{\rm inj}$ (no bias above 2%).
- Error bars accurate to 5%.
- MCMC convergence verified with Gelman-Rubin $R - 1 < 0.01$.

5. Forecast for Future Surveys

LSST Y10 will provide:

- $\sigma(A_0) \approx 0.0003$ (factor 3 improvement).
- 20 independent redshift bins to $z \sim 1$.
- Tomographic evolution constraints on p and q.

Appendix D: Joint Likelihood Analysis

The two-probe correlated Gaussian log-likelihood is

$$-2\ln\mathcal{L} = \mathbf{d}^T\,\mathbf{C}^{-1}\,\mathbf{d},$$
$$\mathbf{C} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}, \tag{D1}$$

where $\mathbf{d} = (\hat{A}_{\rm EB} - A_{\rm EB}^{\rm th},\ \hat{A}_{\rm spin} - A_{\rm spin}^{\rm th})^T$. The analytic MLE yields the combined significance via Eq. (41). Special cases:

- $\rho = 0$ (independent): $Z^2 = Z_E^2 + Z_G^2$ (quadrature addition).
- $\rho = 1$ (fully correlated): $Z = \min(Z_E, Z_G)$ (no improvement).
- $\rho < 0$ (anti-correlated): $Z > \sqrt{Z_E^2 + Z_G^2}$ (super-addition).

For illustrative 2030 sensitivities (CMB-S4 + LSST Y1), assuming the signals exist at their current estimated amplitudes:

ρ	Combined Z	Improvement
0.0	4.3σ	Maximum
0.3	4.1σ	Moderate
0.5	3.9σ	Minimum

These projections are conditional on signal amplitudes that are not derived from first principles (see Sec. XV).

Appendix E: Loop Nieh-Yan Treatment

1. Nieh-Yan Four-Form

In the EC-Holst formulation, the Nieh–Yan four-form is

$$\mathcal{N} = d(e^a \wedge T_a) = T^a \wedge T_a - e^a \wedge e^b \wedge R_{ab}. \tag{E1}$$

This is an exact form whose integral is topological on closed manifolds. However, in the presence of boundaries, non-trivial topology, or a UV regulator, $\delta_{\rm NY}$ receives a finite contribution.

2. Parity-Even Channel

Integrating out torsion in the EC-Holst action gives the parity-even four-fermion interaction (Eq. 4). The $\gamma^2/(\gamma^2+1)$ factor is the standard torsion coupling strength.

3. Parity-Odd Channel: Holst vs. Nieh-Yan

The Holst term contributes to the parity-odd channel through the one-loop diagram sketched in Eq. (27). The Nieh-Yan density provides the counterterm structure: in dimensional regularization, the pole at $d = 4$ is proportional to $\mathcal{N}$. However, the finite remainder depends on the γ_5 prescription and on the precise Nieh–Yan subtraction convention (Sec. IV A). For this reason, α/M is treated as a phenomenological parameter constrained by data, with Eq. (8) providing the order-of-magnitude motivation.

4. Connection to Dark Energy

The inflationary suppression factor $\Xi = [(\alpha/M)\,M_{\rm Pl}] \times \mathcal{D}_{\rm inf}$ sets the dark energy scale because:

1. The effective action Eq. (7) has coefficient α/M (mass dimension -1); the tetrad-curvature combination $\varepsilon^{\mu\nu\rho\sigma} e^I_\mu e^J_\nu \mathcal{F}_{IJ\rho\sigma}$ has mass dimension $+2$ off-shell (Appendix H), giving an operator of mass dimension $+1$—three short of the required $+4$. The missing powers arise through on-shell evaluation at Planck densities (Sec. H 4). The factor of $M_{\rm Pl}$ in the dimensionless combination $[(\alpha/M)\,M_{\rm Pl}]$ arises from this on-shell evaluation ($K \sim M_{\rm Pl}$), not from an additional prefactor in the action itself.

2. The dimensionless combination $[(\alpha/M)\,M_{\rm Pl}] \sim 10^{-2}$ is the phenomenological best-fit value, consistent with the one-loop order-of-magnitude estimate $g^2\gamma/(32\pi^2) \sim 10^{-2}$.

3. Inflationary dilution gives $\mathcal{D}_{\rm inf} = e^{-3N_{\rm tot}}(T_{\rm reh}/M_{\rm GUT})^{3/2}$ (Eq. 17), where $e^{-3N_{\rm tot}}$ arises from dilution of the spin-density source $K_{ab} \propto a^{-3}$.

4. The vacuum energy density is $\rho_\Lambda = \Xi\,M_{\rm Pl}^4 = [(\alpha/M)\,M_{\rm Pl}] \times \mathcal{D}_{\rm inf} \times M_{\rm Pl}^4 \sim 10^{-2} \times 10^{-121} \times M_{\rm Pl}^4 \sim 10^{-123}\,M_{\rm Pl}^4 \approx (2.3\ \text{meV})^4$.

Appendix F: Covariant Rotation Framework

1. $1+3$ Decomposition

The spacetime metric is split via a timelike congruence u^a ($u^a u_a = -1$):

$$g_{ab} = -u_a u_b + h_{ab}, \tag{F1}$$

where h_{ab} is the spatial projection tensor. The covariant derivative of u_a decomposes into:

$$\nabla_b u_a = -u_b \dot{u}_a + \frac{1}{3}\Theta h_{ab} + \sigma_{ab} + \omega_{ab}, \tag{F2}$$

where $\Theta = \nabla_a u^a$ is the expansion scalar, $\sigma_{ab} = D_{\langle a} u_{b\rangle}$ is the shear tensor, $\omega_{ab} = D_{[a} u_{b]}$ is the vorticity tensor, and $\dot{u}_a = u^b \nabla_b u_a$ is the acceleration.

2. Generalized Friedmann Equation

The Gauss equation of the $1+3$ decomposition gives Eq. (12). The Raychaudhuri equation gives Eq. (13). Setting $H = \Theta/3$ and absorbing anisotropies into $\Lambda_{\rm eff}$:

$$\Lambda_{\rm eff} = \Lambda + \sigma^2 - \omega^2. \tag{F3}$$

For our almost-FLRW universe with negligible shear: $\Lambda_{\rm eff} \approx \Lambda - \omega^2 = \Lambda + c_\omega \omega^2$ with $c_\omega = -1$.

3. Sign Verification

The coefficient $c_\omega = -1$ is verified by:

1. Direct calculation from the constraint equation (this appendix).
2. Comparison with exact Bianchi V solutions.
3. Consistency with Bianchi VII$_h$ rotating cosmologies [23].
4. Dimensional analysis and symmetry arguments.

4. Isotropy Bound

The Planck 2016 analysis [23] constrains: $(\omega/H)_0 < 5\times10^{-11}$ (95% CL). This implies:

- $\omega_0 < (5\times10^{-11})\,H_0 = (5\times10^{-11})(2.2\times10^{-18}\ \mathrm{s}^{-1}) \approx 1.1\times10^{-28}$ rad/s
- $|c_\omega \omega^2|/H_0^2 < 2.5\times10^{-21}$
- Rotation completely negligible for background expansion
- Detectable only through parity-odd observables

Appendix G: Error Analysis

1. Fisher Matrix

Parameter uncertainties are propagated through the model using a Fisher matrix approach:

$$F_{ij} = \sum_\alpha \frac{1}{\sigma_\alpha^2} \frac{\partial O_\alpha}{\partial \theta_i} \frac{\partial O_\alpha}{\partial \theta_j}, \tag{G1}$$

where O_α are observables (H_0, σ_8, C_ℓ^{EB}, $A_{\rm dipole}$) and θ_i are model parameters.

2. Key Correlations

- $r(H_0, \sigma_8) = -0.45$: Higher H_0 correlates with lower σ_8 through modified growth.
- $r(A_0, p) = -0.62$: Amplitude-slope degeneracy in spin asymmetry.
- $r(\alpha/M, \mathcal{D}_{\rm inf}) = -0.89$: Strong anticorrelation; only the product Ξ is well-constrained.
- $r(\Delta N_{\rm eff}, H_0) = +0.72$: Extra species shift H_0 upward.

3. Error Propagation for H_0

$$\frac{\delta H_0}{H_0} = \sqrt{\left(\frac{\partial \ln H_0}{\partial \Omega_m}\right)^2 \delta\Omega_m^2 + \left(\frac{\partial \ln H_0}{\partial \Delta N_{\rm eff}}\right)^2 \delta(\Delta N_{\rm eff})^2 + \cdots} = 1.2\%. \tag{G2}$$

4. Goodness of Fit

Total $\chi^2/\mathrm{dof} = 1148.3/1142 = 1.006$, indicating an excellent fit with no evidence for overfitting despite the additional parameters.

Appendix H: Dimensional Analysis and Operator Normalization

This appendix provides a self-contained dimensional audit of the parity-odd operator and the dark energy scale derivation.

1. Mass Dimensions of Basic Objects

In natural units ($c = \hbar = 1$) with $8\pi G = M_{\rm Pl}^{-2}$:

Object	Symbol	[mass]
Reduced Planck mass	$M_{\rm Pl}$	+1
Tetrad (components)	e^I_μ	0
Tetrad determinant	$e = \det(e^I_\mu)$	0
Riemann curvature	$R_{\mu\nu\rho\sigma}$	+2
Curvature 2-form	$R^{IJ} = R^{IJ}{}_{\mu\nu} dx^\mu \wedge dx^\nu$	0
Contorsion	$K^I{}_{J\mu}$	+1
Levi-Civita symbol	$\varepsilon^{\mu\nu\rho\sigma}$	0
Integration measure	d^4x	−4
Lagrangian density	$\mathcal{L}$	+4

2. Holst Term Dimensional Check

The standard Holst action (Eq. 5):

$$S_{\rm Holst} = \frac{M_{\rm Pl}^2}{\gamma} \int e_I \wedge e_J \wedge R^{IJ} \tag{H1}$$

has dimension $[M_{\rm Pl}^2] + [e \wedge e \wedge R] = 2 + (-1 - 1 + 0) = 0$ in form language (the action is dimensionless). In component form: $[M_{\rm Pl}^2 \cdot R_{\mu\nu\rho\sigma}] = 2 + 2 = 4 = [\mathcal{L}]$. ✓

3. Parity-Odd Operator (Scaling Ansatz)

The effective action (Eq. 7) is:

$$S_{\rm eff} = \int d^4x \sqrt{-g}\, \frac{\alpha}{M}\, \varepsilon^{\mu\nu\rho\sigma} \times e^I_\mu\, e^J_\nu\, \mathcal{F}_{IJ\rho\sigma}, \tag{H2}$$

where $[\alpha/M] = -1$ and α is dimensionless ($[\alpha] = 0$, $[M] = +1$). The combined object $\mathcal{F}_{IJ\rho\sigma} \equiv \mathring{D}_{[\rho} K_{IJ\sigma]} + K_I{}^K{}_{[\rho} K_{KJ\sigma]}$ has contributions:

- $[\mathring{D}_\rho K_{IJ\sigma}]$: derivative (+1) × contorsion (+1) = +2
- $[K_I{}^K{}_\rho\, K_{KJ\sigma}]$: contorsion (+1) × contorsion (+1) = +2

So $[\mathcal{F}_{IJ\rho\sigma}] = +2$. The full operator in component form:

$$\left[\frac{\alpha}{M}\, \varepsilon^{\mu\nu\rho\sigma}\, e^I_\mu\, e^J_\nu\, \mathcal{F}_{IJ\rho\sigma}\right] = -1+0+0+0+2 = +1. \tag{H3}$$

This is mass dimension +1, not +4. The operator *by itself* does not produce a Lagrangian density of the correct dimension.

4. How $M_{\rm Pl}$ Enters: On-Shell Evaluation

The missing mass dimensions arise through the on-shell evaluation. At the bounce epoch ($\rho \to \rho_{\rm crit} \simeq 0.27\, \rho_{\rm Pl}$):

1. The contorsion is sourced by the fermion spin density: $K_{ab} \sim 8\pi G\, S_{ab} \sim S_{ab}/M_{\rm Pl}^2$, where S_{ab} is the spin angular momentum density (dimension +3 in natural units). Thus $[K_{ab}] = +3 - 2 = +1$. ✓
2. At Planck-scale densities, the spin density evaluates to $S_{ab} \sim M_{\rm Pl}^3$, giving $K_{ab}^{\rm bounce} \sim M_{\rm Pl}^3/M_{\rm Pl}^2 = M_{\rm Pl}$.
3. The Riemann curvature near the bounce is $R_{\mu\nu\rho\sigma} \sim H^2_{\rm bounce} \sim \rho_c/M_{\rm Pl}^2 \sim M_{\rm Pl}^2$.
4. The $\mathcal{F}$ operator evaluates to $\mathcal{F}^{\rm bounce} \sim K \cdot R \sim M_{\rm Pl} \cdot M_{\rm Pl}^2 = M_{\rm Pl}^3$.
5. The full on-shell energy density contribution is therefore:

$$\rho_\Lambda^{\rm bounce} \sim \frac{\alpha}{M} \cdot M_{\rm Pl}^3 \sim \frac{\alpha\, M_{\rm Pl}^3}{M} = [(\alpha/M)\, M_{\rm Pl}] \cdot M_{\rm Pl}^4 \sim 10^{-2}\, M_{\rm Pl}^4. \tag{H4}$$

5. Inflationary Dilution to Present

The contorsion source K_{ab} dilutes as a^{-3} during inflation (it is sourced by the fermion spin density, which dilutes with volume). After $N_{\rm tot}$ e-folds:

$$\rho_\Lambda^{\rm today} = \rho_\Lambda^{\rm bounce} \times \mathcal{D}_{\rm inf} = [(\alpha/M)\, M_{\rm Pl}] \times \mathcal{D}_{\rm inf} \times M_{\rm Pl}^4 = \Xi\, M_{\rm Pl}^4, \tag{H5}$$

where $\Xi \equiv [(\alpha/M)\, M_{\rm Pl}]\, \mathcal{D}_{\rm inf}$ is dimensionless. The corresponding cosmological constant is:

$$\Lambda_{\rm eff} = \frac{8\pi G}{3}\, \rho_\Lambda = \frac{\rho_\Lambda}{3M_{\rm Pl}^2/(8\pi)} = \Xi\, M_{\rm Pl}^2 \times \frac{8\pi}{3}. \tag{H6}$$

Thus $[\Lambda_{\rm eff}] = 0 + 2 = +2$, as required in the Friedmann equation $H^2 = \Lambda_{\rm eff}/3 + \cdots$. ✓

6. Summary of Dimensional Chain

$$\underbrace{\frac{\alpha}{M}}_{[-1]}\ \underbrace{\mathcal{F}^{\text{on-shell}}}_{[+3 \text{ at bounce}]} \xrightarrow{\text{dilution}} \underbrace{\Xi}_{[0]} \times \underbrace{M_{\rm Pl}^4}_{[+4]} = \underbrace{\rho_\Lambda}_{[+4]}. \tag{H7}$$

The factor of $M_{\rm Pl}$ in $\Xi = [(\alpha/M)\, M_{\rm Pl}]\, \mathcal{D}_{\rm inf}$ arises from the on-shell evaluation of the operator at Planck-scale densities, not from an additional prefactor in the action. This is the key dimensional bridge between the coefficient α/M (mass^{-1}) and the observed vacuum energy density (mass^4).

Off-shell status.—Away from the bounce epoch, the operator $(\alpha/M)\varepsilon^{\mu\nu\rho\sigma} e^I_\mu e^J_\nu \mathcal{F}_{IJ\rho\sigma}$ has mass dimension +1, not the +4 required for a Lagrangian density. The relation $\rho_\Lambda = \Xi\, M_{\rm Pl}^4$ therefore relies on evaluating the operator at the specific kinematic regime of the bounce

($\rho \to \rho_c$, $K \sim M_{\rm Pl}$, $R \sim M_{\rm Pl}^2$) where on-shell substitution supplies the missing three powers of mass. This is a *scaling ansatz*: it gives the correct dimension and numerical value at the bounce, and the subsequent inflationary dilution is a standard classical calculation, but the chain does not constitute a derivation from a renormalizable off-shell effective field theory. Promoting $\rho_\Lambda = \Xi\, M_{\rm Pl}^4$ to a first-principles result would require either (a) constructing a manifestly dimension-4 effective Lagrangian whose vacuum sector reproduces $\Xi\, M_{\rm Pl}^4$, or (b) demonstrating that a non-perturbative condensate freezes the on-shell value into a true vacuum energy. Both remain open problems (Sec. XV).

Appendix I: Reproducibility Materials

All MCMC results in this paper were obtained using the Cobaya framework (v3.5 for the original analysis; v3.6.1 with CAMB v1.6.5 for independent verification) [49] with stock CAMB as the theory engine. The cosmological model is ΛCDM extended by $\Delta N_{\rm eff}$ as a free parameter; no custom CAMB modifications are required. The full reproducibility package is available at:

`https://github.com/Hubify-Projects/bigbounce/tree/v2.0.0/reproducibility`

(pinned to tag `v2.0.0`).
The package includes:

- Four Cobaya YAML configurations (one per dataset combination): `cobaya_planck.yaml`, `cobaya_planck_bao.yaml`, `cobaya_planck_bao_sn.yaml`, `cobaya_full_tension.yaml`.
- `reproduce_cosmology.sh` — One-command script to regenerate all MCMC chains (∼4–12 h per configuration on 4 CPU cores).
- `galaxy_spins/spin_fit_stan.py` — Hierarchical Bayesian model fitting $A(z) = A_0(1+z)^{-p}e^{-qz}$ to published aggregate CW/CCW counts [35], using CmdStanPy.
- `data_build/build_galaxy_spin_dataset.py` — Downloads Galaxy Zoo DECaLS [69] from Zenodo and builds an object-level spiral galaxy catalog.
- `docs/IMPLEMENTATION_MAP.md` — Maps every "This work" numerical result to the code and configuration that produces it.
- `docs/KNOWN_GAPS.md` — Honest disclosure of reproducibility gaps.
- `results/mcmc_posterior_summary.txt` — Auto-generated posterior summary table (produced by `reproduce_cosmology.sh`).

Pre-computed MCMC chains are not included due to file size (∼1 GB per run). Bayes factors (Table VI) were estimated via the Savage-Dickey density ratio; dedicated nested sampling would provide more robust estimates but is not included. No CNN galaxy classifier or CMB polarization map analysis code is included; see the Data and Code Availability statement (Sec. XVI) for details.

Appendix J: Claims Classification

Table XIV classifies every major claim in this paper as *Derived* (follows from the formalism with explicit calculation shown), *Assumed* (input assumption not derived in this work), or *Fit/Inferred* (value determined by fitting to data).

[1] Planck Collaboration, N. Aghanim, *et al.*, Planck 2018 results. VI. cosmological parameters, Astronomy & Astrophysics **641**, A6 (2020), arXiv:1807.06209 [astro-ph.CO].

[2] S. Weinberg, The cosmological constant problem, Reviews of Modern Physics **61**, 1 (1989).

[3] A. G. Riess, W. Yuan, L. M. Macri, *et al.*, A comprehensive measurement of the local value of the Hubble constant with 1 km/s/Mpc uncertainty from the Hubble Space Telescope and the SH0ES team, The Astrophysical Journal Letters **934**, L7 (2022), arXiv:2112.04510 [astro-ph.CO].

[4] M. Asgari *et al.*, KiDS-1000 cosmology: Cosmic shear constraints on the amplitude of matter fluctuations, Astronomy & Astrophysics **645**, A104 (2021), arXiv:2007.15633 [astro-ph.CO].

[5] DES Collaboration, T. M. C. Abbott, *et al.*, Dark Energy Survey Year 3 results: Cosmological constraints from galaxy clustering and weak lensing, Physical Review D **105**, 023520 (2022), arXiv:2105.13549 [astro-ph.CO].

[6] DESI Collaboration, A. G. Adame, *et al.*, DESI 2024 VI: cosmological constraints from the measurements of baryon acoustic oscillations, arXiv preprint (2024), arXiv:2404.03002 [astro-ph.CO].

[7] DESI Collaboration, M. Abdul-Karim, *et al.*, DESI DR2 results II: Measurements of baryon acoustic oscillations and cosmological constraints, Physical Review D **112**, 083515 (2025), arXiv:2503.14738 [astro-ph.CO].

[8] A. Ashtekar and P. Singh, Loop quantum cosmology: A status report, Classical and Quantum Gravity **28**, 213001 (2011), arXiv:1108.0893 [gr-qc].

[9] F. W. Hehl, P. von der Heyde, G. D. Kerlick, and J. M. Nester, General relativity with spin and torsion: Foundations and prospects, Reviews of Modern Physics **48**, 393 (1976).

[10] N. J. Popławski, Cosmological constant from quarks and torsion, Annalen der Physik **523**, 291 (2011),

TABLE XIV. Classification of claims: Derived vs. Assumed vs. Fit/Inferred. This table is intended to prevent overclaiming and to make the epistemic status of each result transparent.

Claim	**Status**	**Where**	**Notes**
Derived (explicit calculation shown)			
LQC bounce at $\rho_{\rm crit} \simeq 0.27$–$0.41\,\rho_{\rm Pl}$	Derived	Eq. (10)	From LQC; range reflects ABCK ($\gamma = 0.274$) vs. DLM ($\gamma =$
Four-fermion interaction from torsion	Derived	Eq. (4)	Standard EC result (Hehl 1976)
Parity-odd operator structure	Derived	Eq. (6)	From Holst term + fermion loop
$\mathcal{D}_{\rm inf} = e^{-3N_{\rm tot}}(T_{\rm reh}/M_{\rm GUT})^{3/2}$	Derived	Eq. (17)	Dilution of $K_{ab} \propto a^{-3}$
$C_\ell^{EB} \approx 2\beta(C_\ell^{EE} - C_\ell^{BB})$	Derived	Eq. (22)	Standard birefringence formula
$\Lambda_{\rm eff} = \Xi\, M_{\rm Pl}^2 + c_\omega \omega^2$	Derived	Eq. (15)	$1+3$ covariant decomposition
Dimensional chain: $\rho_\Lambda = \Xi\, M_{\rm Pl}^4$	Derived	App. H	On-shell evaluation
Assumed (input, not derived in this work)			
$w = -1$ at late times	Assumed	Sec. II C 2	IR effective action not computed
$\gamma = 0.274 \pm 0.020$	Assumed	Eq. (2)	LQG black hole entropy
Parent rotating black hole origin	Assumed	Sec. II B	Popławski scenario
Smooth bounce $\rightarrow$ slow-roll transition	Assumed	Sec. XV D	Not numerically simulated
$A(z) = A_0(1+z)^{-p}e^{-qz}$ functional form	Assumed	Eq. (20)	Phenomenological ansatz
Fit / Inferred from data			
$H_0 = 69.2 \pm 0.8$ km/s/Mpc	MCMC fit	Eq. (24)	Original (superseded; see Sec. III D)
$H_0 = 67.68 \pm 1.06$ km/s/Mpc	Verification	Table III	Full-tension, frozen chains
$\sigma_8 = 0.785 \pm 0.016$	MCMC fit	Eq. (25)	Original (tension dataset)
$\sigma_8 = 0.803 \pm 0.008$	Verification	Table III	Full-tension, frozen chains
$\Delta N_{\rm eff} = -0.020 \pm 0.169$	Verification	Table III	Full-tension; consistent with zero
$\Delta N_{\rm eff} = +0.065 \pm 0.17$	Verification	Table III	Planck+BAO+SN; consistent with zero
$\alpha/M \sim 10^{-21}$ GeV^{-1}	Fit	Table XIII	One-loop motivates order of magnitude
$A_0 \sim 0.003$	Empirical fit	Sec. III B	Contested; 9–12 order gap from α/M
$\beta = 0.242^\circ \pm 0.061^\circ$	Literature combination	Sec. XIV D	Summary likelihood of Planck + ACT
$f_{\rm photon} \times C_0 = 1.73 \pm 0.44$	Consistency check	Sec. XIV D	Derived from combined β; $\mathcal{O}(1)$
$\ln B = +4.8$ (tension dataset)	Inferred	Eq. (40)	Dataset-dependent

arXiv:1005.0893 [gr-qc].

[11] N. J. Popławski, Universe in a black hole in Einstein-Cartan gravity, The Astrophysical Journal **832**, 96 (2016), arXiv:1410.3881 [gr-qc].

[12] N. J. Popławski, Cosmology with torsion: An alternative to cosmic inflation, Physics Letters B **694**, 181 (2010), arXiv:1007.0587 [astro-ph.CO].

[13] N. J. Popławski, Four-fermion interaction from torsion as dark energy, General Relativity and Gravitation **44**, 491 (2012), arXiv:1102.5667 [gr-qc].

[14] R.-Y. Liu *et al.*, Einstein-Cartan torsion and the S_8 tension, European Physical Journal C **85**, 112 (2025).

[15] L. Freidel, D. Minic, and T. Takeuchi, Quantum gravity, torsion, parity violation and all that, Physical Review D **72**, 104002 (2005), arXiv:hep-th/0507253 [hep-th].

[16] S. Mercuri, Fermion coupling to the Holst action, Physical Review D **73**, 084016 (2006), arXiv:gr-qc/0601013 [gr-qc].

[17] S. Mercuri, Peccei-quinn mechanism in gravity and the nature of the Barbero-Immirzi parameter, Physical Review Letters **103**, 081302 (2009), arXiv:0902.2764 [gr-qc].

[18] I. L. Shapiro and P. M. Teixeira, Quantum Einstein-Cartan theory with the Holst term, Classical and Quantum Gravity **31**, 185002 (2014), arXiv:1402.4854 [gr-qc].

[19] A. Ashtekar, J. C. Baez, A. Corichi, and K. Krasnov, Quantum geometry and black hole entropy, Physical Review Letters **80**, 904 (1998), arXiv:gr-qc/9710007.

[20] M. Domagała and J. Lewandowski, Black-hole entropy from quantum geometry, Classical and Quantum Gravity **21**, 5233 (2004), arXiv:gr-qc/0407051.

[21] K. A. Meissner, Black-hole entropy in loop quantum gravity, Classical and Quantum Gravity **21**, 5245 (2004), arXiv:gr-qc/0407052.

[22] G. F. R. Ellis and M. Bruni, Covariant and gauge-invariant approach to cosmological density fluctuations, Physical Review D **40**, 1804 (1989).

[23] D. Saadeh, S. M. Feeney, A. Pontzen, H. V. Peiris, and J. D. McEwen, How isotropic is the universe?, Physical Review Letters **117**, 131302 (2016), arXiv:1605.07178 [astro-ph.CO].

[24] H. Golden, Systematic closure of minimal first-principles routes to dark energy in Einstein-Cartan-Holst gravity (2026), companion technical note, available as ancillary material with the arXiv submission.

[25] Y. Minami and E. Komatsu, New extraction of the cosmic birefringence from the Planck 2018 polarization data, Physical Review Letters **125**, 221301 (2020), arXiv:2011.11254 [astro-ph.CO].

[26] J. R. Eskilt and E. Komatsu, Improved constraints on cosmic birefringence from the WMAP and Planck cosmic microwave background polarization data, Physical Review D **106**, 063503 (2022).

[27] S. M. Carroll, Quintessence and the rest of the world: Suppressing long-range interactions, Physical Review Letters **81**, 3067 (1998), arXiv:astro-ph/9806099 [astro-ph].

[28] P. Diego-Palazuelos and E. Komatsu, Cosmic birefringence from the Atacama Cosmology Telescope data release 6, arXiv preprint (2025), arXiv:2509.13654 [astro-ph.CO].

[29] L. Yin, G.-H. Du, T.-N. Li, and X. Zhang, Joint constraints on cosmic birefringence and early dark energy from ACT, Planck, DESI, and PantheonPlus, arXiv preprint (2026), arXiv:2601.13624 [astro-ph.CO].

[30] SPIDER Collaboration, Constraints on cosmic birefringence from SPIDER, Planck, and ACT observations, arXiv e-prints (2025), arXiv:2510.25489 [astro-ph.CO].

[31] ACT Collaboration, F. J. Qu, *et al.*, The Atacama Cosmology Telescope: DR6 gravitational lensing map and cosmological parameters, The Astrophysical Journal **962**, 112 (2024), arXiv:2304.05203 [astro-ph.CO].

[32] LiteBIRD Collaboration, E. Allys, *et al.*, Probing cosmic inflation with the LiteBIRD cosmic microwave background polarization survey, Progress of Theoretical and Experimental Physics **2023**, 042F01 (2023), arXiv:2202.02773 [astro-ph.IM].

[33] L. Shamir, Handedness asymmetry of spiral galaxies with $z < 0.3$ shows cosmic parity violation and a dipole axis, Physics Letters B **715**, 25 (2012), arXiv:1207.3674 [astro-ph.CO].

[34] L. Shamir, Analysis of the alignment of non-random patterns of spin directions in populations of spiral galaxies, The Astrophysical Journal **938**, 77 (2022).

[35] L. Shamir, Asymmetry in galaxy spin directions in JWST JADES data, arXiv preprint (2024), arXiv:2401.09450 [astro-ph.GA].

[36] D. Patel and H. Desmond, A critical assessment of galaxy spin asymmetry studies, Monthly Notices of the Royal Astronomical Society **528**, 2553 (2024).

[37] O. H. E. Philcox and J. Ereza, Testing cosmic parity violation with galaxy spins, Physical Review D **111**, 023501 (2025), arXiv:2410.18185 [astro-ph.CO].

[38] M. J. Longo, Detection of a dipole in the handedness of spiral galaxies with redshifts $z \sim 0.04$, Physics Letters B **699**, 224 (2011), arXiv:1104.2815 [astro-ph.CO].

[39] ACT Collaboration, ACT DR6 constraints on extended cosmological models, arXiv e-prints (2025), $N_{\rm eff} = 2.86 \pm 0.13$ (ACT DR6 alone), arXiv:2503.14454 [astro-ph.CO].

[40] T. Liu, X. Li, T. Xu, M. Biesiada, and J. Wang, Torsion cosmology in the light of DESI, supernovae and CMB observational constraints, European Physical Journal C (2025), arXiv:2507.04265 [gr-qc].

[41] I. Pantos and L. Perivolaropoulos, Dissecting the Hubble tension: Insights from a diverse set of sound horizon-free h_0 measurements, arXiv e-prints (2026), sHF: $H_0 = 69.37 \pm 0.34$; 6.5σ tension between distance ladder and other methods, arXiv:2601.00650 [astro-ph.CO].

[42] ACT Collaboration, ACT DR6 power spectra, likelihoods and ΛCDM parameters, arXiv e-prints (2025), $H_0 = 68.22 \pm 0.36$ (P-ACT + DESI DR1), arXiv:2503.14452 [astro-ph.CO].

[43] K. C. Wong *et al.*, H0LiCOW – XIII. a 2.4% measurement of H_0 from lensed quasars: 5.3σ tension between early- and late-universe probes, Monthly Notices of the Royal Astronomical Society **498**, 1420 (2020), arXiv:1907.04869 [astro-ph.CO].

[44] D. W. Pesce *et al.*, The Megamaser Cosmology Project. XIII. combined Hubble constant constraints, The Astrophysical Journal Letters **891**, L1 (2020), arXiv:2001.09213 [astro-ph.CO].

[45] W. L. Freedman, Measurements of the Hubble constant: Tensions in perspective, The Astrophysical Journal **919**, 16 (2021), arXiv:2106.15656 [astro-ph.CO].

[46] X. Li *et al.*, Hyper Suprime-Cam Year 3 results: Cosmology from cosmic shear two-point correlation functions, Physical Review D **108**, 123518 (2023), arXiv:2304.00702 [astro-ph.CO].

[47] V. Ghirardini *et al.*, The SRG/eROSITA all-sky survey: Cosmology constraints from cluster abundances in the western galactic hemisphere, Astronomy & Astrophysics **689**, A298 (2024), arXiv:2402.08458 [astro-ph.CO].

[48] H. T. Nieh and M. L. Yan, An identity in Riemann-Cartan geometry, Journal of Mathematical Physics **23**, 373 (1982).

[49] J. Torrado and A. Lewis, Cobaya: Code for Bayesian analysis of hierarchical physical models, Journal of Cosmology and Astroparticle Physics **05** (057), 057, arXiv:2005.05290 [astro-ph.IM].

[50] G. Cowan, K. Cranmer, E. Gross, and O. Vitells, Statistics, Phys. Rev. D **110**, 030001 (2024), sec. 40 in the Review of Particle Physics. See Sec. 40.2.2 for combining measurements.

[51] Y.-F. Cai, W. Xue, R. Brandenberger, and X. Zhang, Non-gaussianity in a matter bounce, JCAP **0905**, 011, arXiv:0903.0631.

[52] H. Golden, Testing the matter bounce with primordial non-gaussianity: Forecasts for spherex and megamapper (2026), in preparation.

[53] K. Gödel, An example of a new type of cosmological solutions of Einstein's field equations of gravitation, Reviews of Modern Physics **21**, 447 (1949).

[54] F. Cabral, F. S. N. Lobo, and D. Rubiera-Garcia, Cosmological bounces, cyclic universes, and effective cosmological constant in Einstein-Cartan-Dirac-Maxwell theory, Physical Review D **102**, 083509 (2020), arXiv:2003.07463 [gr-qc].

[55] A. Perez and D. Sudarsky, Dark energy from quantum gravity discreteness, Physical Review Letters **122**, 221302 (2019), arXiv:1711.05183 [gr-qc].

[56] F. Izaurieta, E. Ramírez, and E. Rodríguez, Spinning dark matter and the Hubble tension, Physics of the Dark Universe **30**, 100662 (2020).

[57] S. Akhshabi and E. Zamani, Effects of torsion on cosmological distance measures, Physical Review D **107**, 064023 (2023).

[58] S. Legner, W. Handley, and W. Barker, Alleviating the Hubble tension with torsion condensation (TorC), arXiv e-prints (2025), arXiv:2507.09228 [astro-ph.CO].

[59] L. Fabbri, Avoiding singularities with propagating torsion, arXiv e-prints (2025), arXiv:2502.17979 [gr-qc].

[60] A. Sanyal, P. K. Dhankar, A. Munyeshyaka, S. Islam, F. Rahaman, and B. Pourhassan, Cosmic hysteresis in reconstructed $f(t)$ bounce models: A torsion-based thermodynamic perspective, arXiv preprint (2026), arXiv:2602.15924 [gr-qc].

[61] S. Alam, S. Sen, and S. Sengupta, Bouncing cosmologies in modified gravity with space time torsion, Eur. Phys. J. C (2025), arXiv:2509.03508 [gr-qc].

[62] P. Chattopadhyay, One-loop effective action in chiral Einstein-Cartan gravity, arXiv preprint (2023), arXiv:2310.10405 [gr-qc].

[63] L. Shamir, The distribution of galaxy rotation in JWST advanced deep extragalactic survey, MNRAS **538**, 76 (2025), arXiv:2502.18781 [astro-ph.GA].

[64] J. Shim *et al.*, Probing vector chirality in the early universe, arXiv e-prints (2025), revised August 2025, arXiv:2406.06080 [astro-ph.CO].

[65] G. 't Hooft, Naturalness, chiral symmetry, and spontaneous chiral symmetry breaking, NATO Sci. Ser. B **59**, 135 (1980).

[66] T. Fujita, K. Murai, H. Nakatsuka, and S. Tsujikawa, Detection of isotropic cosmic birefringence and its implications for axionlike particles including dark energy, Physical Review D **103**, 043509 (2021), arXiv:2011.11894 [astro-ph.CO].

[67] C. Heinrich, O. Dore, and E. Krause, Measuring $f_{\rm nl}$ with the spherex multi-tracer redshift space bispectrum, (2023), arXiv:2311.13082.

[68] H. Golden, Spin-torsion cosmology and the search for geometric dark energy: Interactive data and supplementary materials (2026), interactive data visualizations, observational comparison charts, and complete paper available online.

[69] M. Walmsley, C. Lintott, T. Géron, *et al.*, Galaxy Zoo DECaLS: Detailed visual morphology measurements from volunteers and deep learning for 314 000 galaxies, Monthly Notices of the Royal Astronomical Society **509**, 3966 (2022), arXiv:2102.08414 [astro-ph.GA].